Appendix I

TFL Amendments

TFL 6 Amendments

Amend No	Date	Schedule A (ha)	Schedule B (ha)	Net Change	Amendment Description	Operation	Local Area
n/a	Oct. 19, 1953	285.3		285.3	T.L. 3013 and R/W lot 320 J.L. are added to Sched. A	Holberg	Winter Hbr
n/a	Oct 19, 1953	127.5		127.5	Pt. Sec. 20 & 21, Tp. 11, Atkins Cove and N.W. 1/4 Sec. 35, Tp. 38, Holberg are added to Sched. A	Jeune Ldg and Holberg	Atkins Cove and Holberg
n/a	Oct. 19, 1953	-259.0	259	0	P.L. 328 is transferred from sched. A to sched. B	Holberg	NE Holberg
n/a	Nov. 30, 1953	258.6		258.6	T.L. 3014 is added to sched. A	Winter Hbr	Winter Hbr
n/a	Dec. 8, 1954	114.1		114.1	Pt. Sec. 10, 21 & 28, Tp. 18 is added to sched. A	Quatsino Sd	Mahatta & Quatsino Village
n/a	May 19, 1956	-0.4		-0.4	L. 1199 (school site) is deleted from sched. A	Mahatta	Mahatta River Camp
1	Nov. 28, 1956		-0.8	-0.8	2 ac. Of L. 13 is deleted from sched. B for private use	Jeune Ldg	Atkins Cove
2	Jan. 26, 1956		-1.2	-1.2	3 ac. Of Sec. 23, Tp. 18, is deleted from sched. B (private use)	Quatsino Sd	Drake Is.
3	Mar. 8, 1956	104.0		104	L. 95 & Pt. 317 are added to sched. A	Holberg	Holberg Townsite
4	Mar. 21, 1956			0	40' R/W between Robson Cove and L. 52 is deleted from sched. B	Holberg	Koprino Hbr
5	Mar 21, 1956	23.1		23.1	Pt. Sec. 8, Tp. 18 Quatsino Sd. Is added to sched. A	Mahatta	Quatsino Sd - near mouth of Johnson Cr
6	Apr. 12, 1956			0	Pt. Of Powell River's road is deleted from sched. B	Coal Harbour	East CH by Frances Lake
7	Apr. 23, 1956	66.6		66.6	Pt. Sec. 9, Tp. 18, Quatsino Sd. is added to sched. A	Mahatta	Quatsino Sd - near mouth of Johnson Cr
8	Aug. 17, 1956		-30.61	-30.61	M.C. Lots 81, 82, 89 & 108 on Neroutsos In. are deleted from sched. B	Jeune Ldg	Neuotsos Inlet NW side (Yreka Mines)
9	Sept. 9, 1957	334.3		334.3	Lots 1 & 2 , Sec. 1, are added to sched. A	Coal Hbr	Coal Hbr. Above townsite
10	Sept. 26, 1957	15.4		15.4	Pt. Sec. 26, Tp. 18, Quatsino Sd. are added to sched. A	Holberg	west of Quatsino village
11	Oct. 9, 1957			0	66' r/w access to I. Folis X70756, Quatsino S. is deleted from sched. B	Holberg	west of Quatsino village
12	Jan. 13, 1958			0	L. 2090, is deleted from sched. B (see #13)	Jeune Ldg	Port Alice south of Pulp Mill
13	Mar. 21, 1958	-15.9		-15.9	Cancels and replaces Amd. 12. Removes L. 2089 & 2090 from sched. A (for Port Alice Townsite expansion)	Jeune Ldg	Port Alice Pulp Mill
14	Mar. 21, 1958		-1.6	-1.6	NE 1/4 Sec. 36, Tp. 29, Quatsino Sd. is deleted from sched. B	Mahatta	Koskimo Bay

Amend No	Date	Schedule A (ha)	Schedule B (ha)	Net Change	Amendment Description	Operation	Local Area
15	May 7. 1958	42.9		42.9	Pt. Sec. 20, Tp. 18, Quatsino Sd. is added to sched. A	Holberg	west of Quatsino village
16	May 21, 1958			0	Clause 2 of the TFL 6 document changed. TFL 6 now appturtenant to all company "Manufacturing Plants" in B.C. Port Alice to be maintained at a capacity not less than its average annual production for the 5 yr. period ending Oct. 26, 1950.	Whole TFL	Whole TFL
					Clause 3 also changed. Wood removed from company TFL's may now be processed in company plants or exchanged for an equivalent volume of wood which in turn must be so processed.		
17	Sept. 9. 1958			0	Cancels and replaces amend. 16. Clause 2 as per #16 accepted. Clause 3 revised to cover wood from TFL 6 only	Whole TFL	Whole TFL
18	July 11, 1960			0	Timber Sale X47566 is added to sched. B	Holberg	NE Holberg along Goodspeed R
19	May 3, 1962			0	B.C. Tel Reflector Site is deleted from sched. B	Holberg	near Mt Byng & Colony Lake
20	Aug. 24, 1961	153.8		153.8	L. 189 & 188 JL: Pcl. A, B & D of Sec. 14, Tp. 37 San Jo are added to sched. A	Jeune Ldg Holberg	JL campsite area Holberg DND base
21	Aug. 24, 1961	259.0		259	TL 621 is added to sched. A	Jeune Ldg	east of JL campsite
22	Mar. 1, 1962			0	40' r/w at Kokwina Cove is deleted from sched. B	Jeune Ldg	Atkins Cove
23	Mar. 9, 1962	-259.0	259.0	0	TL 621 is transferred from sched. A to B	Jeune Ldg	east of JL campsite
24	Apr. 2, 1962	-62.7		-62.7	L. 95 is deleted from sched. A (see also #3)	Holberg	Holberg Townsite
25	Apr. 30, 1962		471.8	471.8	T.S. X47578, X43941, X46282, X47143, X45711 & X44635 are added to sched. B	Various Operations	scattered throughout TFL
26	Aug. 21, 1962	-420.8	420.8	0	TL 10947 & TL 2057 are transferred to sched. B	Jeune Ldg	Alice Lake
27	Dec. 12, 1962	-3.1		-3.1	Road r/w within TL 10282 is deleted from TFL 6	Holberg	near Nahwitti Lake
28	Dec. 12, 1962		-2.2	-2.2	Road r/w is deleted from sched. B	Holberg	Nahwitti L
29	Apr. 8, 1963	-33.5		-33.5	Pipeline r/w access road and powerline is deleted from TFL 6	Port Alice	Pipeline from mill to Victoria L.
30	Apr. 8, 1963	73.5		73.5	Fr. E 1/2 Sec. 10, Tp. 18, added to sched. A	Mahatta	mouth of Ingersoll R
31	May 31, 1963	63.8		63.8	L. 95, returned to sched. A (see also #24 & 3)	Holberg	Holberg Townsite

Amend No	Date	Schedule A (ha)	Schedule B (ha)	Net Change	Amendment Description	Operation	Local Area
32	Nov. 1, 1963	47.3		47.3	L. 1382 is added to sched. A	Jeune Ldg	Thurburn Bay
33	Mar. 20, 1964	56.6		56.6	Westerly 17.5 chains of Sec. 52 is added to sched. A	Holberg	Ahwachaolto In.
34	May 22, 1964	-190.2	16.2	-174	L. 2137 is deleted from schedule A & B.	Jeune Ldg	Port Alice townsite
35	Sept. 8, 1964			0	Road r/w between Robson cove and Sec. 52 is removed from TFL	Botel Lake	Robson Cove (Koprino)
36	Jan. 12, 1965			0	100' Tramway r/w is deleted from TFL	Jeune Ldg	Neroutsos Inlet (Yreka Mines)
37	Apr. 28, 1965			0	300 ' Tramway r/w is deleted from TFL	Jeune Ldg	Neroutsos Inlet (Yreka Mines)
38	May 14, 1965	-259.0	259.0	0	TL 12174 transferred to sched. B	Holberg	NE Holberg near Goodspeed R
39	June 25, 1965	45.3		45.3	Frac. SW 1/4, Sec. 35, Tp. 11, is added to sched. A	Jeune Ldg	Varney Bay
40	July 22, 1965	4.0		4	Pcl. 1 of Pcl. A of Sec. 14, Tp. 37, is added to sched. A	Holberg	near DND base
41	Oct. 12, 1965	-0.4		-0.4	Passive Repeater site in L. 2, Sec. 10 is deleted from sched. A	Coal Hbr	south of CH townsite
42	Mar. 10, 1966	-44.0	-171.5	-215.5	D.N.D.site & r/w removed from sched. A & B	Holberg	DND site
43	Mar. 14, 1966	-259.0	259.0	0	TL 10805 transferred to sched. B	Holberg	NE Holberg
44	July 11, 1966	-1.1		-1.1	Telephone cable r/w within PLL 193 is deleted from sched. A	Quatsino Peninsula	Colony L
45	July 11, 1966			0	Road r/w within Sec. 19, Tp. 11, is deleted & removed from sched. B	Jeune Ldg	Atkins Cove
46	Mar 6, 1967			0	Site 50'x50' is deleted from sched. B (for nautical beacon)	Mahatta	Cliffe Pt. (westerly tip)
47	Jan. 15, 1968			0	Clause 15A is added to indenture		
48	Jan. 30, 1968		-4.9	-4.9	Pt. Rainbow M.C. is deleted from sched. B	Coal Hbr	Apple Bay
49	July 30, 1968	231.8		231.8	E 1/2 of E 1/2 , S 9, Tp 37 except pt in plan 174105; SE 1/4, S 10, Tp 37; NW 1/4 of NW 1/4, Tp 37; SW 1/4 of NW 1/4 and NW 1/4 of SW 1/4, S 14, Tp 37 except pt shown in erd on plan 1762-R being CT 6441-W; E 1/2 of E 1/2, S 6 Tp 37	Holberg	San Josef Valley
50	Nov. 5, 1968		-0.2	-0.2	L. 2249 transmitter tower is deleted from sched. B	Jeune Ldg	Teeta Cr
51	Sept. 25, 1969		0.1	0.1	Lighthouse reserve is deleted from sched. B	Quatsino Peninsula	Quatsino Narrows

Amend No	Date	Schedule A (ha)	Schedule B (ha)	Net Change	Amendment Description	Operation	Local Area
52	Nov. 27, 1969			0	The word "northerly" in the 25th line on p. 2 of the description of sched. B lands is deleted and northern bdy. of TL 7636 moved north.		
53	Aug. 24, 1970		-88.6	-88.6	Pt. of Sec. 24, Tp. 10 is deleted from sched. B (for Utah Mining site)	Coal Hbr.	
54	May 22, 1970		0.8	0.8	2 ac of L. 13, is added to sched. B.		Atkins Cove
55	Dec. 3, 1970	-15.1		-15.1	L. 1 of L. 202 is deleted from sched. A	Jeune Ldg	Alice L
56	Mar. 23, 1971			0	Clause 32 changed. Provides issuance of CPs by the District Forester		
57	Mar. 8, 1971	-8.9		-8.9	L. 1 of Sec. 14, Tp. 10 is deleted from TFL 6	Coal Hbr.	Quatsino Band Village
58	June 8, 1971		-1.2	-1.2	Pt. of TL 620 is deleted from sched. B'	Jeune Landing	Jeune Ldg campsite dump
59	Aug. 20, 1971	17.3		17.3	L. 187 is added to sched. A	Jeune Landing	QDLS
60	July 30, 1971	-0.9		-0.9	Pt. of L. 176 is deleted from sched. A	Jeune Landing	north of Alice L.
61	Sept. 16, 1971			0	Pt. (12.1m2) of L. 1124 & 1198 is deleted from sched. B (for nautical beacons)	Rupert Inlet	Quatrsino Narrows
62	nov, 29, 1971			0	Pt. (12.1 m2) of Sec. 28 & 33, Tp. 11 is deleted from sched. B (for nautical beacons)	Rupert Inlet	Quatrsino Narrows
63	Sept. 28, 1972			0	Sketch for amendment #62 amended	Rupert Inlet	Quatrsino Narrows
64	Feb. 26, 1973	247.3		247.3	Koprino IR 10 (of secs. 16, 17, 20 & 21, Tp. 27) and the East 62.5 ch. of sec. 52 are added to sched. A	Holberg	Koprino Hbr
65	July 4, 1974		-60.7	-60.7	Additional area is deleted from TFL 6 (for Utah Mine site)	Coal Hbr.	Utah Mine site
66	Oct. 24, 1977		0.1677	0.1677	Blk A of L. 317 is deleted.	Holberg	Holberg Townsite
67	Apr. 17, 1978	13.9		13.9	Sec. 50, Schloss Is. is added to sched. A		Koprino Hbr
68	Feb. 8, 1979		-0.9	-0.9	L. 347 is deleted from TFL 6 (school site)	Holberg	Holberg Townsite
69	Sept. 10, 1980	-3.0		-3	Area is deleted from		Port Alice
70	June 13, 1985	-.83		-0.83	Lot 450 withdrawn from TFL 6	Holberg	Holberg Townsite
71	Nov. 1, 1985			0	Adds District Manager to part 3 of document		
72	Aug. 19, 1986	0.4		0.4	Passive Repeater site in L. 2, Sec. 10 is added from sched. A (cancels amend 41)	Coal Hbr	south of CH townsite
73	Feb. 9, 1987	33.6		33.6	Lot 104 is added to sched A	Jeune Landing	north of QDLS

Amend No	Date	Schedule		Net Change	Amendment Description	Operation	Local Area
		A (ha)	B (ha)				
74	Sept. 28, 1988			0	Reduction of the crown portion of the AAC by 5% due to a change in Corporate control.	whole TFL	
75	Oct. 30, 1989		405.0	405	Crown area is deleted for purposes of Raft Cove Prov. Park	Holberg	Raft Cove - Macjack Rive
76	July 16, 1992	312.8		312.8	NW 1/4 S. 1, Tp 41; SE 1/4 S. 11, Tp 41; E 159 ac frac 1/2 S. 12, Tp 41 except that lying in the NW 1/4; SE 1/4 S. 12, Tp 41 (160 ac); and NE 1/4 S. 31, Tp 38.	Holberg	head of Ronning Cr
77	Jan. 1, 1996			0	subparagraph 1.07 (a) reference to 100,620 m3 is changed to 21,120 m3	whole TFL	
78	Nov. 10, 1998		-0.23	-0.23	BC Hydro substation is deleted from L. 189	Jeune Landing	Port Alice townsite
79	Dec. 31, 1998	-14.8		-14.8	Additional Quatsino Band townsite is deleted from L. 198	Coal Hbr	north of CH townsite
80	Jan. 17, 1997			0	Added the 50% vol. of undivided lot 305 previously owned by the Prov Govt	Holberg	Botel Lake peninsula
81	n/a	n/a			an addition to Port Alice townsite is in the works but status is unknown	Jeune Ldg	Port Alice townsite
82	Oct. 1, 1998		-1869.1		south Mahatta River drainage area is deleted from Sched B for land trade purposes as part of the Strathcona Exchange	Mahatta	south Mahatta R
83	Oct. 1, 1998	11572.5	19669.8	31242.3	Port McNeill TFL 25 Bk 4 is transferred to TFL 6 as Bk 2; additionally subparagraph 1.07 (a) is changed.	Port McNeill	whole block
84	n/a		n/a - in process		Cluxewe Salt Marsh will be deleted from sched A	Port McNeill	Cluxewe Salt Marsh
		12653.9	19788.1	34309.1			

Appendix II

Timber Supply Information Package

WFP
Western Forest Products Limited

TIMBER SUPPLY ANALYSIS INFORMATION PACKAGE

IN PREPARATION OF

MANAGEMENT PLAN 9

FOR

TREE FARM LICENCE 6

Revised and Submitted to the Timber Supply Branch,
Ministry of Forests

August 2000

REGISTERED PROFESSIONAL FORESTER
PROVINCE OF
DAVID A. BYNG
BRITISH COLUMBIA
NO. 3161

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Planning and Resource Inventory Forester
Western Forest Products Limited

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APPENDICES

1.0 INTRODUCTION

This Information Package provides a comprehensive listing of data, assumptions and modelling procedures that will be used in preparing the Timber Supply Analysis for Management Plan #9. The information and factors related to the Timber Supply Analysis are presented in a manner that will allow for a spatially explicit volume-based timber supply analysis.

Western Forest Products Limited (WFP) will undertake a timber supply analysis for TFL 6 to provide estimates of timber harvest levels over a 200-year planning horizon that are a function of the land base, timber volumes and growth rates, along with constraints designed to protect and enhance non-timber resource values. An initial harvest forecast will examine the impacts of current management practices and operational requirements of the Forest Practices Code and other current regulations and guidelines. Sensitivity analyses will investigate the expected impacts of various management scenarios designed to meet specified goals and to determine the relative importance of variations in assumptions. Varying the harvest forecast through option analyses will be modelled either through the removal of operable areas from the timber harvesting land base, through the imposition of forest cover harvest constraints or through changes in growth and yield assumptions.

Each timber supply forecast is guided by the objectives of sustaining harvest level, achieving the potential long-term harvest level and planning an orderly transition from the current level of harvest to the long-term sustainable level. In meeting these objectives WFP will continue to harvest to the timber inventory profile within the constraints set by cut control regulations, approved harvesting plans, market demand, objectives for other resources and maintenance of long-term productivity. Due to the current age class structure in TFL 6 it is expected that the majority of the cut in the short to medium term can be concentrated in mature and over-mature stands without compromising the objectives stated above.

The forest estate models, COMPLAN and/or ATLAS, will be used to complete the timber supply analysis. COMPLAN and ATLAS are spatial simulation models that allow the effects of adjacency to be modelled and incorporated into the timber supply analysis. They also permit the incorporation of existing Forest Development Plans (FDP) to bring greater operational relevance to the results. In addition, future cutblock design generated using a computer simulated block-building algorithm, will be completed and incorporated into the analysis. The end result is a detailed analysis that can guide operational planning and be checked and verified as planning proceeds.

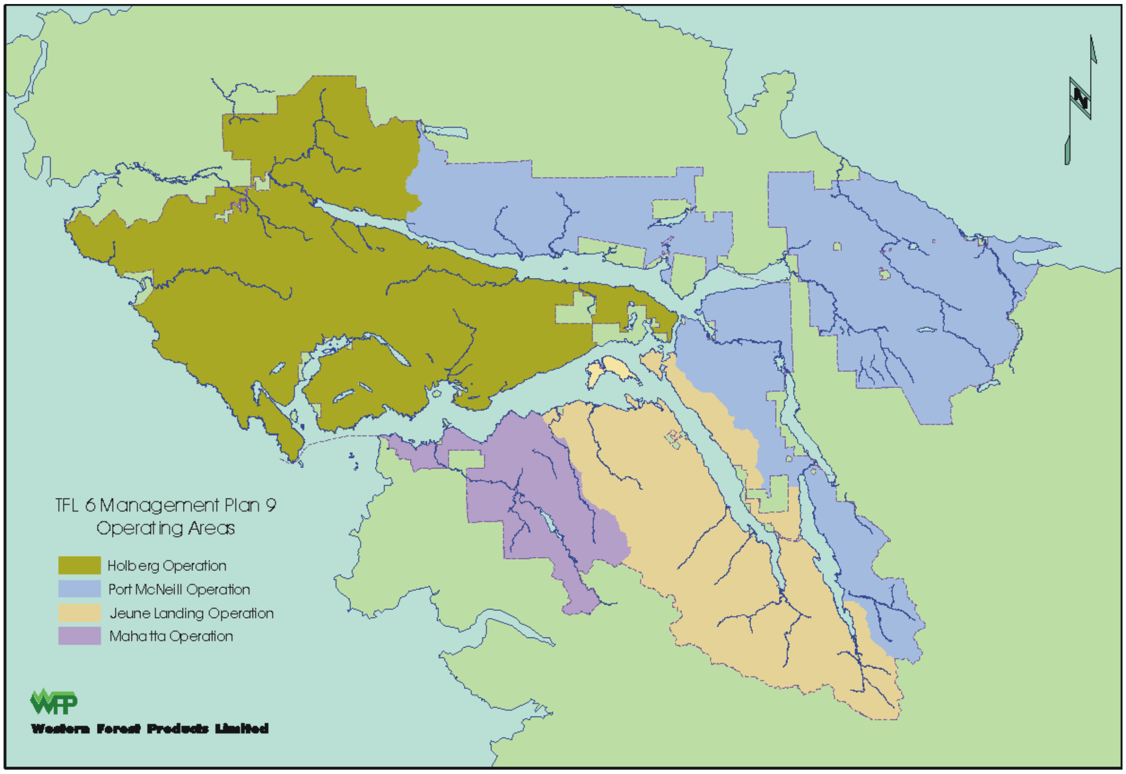


Figure 1: TFL 6 Operating Areas

2.0 PROCESS

This Information Package has been reviewed by the Timber Supply Forester at the Timber Supply Branch, subsequent to the acceptance of the Statement of Management Objectives, Options and Procedures, and has been amended to address comments and concerns originating from this review. This document includes all of the information considered relevant for the purposes of the timber supply analysis for Management Plan 9. This approved Information Package is included as an Appendix to the Timber Supply Analysis Report of TFL 6 Management Plan 9.

Due to the nature of the spatially explicit volume-based timber supply analysis some information is not relevant and has been left out intentionally. Otherwise, every effort has been made to follow the provincial guide for the submission of information packages while preparing this Timber Supply Information Package.

3.0 TIMBER SUPPLY FORECASTS/OPTIONS/SENSITIVITY ANALYSES

TABLE 2 lists the current management harvest forecast and associated sensitivity analyses that will be modelled in the timber supply analysis. Additional options to be modelled are shown in TABLE 3. A brief description of assumptions and issues related to each option is also included in the tables.

3.1 Current Management

The initial harvest forecast will use current operational requirements and management practices on the TFL to forecast timber supply. This forecast incorporates existing land use designations, including Special Management Zones[3], and currently enforced regulations and guidelines, particularly the Forest Practices Code. This option represents the current management within the TFL and is used as a basis of comparison for assessing other options.

Current Management is as follows:

- The operable land base includes those forested areas accessible using conventional (Oc) and non-conventional (Oh) harvesting techniques.
- Pure deciduous stands and mixed deciduous-coniferous stands are included in the timber harvesting land base but only the coniferous volume, if any, will contribute to the analysis.
- Basic silviculture is carried out, planting will be restricted to 100% of the S2, S4, S5, S6, S13, M2-5 sites and ~70-80% of S1ha, S12ha, S12ch, M1 sites.
- Fertilization of 100% of the S1CH and S2 stands harvested each year
- Spacing of 300 ha of S1HA and S12HA stands
- Tree Improvement gains will be applied to future plantations:
    - Fd, Hw and Cy – 5% gain for first decade and 15% gain for plantation established after that;
    - Cw and Ss – 3 % gain for all plantations
- Visual quality objectives are modelled based on known scenic areas with maximum denudation objectives.
- Wildlife Tree Patch retention is modelled.
- Biodiversity and Landscape Units – seral stage targets for only Old Seral will be applied to each landscape unit based on target proportions of 10/45/45, high, intermediate and low.

The area available for timber production under Management Plan 9 will be 149,747 ha. With the timber harvesting land base currently available under Management Plan 8 being 132,454 hectares, there has been a gain of 17,293 ha of forestland available for timber

[3] Although not officially approved by Government, the spirit and intent of the SMZs have been reflected in the current management of TFL 6 since they have been recommended for designation.

production since the last Management Plan. This gain is attributable to the consolidation of the former TFL 25, Block 4 into TFL 6 in 1998. This consolidation resulted in a total area increase of approximately 31,000 ha, of which ~24,500 ha was identified as the timber harvesting landbase for Block 4. However, landbase deletions in both TFL 6 and the newly amalgamated area due to land trades and transfers for the Small Business Forest Enterprise Program (SBFEP), and new protected areas have offset this increase somewhat.

TABLE 1

LANDBASE COMPARISON TO PREVIOUS MANAGEMENT PLAN

	TFL 6 MP 9	TFL 6 MP 8	TFL 25 Block 4 MP 9	Consolidation Total	Difference
Total Area	198,113	170,213	31,300	201,513	(3,400)
THLB Area	149,747	132,454	24,558	157,012	(7,265)

3.2 Sensitivity Analyses

Sensitivity analyses will be conducted on the current management option to examine the uncertainty around the components of the analysis, namely the resource inventories, timber growth and yield and management practices. The timber supply impacts from this uncertainty will be factored into the analysis either through the removal of area from the timber harvesting land base or through the incorporation of temporal constraints on the harvest levels. Sensitivity analyses include the following:

1. Land Base: The land base for TFL 6 will be reduced by approximately 10% to determine how sensitive the harvest forecast is to land reduction. This will be done spatially by removing all Terrain Stability Class 4 and 5 land.
2. Operability: Various operability classes have been developed for the TFL 6 land base that reflect the harvesting system, timber quality, terrain stability and economic accessibility. The purpose of the analysis is to quantify the impacts on timber supply under a range of economic conditions by including as part of the operable land base a range of operability classes. The current management option includes areas that can be harvested
    - Conventionally (cable, ground-based) (Oc), and;
    - Non-conventional (helicopter) (Oh),

    and therefore excludes from the productive land base those areas classified as:
    - physically inoperable (I), and;
    - operable only during favourable economic conditions (Oce and Ohe).

    Sensitivity analyses will model the impacts of i) removing the non-conventional area (Oh) and ii) including areas that are considered economically marginal (inclusion of Oce and Ohe areas).
3. Volumes: The impact on harvest forecasts of adjusting the available timber supply in the current management option for both the mature (140+ years) and second growth components by ±10% will be tested.

4. Site Productivity: Site indices (SI 50) assigned on the basis of ecosystem classification and mapping will be varied by ±10% from the current management option.

5. Harvest Age: The effect of rotation length on harvest forecasts will be evaluated by assigning the age at which an analysis unit's mean annual increment has reached within 5% of its culminated value.

6. Green-Up and Adjacency: Green-up heights will be varied ± 2m to assess the impact of time to green up. The current management green-up height will vary in accordance with the visual classification of the area.

7. Visual Quality Objectives: The current management option incorporates constraints resulting from Visual Quality Objectives (VQO) assigned by the Landscape Inventory completed for the TFL in 1994 that are located within "known" Scenic Areas, as defined by the MOF in early 1999. As a comparison to the current management option, a sensitivity analyses will be used to quantify the impacts by varying the percentage of area below Visually Effective Green-up (VEG) to the mid range percent denudation limit recommended for the VQO class.

8. Deciduous Stands: Sensitivity analyses will look at the impact on the timber supply of including the deciduous volume component from pure deciduous and mixed deciduous-conifer stands. The analyses will examine the differences of harvesting these stands as a conversion opportunity (deciduous stands will be harvested at about 50 years) or as part of regular conifer harvesting activities (harvest age based on the associated conifer species). Stands would be reforested to conifers after harvest in both scenarios. Pure deciduous stands not harvested will be left to naturally convert to a coniferous stands.

9. Biodiversity and Landscape Units: The current management option seral stage constraints for biodiversity will be expanded to include targets for early and mature plus old seral stages. This analysis will be used to determine the overall sensitivity of applying full biodiversity seral stage targets to TFL 6.

10. Biodiversity Emphasis Options: Interim Biodiversity Emphasis Options (BEO) ratings have been assigned to the Landscape Units of TFL 6. The current management option does not consider interim BEO ratings. Interim BEO ratings on Landscape Units will be considered in a sensitivity analysis in order to study the implications of managing for the maintenance of biodiversity at the landscape level. Old seral targets will be modelled within each Landscape Unit. Old forest in designated Special Management Zones will receive first priority for biodiversity reserves.

11. Commercial Thinning: Operational programs examining the economic and biological feasibility of commercial thinning have been carried out during the past 5 years. Results suggest that the TFL contains only a limited number of sites suitable for undertaking a commercial thinning program. The analysis will restrict itself to examining commercial thinning on the most favourable sites and stands, namely:
    - Low access costs
    - Level topography with slope <20%

- 40 and 70 years old, any species
- Medium and high sites
- Well-drained soils.

12. Silvicultural Opportunities: The impacts of not fertilizing or spacing and eliminating the use of genetically improved stock will be assessed.

During preparation of the analysis, more issues that warrant sensitivity analyses may become apparent. The licensee will use professional judgement to add additional analyses for consideration in determining an AAC. These will be included in the Timber Supply Analysis Report.

3.3 Alternate Harvest Flows over Time

The timber supply analysis will present two approaches to harvest flows: 1) even flow from present at long term level and 2) transition from current harvest level to long term harvest level in increments of change not to exceed 10% per decade. This will allow the Chief Forester to assess the short, medium and long-term impacts of harvesting rates.

3.4 Options

An unconstrained, current management option (operability the only constraint) will show the impact of constraints on AAC and illustrate the magnitude of trade-offs society is making for the protection of non-timber values.

This option will be repeated by adding in full silviculture treatments (100% planting, all stock to be from improved seed, fertilization of all cedar sites, commercial thinning wherever a positive AAC effect) to approximate the ultimate timber producing potential of TFL 6.

TABLE 3 summarizes these additional options.

TABLE 2

CURRENT MANAGEMENT SCENARIO ANALYSIS OPTIONS

Issue to be Tested	Proposed Options / Sensitivity Analysis	
	Title	**Reason for Analysis and Range to be tested**
To project the timber supply based on current management practices, performance, operational requirements and currently enforced guidelines while meeting the objective of maintaining a timber supply which is not excessively variable over time and which maintains the long-term productivity of the TFL.	**Current Management Option**	Current Management Option includes the following: • Visual Quality based on known scenic areas within the TFL inventory • WTP - 3.25% volume net down to meet WTP requirements (current WTP retention is at 13%; however we are assuming that 75% of the WTP designated will be previously constrained areas) • Riparian reserves based on FPC requirements • Conventional and helicopter areas included. • Silviculture practices as described in Section 3.1 • Biodiversity targets for old seral based on the 10/45/45, high/intermediate/low proportions for all landscape units • Parks excluded, major recreational sites excluded; UWR excluded
	(1) Land Base	The impact on reducing the land base by approximately 10% will be evaluated by removing all Terrain Stability Class 4 and 5 terrain from the Timber Harvesting Landbase.
	(2) Operability	The impact on the harvest flow will be evaluated by including different operability classes in the timber harvesting land base as follows (current management practices for all): • Conventional operable land base included only. • Conventional, helicopter and economically marginal areas included.
	(3) Volumes	The impact on the harvest flow will be evaluated by varying stand yields as follows: • ±10% mature volume. (Mature volume as in current management option) • ±10% second growth volume. (Second growth volume as in current management option)
	(4) Site Productivity	Site Indices (SI 50) will be varied by ±10% to model the uncertainties associated with assigned SI.
	(5) Harvest Age	The effect of minimum harvest criteria on harvest forecasts will compare the current management options of: • Assuming 42cm, 37cm and 35cm average dbh for Productivity Groups 1, 2/3 and 4 respectively, To the option of: • Assuming 95% of maximum mean annual increment. • Assuming 45cm, 40cm and 35cm average dbh for Productivity Groups 1, 2/3, and 4 respectively.
	(6) Green-Up and Adjacency	The impact green-up periods have on harvest forecasts will be assessed by assuming green-up heights of ± 2m, with the base height used varying in accordance with the VQO classification of the area.
	(7) VQO	The effects on 1) varying the percent-denudated limit to the mid range and 2) including only those VQOs in areas designated as “known” Scenic Areas during the mitigation strategy review.
	(8) Deciduous Stands	The deciduous volume component in pure deciduous and deciduous-coniferous mixed stands will be included as harvested volume in the analysis.
	(9) Biodiversity	The implications on timber supply associated with managing to early seral and mature plus old seral stage targets in combination with the biodiversity assumptions used within the CMO.
	(10) Interim BEO	The implications on timber supply associated with managing for biodiversity as dictated by the interim Biodiversity Emphasis Options (old seral targets requirements).
	(11) Commercial Thinning	Commercial thinning on suitable stands and sites will be evaluated. 30% of the basal area will be removed in stands between 40 and 70 years of age on S1ha and S3 ecosites on slopes less than 20%.
	(12) Silviculture Opportunities	The impact of not fertilizing, spacing or using genetically improved stock will be assessed.

TABLE 3

OTHER OPTIONS

Option	Issue to be Tested	Constraints
Unconstrained Run	To examine the timber supply impacts of having an unconstrained run as a basis of comparison to current management requirements.	No constraints will be imposed upon this run with the exception of operability.
		As above with full silviculture treatment activities being applied

4.0 HARVEST MODELS

The proprietary forest estate simulation model COMPLAN will be employed in TFL 6 to determine the AAC based on spatially explicit, volume-based cut control. In addition, runs will also be undertaken as a check to the COMPLAN results using the spatial, forest level planning model ATLAS.

4.1 COMPLAN

COMPLAN is a spatially explicit forest estate model that schedules harvests at the cutblock or stand level subject to adjacency (green-up) and non-timber resource constraints (cover constraints). There is a great deal of flexibility built into the model so it is possible to evaluate many different scenarios with a large degree of realism.

COMPLAN uses a hierarchical data structure that takes advantage of a compartmental management approach to spatial data organization. Advantages of this approach include easy integration with GIS systems, adaptation to a variety of tenure administration structures, and integration of both strategic and operational planning.

Tests have been completed which compare results of COMPLAN with those from the B.C. Ministry of Forests' model FSSIM. These tests, done in cooperation with the MoF showed that COMPLAN could produce results that are extremely similar to that of FSSIM. The differences are insignificant but the reasons for the differences are well understood and documented.

<u>Key Features</u>

COMPLAN offers a number of key features that make it ideally suited for both strategic and operational planning:

- Annual internal time increment allows accurate representation of growth, harvest, adjacency and constraint status.
- Yield table structures allow for many additional variables other than volume to be modelled.
- Initial inventory values for volume, height and other yield table columns are used and then trended with the yield tables allowing for polygon-specific values to be used.
- Constraints are localized to site-specific conditions (e.g. green-up time will be longer for cutblocks on poor sites compared with cutblocks on good sites).
- Cover constraints that address non-timber values can overlap so that it is not necessary to divide the area into management zones according to which constraint is most restrictive.

- All forested land base is retained in the simulation and contributes to cover requirements even if it is not part of the timber harvesting land base.

- Alternative silviculture systems such as retention, selection and shelterwood may be modelled.

- Commercial thinning is supported.

- Spatially explicit nature allows harvest schedules to be easily mapped and verified.

- Flexible yield table columns and the ability to shift yield tables at different ages allow for modelling of succession as well as alternative silvicultural strategies.

- Several different prioritization algorithms are available, including minimize growth loss, oldest first, geographic priority and analysis unit priority.

- Cutblock aggregation can be used.

- Several options exist for “harvesting the profile”.

- Revenues and costs can be tracked and reported.

- Road networks can be modelled.

- There are no artificial limitations on numbers of polygons, yield tables, or other model inputs.

4.2 ATLAS

Like COMPLAN, ATLAS is a spatially explicit forest estate model that schedules harvest units according to a range of spatial and temporal constraints.

It uses a hierarchical data structure, with polygons the basic spatial element containing the inventory data that are essential for forest level modelling. Silviculture treatments and systems and growth and yield data are applied to each polygon through the stand group. Polygons also provide a link to the road network.

Constraints are typically applied to groupings of polygons, called cliques, which in turn can be aggregated to form zones. Although constraints can be applied to both cliques and zones, harvest priorities can only be established for zones. Setting harvest priorities and establishing constraints on larger geographic scales is accomplished through the access unit, which are groupings of zones covering a large portion or all of a forest.

Project and run parameters (such as time horizon, planning period, harvest flow targets, etc.) are defined by the user for each simulation. Unique collections of individual constraints, or constraint sets, can be defined and applied to cliques, zones and access units. Constraints include adjacency/green-up rules, seral stage constraints, within block reserves and equivalent clearcut area rules for modelling hydrological parameters.

Harvest priorities are set by the user at the access unit and zone levels and are available at the polygon level as either oldest polygon first or minimum road network distance. The ATLAS algorithm simulates a harvest scenario in the following sequential manner:

1. go to the access unit with the highest harvest priority and which is eligible for harvest. If no access units are eligible, go to step 6, otherwise go to step 2.

2. go to the zone within this access unit that has the highest harvest priority, and which is eligible for harvest. If no zones are eligible for harvest, close this access unit, and return to step 1, otherwise go to step 3.

3. go to the polygon within this zone that has the highest harvest priority, and is eligible for harvest (not reserved, meets minimum age requirement, not excluded due to adjacency, etc.). If no units are eligible, close this zone, and go to step 2, otherwise go to step 4.

4. Temporarily harvest the block and check all constraints that have been applied to relevant cliques, the zone, and the access unit. If successful, go to step 5. If unsuccessful, reject the temporary harvest and pursue the three possibilities:
   - if the polygon fails a clique constraint, make the polygon ineligible for harvest and return to step 3.
   - if the polygon fails a zone constraint, close the zone and return to step 2.
   - if the polygon fails an access unit constraint, close the access unit and return to step 1.

5. Permanently accept the harvest of this polygon. Update attributes and eligibility of polygon, zone, and access unit resulting from this harvest. Apply the harvest volume to the periodic harvest target. If the harvest target has been met go to step 6, otherwise return to step 3.

6. Summarize and report harvest statistics for this period. If the planning horizon has been reached, stop, otherwise age the forest by one planning period, update attributes and constraints, and return to step 1.

5.0 CURRENT FOREST COVER INVENTORY

The most recent forest cover inventory for TFL 6 was completed in 1970 based on aerial photographs taken in 1967. Since then, it has been maintained and updated to account for forest cover changes due to harvesting and reforestation activities. The data used in this analysis has been updated to January 1, 1998 and reflect area and volume changes in the land base due to logging, reforestation, growth and natural depletions occurring up until that date. Coverages depicting actual logging depletions occurring in 1998 and 1999 will be used, along with stand growth, to update the inventory to January 1, 2000.

The forest inventory has been digitized to a Geographic Information System (PAMAP GIS). The computer graphics operate interactively with the inventory database for forest and non-forest cover, productivity groupings, operability, ownership, ecosystem classification, terrain designations and stability classes and other resource designations (e.g. Scenic Areas and VQOs, Special Management Zones, Ungulate Winter Range, Biodiversity Emphasis Option ratings, eagle nests, etc.). The GIS for TFL 6 is maintained annually. Currently, the GIS map base is in a NAD 83 UTM projection.

The current forest cover inventory meets the business needs of Western Forest Products. Because of its age, it has no formal letter of approval from the MoF, although at the time it was completed the methodology followed MoF recommendations.

An inventory audit conducted by the Inventory Branch in 1980 on the TFL found the inventory acceptable. Although the Inventory Branch scheduled an inventory audit for 1997, it was stopped following a small field data collection and was later switched to Phase 1 of the Vegetation Resources Inventory (VRI). WFP has recently begun the new VRI project for the TFL and has undertaken discussions with the MoF Region to apprise them of its plans. The photo interpretation phase is currently being carried out in 1999/2000 and the ground-sampling phase will begin in 2000/2001.

6.0 DESCRIPTION OF LAND BASE

This section describes the TFL land base and the methodology used to determine the way in which land contributes to the analysis. Some portions of the productive land base, while not contributing to harvest, may be available to meet other resource needs.

6.1 Timber Harvesting Land Base Determination

The Timber Harvesting Land Base and the Total Long-term Land Base in TFL 6 are computed in TABLE 4. Note that areas are reported as net areas by Schedule A and Schedule B lands. Areas and volumes have been extracted from a stand database constructed for the preparation of this Information Package. Appendix I has detailed current volume summaries for the timber harvesting land base; mature stand volumes are from inventory information while immature volumes have been derived from yield curve information.

In 1992, in MP 8 the total area of reductions applied against the forest land base amounted to 23,793 ha. Forest land reduction in the amalgamation TFL 25 Block 4 area amounted to 3,555 ha in the 1996 MP 9 for TFL 25. In 1999, the area of reductions has increased from 27,348 ha to 35,744 ha.

The following sections provide the total area classified in each category noted in TABLE 4 above and serve to summarize the area deducted from the timber harvesting land base including overlaps.

6.2 Total Area

The total area of the TFL is 198,113 ha as shown in TABLE 4. The total area in 1992 was 170,213 ha, a net increase in total area of 27,900 ha. As previously mentioned, this increase is largely attributable to the addition of the former Block 4 of TFL 25. This consolidation has resulted in two blocks in the TFL: Block 1 (original TFL 6) totalling 166,871 ha and Block 2 (formerly TFL 25, Block 4) totalling 31,242 ha. Land trades and transfers for the SBFEP and deletions for protected areas since the last MP have restrained this increase to some degree.

TABLE 4
TIMBER HARVESTING LAND BASE DETERMINATION FOR TFL 6

Classification	Area (ha)			Mature Volume (m³)		
	Schedule A	Schedule B	Total	Schedule A	Schedule B	Total
Total Area	41,344.4	156,768.9	198,113.3	15,623,738.4	48,786,233.6	64,409,972.0
Less: Non-Forest	1,864.1	10,291.0	12,155.1	0	0	
Less: Non-Productive Forest	72.5	394.8	467.3	0	0	
Total Productive Forest	39,407.8	146,083.1	185,490.9	15,623,738.4	48,786,233.6	64,409,972.0
Less Reductions to Total Productive Forest: Non-Commercial (NP Br)	78.8	335.9	414.7	0	0	0
Low Site	1,851.3	10,096.1	11,947.4	949,560.0	4,729,139.2	5,678,699.2
Inoperable / Inaccessible (I, Oce, Ohe)	1,771.7	10,807.6	12,579.3	1,508,351.6	9,113,339.1	10,621,690.6
Riparian Reserves	1,576.6	4,725.7	6,302.3	592,829.4	1,058,842.6	1,651,672.0
Wildlife Habitat Reserves (e.g. UWR)	402.9	1,273.9	1,676.8	360,263.7	986,753.4	1,347,017.1
Recreational Areas	22.8	157.3	180.1	17,178.2	138,279.0	155,457.2
Ecological Reserve	0.0	120.0	120.0	0	52,578.0	52,578.0
Problem Forest Types (Deciduous)	0.0	0.0	0.0[1]	348.4	731.1	1,079.6
Unclassified Roads, Trails and Landings	556.8	1,966.7	2,523.5	64,773.7	114,190.0	178,963.7
Total Reductions to Productive Forest	6,260.7	29,483.2	35,743.9	3,493,305.0	16,193852.4	19,687,211.4
Less: Volume Reductions **Total Reduced Land Base**	33,147.1	116,599.9	149,747.0	12,130,433.4	32,592,381.2	44,722,814.6
Less: Not Sufficiently Restocked Areas	684.3	1,904.0	2,588.3	0	0	0
Add: Not Sufficiently Restocked Areas	684.3	1,904.0	2,588.3	0	0	0
Timber Harvesting Land Base	33,147.1	116,599.9	149,747.0	12,130,433.4	32,592,381.2	44,722,814.6
Less: Future Roads, Trails and Landings	756.4	2081.3	2,837.7	0	0	0
Total Long Term Land Base	32,390.7	114,518.6	146,909.3	12,130,433.4	32,592,381.2	44,722,814.6

[1] Deciduous leading stands contribute to the timber harvesting landbase, but only their conifer volume is applied to timber supply (See Section 3.1)

6.3 Non-Forest

The non-forest portion includes those portions of the TFL land base on which merchantable tree species are largely absent. TABLE 5 summarizes by category the total non-forest area within TFL 6 by operating area. The total non-forest area deducted from the timber harvesting land base is 12,155 ha.

TABLE 5

TOTAL AREA OF NON-FOREST WITHIN TFL 6

Type	Non-Forest Area (ha)				
	Operating Area				Total
	Port McNeill	Holberg	Jeune Landing	Mahatta	
Alpine	855.5	49.8	2530.0	46.3	3481.6
Swamp, Marsh, Creek, River, Lake	4023.8	1512.1	837.0	347.3	6720.2
Fish Farm	0	0	2.1	1.6	3.7
Rock and Slides	399.2	297.5	360.9	54.7	1112.2
Classified Roads and Pits	146.3	198.8	86.1	41.5	472.7
Hydro and Telephone R-of-Way	253.5	8.3	6.8	0	268.5
Dump, Camps and Sort	19.2	51.0	23.2	2.7	96.1
TOTAL	**5697.5**	**2117.5**	**3846.1**	**494.0**	**12,155.1**

6.4 Non-Productive Forests

There are 467 ha of non-productive land in the TFL. TABLE 6 provides a summary of this area by operating area. These are areas occupied by brush and grass.

TABLE 6

TOTAL AREA OF NON-PRODUCTIVE FOREST WITHIN TFL 6

Criteria	Operating Area				Total
	Port McNeill	Holberg	Jeune Landing	Mahatta	
Brush	83.5	267.4	81.8	12.7	445.4
Grass	0	21.9	0	0	21.9
Total	**83.5**	**289.3**	**81.8**	**12.7**	**467.3**

6.5 Non-commercial Cover

Approximately 415 hectares (TABLE 7) have been identified as non-commercial deciduous (NCD) within the forest cover inventory of the TFL and, as such, have been removed from the THLB.

TABLE 7

AREA CLASSIFIED AS NON-COMMERCIAL

Non-Commercial	Total Area (ha)	Operating Area				Total Area Reduced
		Port McNeill	Holberg	Jeune Landing	Mahatta	
NCD	414.7	47.2	220.3	90.5	56.7	414.7

Note: Total area refers to the entire productive forest area covered by this classification including other, overlapping classification.

6.6 Low Site

A study undertaken by J.S. Thrower in 1997 confirmed the validity of using ecological information for estimating site productivity, an approach used in MP 8. The Research Branch approved the Thrower report on April 18, 1997. The approach used assigned site index at 50 years at 1.3m (SI 50) on the basis of ecosystem classification and mapping completed for the TFL in 1985. Thrower has since enhanced the classification by identifying five productivity classes for the ecosystems of the TFL and assigning by tree species the appropriate SI 50 values to each class. For this analysis, stands within productivity group 5, having an average SI50 of 12m were removed. TABLE 8 provides a breakdown of the gross productive areas reduced for low sites by operating area.

TABLE 8

AREA CLASSIFIED AS LOW SITE

Low Site Criteria	Total Area (ha)	Operating Area				Total Area Reduced
		Port McNeill	Holberg	Jeune Landing	Mahatta	
Productivity Group 5 (CP, MH, MH1 S7, S8, S9, S11)	11,947.4	8075.3	2364.0	1353.5	154.6	11,947.4

Note: Total area refers to the entire productive forest area covered by this classification including other, overlapping classification.

6.7 Inoperable/Inaccessible

TABLE 9 provides information on the area in the TFL that is classified as *inoperable*. The majority of this area is in the Holberg Operating Area of TFL 6.

Operability classes have been developed for the TFL 6 land base that reflect the harvesting system, timber quality, terrain stability and economic accessibility. A terrain stability inventory was completed for Block 1 to MoF standards in 1995 and for Block 2 in 1992, since updated to MoF standards in 1998.

The reclassification of the TFL's operability was completed in early 1999 based on a methodology reflecting the elements noted above and approved by the MoF Port McNeill Forest District in September 1998. The classification resulted in two categories of inoperable areas being defined. The first category strictly relates to the area being physically inaccessible and thus, not available for timber harvesting (I). The second category uses an economic criterion to determine operability (Oce/Ohe). In this case, timber harvesting under current market conditions is not justifiable given the costs of harvesting and the expected value of the timber. Physical inoperability relates to the presence of a physical barrier or terrain constraint leaving accessibility virtually impossible. Classifying areas as operable with an economic constraint relates to the inability to harvest stands in a cost-effective manner given the low market value of the timber at the current time. Two classes are recognized in this analysis: (1) Oce for areas that could be logged profitably by conventional harvesting systems should markets improve sufficiently and (2) Ohe for areas that could be heli-logged profitably should markets improve sufficiently.

Of the net inoperable land base, 949 ha are currently classified as Oce/Ohe and 11,630 ha are currently classified as I. The area classified as inoperable and therefore deducted from the productive forest land base is 12,579 ha.

TABLE 9

AREA CLASSIFIED AS INOPERABLE

Criteria	Total Area (ha)	Operating Area				Total Area Reduced
		Port McNeill	Holberg	Jeune Landing	Mahatta	
I – Physically Inoperable	18,433.9	1714.6	5196.3	3749.5	969.6	11,630.0
Oce – Operable for conventional logging with economic constraints removed	549.9	87.3	150.2	172.2	42.1	451.8
Ohe – Operable for heli-logging with economic constraints removed	557.0	33.3	114.3	270.2	79.6	497.45
Total	**19,540.8**	**1835.3**	**5460.8**	**4191.9**	**1091.4**	**12,579.3**

Note: Total area refers to the entire productive forest area covered by this classification including other, overlapping classification.

6.8 Environmentally Sensitive Areas (ESAs)

Areas assessed as sensitive or valuable for harbouring other resource values have been delineated through inventories conducted before and since MP 8. Land base reductions reflecting the presence of these delineated areas are captured in other sections of the Information Package, including for terrain stability and soil sensitivity, which has been factored into the development of operability classes (Section 6.5), for wildlife habitat (Section 6.11) and recreational reserves (Section 6.12). In addition, productive area net downs for riparian reserves (Section 6.9) and forest cover constraints applied to capture the reservation of future Wildlife Tree Patches (WTP) are also designed to protect non-timber resource uses. As a result, no further reductions are applied here.

6.9 Riparian Reserves - Streams

Although overview mapping is available, a comprehensive riparian inventory has not yet been completed for TFL 6. However, operational stream inventories associated with development planning have been conducted since 1988 and a Reconnaissance

(1:20000) Fish and Fish Habitat Inventory project within the Special Management Zones of TFL 6 has been underway since 1998.

This detailed information in conjunction with GIS modelling helped to obtain an overall estimate of the riparian classes and reserve areas for the TFL. The approach employed in the timber supply analysis was to utilize the available stream classification in the GIS to apply reserves to all known and predicted fish bearing streams, in accordance to specifications in the FPC's Riparian Management Area Guidebook. Detailed calculations (TABLE 10) are described below.

TABLE 10

RIPARIAN RESERVE ZONES

Riparian Feature Class	Feature Perimeter or Length (km)		% Of Class relative to total streams classified	Total Riparian reserve width From FPC (metres)	Weighted Average Riparian reserve zone for S7 and Unclassified Streams
	Topography <30% gradient	Topography > 30% gradient			
Double Line Streams					
S1	21.2		43%	50	
S2	27.4		55%	30	
S3	1.1		2%	20	
s7	439.8				38.4 (40)
Unclassified	319.5				38.4 (40)
Single Line streams					
S2	83.1	0	28%	30	
S3	88.8	0	30%	20	
S4	18.0	0	6%	0	
S5	20.9	58.5	7%	0	
S6	56.3	143.6	19%	0	
s7	196.0	0			14.4 (20)
s8	33.9	76.5	10%	0	
Unclassified	2038.4	4153.3			11.2 (10)
Lakes and Wetlands					
L1 (> 5 ha)	176.3			10	
W1 (>5 ha)	190.9			10	

Currently within the GIS streams are classed as S1 to S6 (as per FPC definitions), s7[4] (which is a known fish bearing stream of unknown width), s8 (which is a known non-fish bearing stream of unknown width) and Unclassified (which are streams of unknown fish presence and width).

Double line streams – riparian reserve zones for s7 and unclassified streams were approximated using a weighted average of classified doubled-line streams relative to FPC specified riparian reserves. Within the GIS all double-lined streams are assigned a riparian reserve based on their classification; however, due to limitation of the GIS each riparian reserve was rounded to the nearest ten metres.

Unclassified single-line streams – a GIS analysis (terrain model) was used to separate, and class streams of less than 30% gradient as being potentially fish bearing. The 30%

[4] S7 and s8 are internal classifications used in the GIS to designate streams that have been classified as A, B or I/II/III (fish bearing) and C or IV (non-fish bearing) in pre-FPC classifications.

gradient parameter is more conservative than the normal assumption of <20% due to the coarse nature of the digital elevation model (TRIM) and because fish have been identified, in some cases, in streams of >20% gradient. Based on the 497.5 km of known S2 to s8 classified single line streams identified as less than 30% gradient, it was determined that 78% of the unclassified single line streams are potentially fish bearing (386.3 km of fish bearing stream/497.5 km of streams <30% gradient). A weighted average riparian width was then calculated for s7 single line streams. This riparian reserve width of 14.4 was than multiplied by 78% to establish the riparian reserve zone for the unclassified single line streams. Within the GIS all single line streams of gradients <30% will be assigned the appropriate riparian reserve width. Due to limitation of the GIS each riparian reserve was rounded to the nearest ten meters; the rounding effect was conservative in that it increased the reserve area.

The reductions for all riparian classes have been applied to operable productive forested areas located within these buffers that have not already been deducted from the productive land base. TABLE 11 provides a summary of the areas reduced within each operating area in the TFL.

TABLE 11

RIPARIAN RESERVE ZONES BY OPERATION

Operating Area	Riparian Class	Total Area (ha)	Reduction Area (ha)
Port McNeill	Double Line Stream Riparian	830.1	678
	Single Line Stream Riparian	1886.9	1640.6
	Lakes and Wetlands Riparian	131.1	86.2
Holberg	Double Line Stream Riparian	786.6	688.0
	Single Line Stream Riparian	2131.0	1994.8
	Lakes and Wetlands Riparian	58.9	32.1
Jeune Landing	Double Line Stream Riparian	486.7	441.7
	Single Line Stream Riparian	368.0	343.0
	Lakes and Wetlands Riparian	21.6	18.7
Mahatta	Double Line Stream Riparian	195.7	193.3
	Single Line Stream Riparian	201.3	178.4
	Lakes and Wetlands Riparian	9.1	7.5
All Operations	*Double Line Stream Riparian*	*2299.1*	*2001.0*
	Single Line Stream Riparian	*4587.2*	*4156.8*
	Lakes and Wetlands Riparian	*220.7*	*144.5*
Total		**7107.0**	**6302.3**

Note: Total area refers to the entire productive forest area covered by this classification including other, overlapping classification.

6.10 Wildlife Habitat Deductions

A number of wildlife inventories have been undertaken or broadened since the last MP in an effort to identify and classify potential wildlife habitat areas suitable for deer and other species.

Distinct areas defined and reserved as wildlife habitat include Ungulate Winter Range having high or moderate ungulate ratings. In Management Plan 8 2,040 ha of gross area was identified as Deer Winter Range, this area was approximate and since that time areas have been refined to a gross area of 2283 ha. Since MP 8, 1,437 gross ha have been grandfathered as UWR, of which 1031 ha are unconstrained. The remaining 825.6 gross ha of non-designated UWRs, of which 646.0 ha are unconstrained, are located within the Koprino River area and will likely be replaced for wildlife habitat areas that focus on Marbled Murrelet nesting needs. All grandfathered areas have been field assessed by wildlife personnel as part of a TFL wide habitat classification project. Because a number of the UWRs were found to be of low value, the spatial analysis will recognize, along with identified moderate and high UWR ratings, non-identified areas that contain more suitable habitat. The intent is to replace areas with low ungulate ratings with an equal area of more suitable habitat. The areas noted in TABLE 12 reflect the intent to retain an equivalent amount of wildlife habitat as was used in Management Plan 8.

Future WTPs will be handled through a volume reduction as described in Section 10.2.1.6 in the timber supply model. It should be noted that at least 75% of the WTPs will be incorporated into riparian reserves or other constrained areas.

A total of 2162 ha of the TFL's productive land base has been specifically reserved as wildlife habitat. This compares to 2,246 ha of wildlife habitat areas identified in MP 8. TABLE 12 summarizes the operable, productive forest area reserved for wildlife habitat.

TABLE 12
AREA CLASSIFIED AS WILDLIFE MANAGEMENT AREAS

Operating Area	Total Area (ha)	Ungulate Winter Range Areas (ha)	Non-Designated UWR Areas (ha)	Total Area Reduction (ha)
Port McNeill	174.0	22.8	78.7	101.5
Holberg	637.5	0	510.3	510.3
Jeune Landing	951.0	718.4	57.0	775.4
Mahatta	399.6	289.6	0	289.6
Total	**2162.0**	**1030.8**	**646.0**	**1676.8**

Note: Total area refers to the entire productive forest area covered by this classification including other, overlapping classification

6.11 Recreational Reserve Area Reductions

Extensive areas of moderately sensitive recreation areas have been identified by the recreation inventory completed in 1990. Four areas (Marble River Recreational Area, Grant Bay Recreational Area, Spruce Bay Old Growth Reserve and Topknot Bay Recreational Area have been reserved for their recreational value (TABLE 13). Recreational opportunities will be reinforced through the integration of forest harvest planning and landscape inventories designed to maintain visual landscape quality and preserve known scenic values.

TABLE 13
AREA CLASSIFIED AS RECRATIONAL RESERVES

Operating Area	Description of Recreational Area	Total Area (ha)	Total Area Reduction (ha)
Port McNeill	Marble River Recreational Area	5.1	5.1
	Spruce Bay Old Growth Reserve	17.6	17.6
Holberg	Grant Bay Recreational Area	149.7	113.9
	Topknot Bay Recreational Area	47.7	43.4
Total		**220.1**	**180.1**

Note: Total area refers to the entire productive forest area covered by this classification including other, overlapping classification

6.12 Ecological Reserves

As one of the most northerly sites of Douglas-fir on Vancouver Island, the 120 ha Varney Gene Pool Reserve has been established. The Spruce Bay genetic reserve and the Misty Lake Ecological Reserve have also been identified in the TFL. The Spruce Bay genetic reserve is included in the Spruce Bay Old Growth Reserve, addressed in the section above, and the Misty Lake Ecological Reserve is a Provincial Park, which has been excluded from the TFL.

6.13 Cultural Heritage Resource Reductions

No reductions have been made for cultural heritage resources in the TFL. Most features have been and can be protected within wildlife tree patches, riparian reserves, and other reserves.

6.14 Other Sensitive Site Reductions

The landscape unit/biodiversity requirements have not been included as area deductions as it is expected that the areas currently reserved from harvesting will be sufficient to fulfil many of these requirements.

As stated above, the timber supply analyses undertaken for the current management option does not assume implementation of the interim Biodiversity Emphasis Options (BEO) for any Landscape Unit within the TFL. A sensitivity analysis will analyze the impacts on timber supply by constraining the harvest to meet the interim BEO. A similar approach is being taken for modelling the impacts associated with the two Special Management Zones. As indicated in draft Vancouver Island Land Use documents, one of the objectives of the SMZs will be to focus retention areas needed to meet Landscape Unit planning requirements within the SMZ itself. In both cases, land base deductions are not required as ecological parameters such as seral stage distributions will be modelled using forest cover constraints.

6.15 Problem Forest Types (Deciduous Stands)

TABLE 14 provides a breakdown of the area occupied by deciduous-leading stands. There have been no reductions to the productive forest land due to these stand types as the timber supply implications pertaining to deciduous stands are purely volume based. The current management option assumes that only the coniferous volume in deciduous-leading stands available for harvesting will be included in the timber supply.

TABLE 14

PROBLEM DECIDUOUS FOREST TYPES

Operating Area	Inventory Type Group	Total Area (ha) By Age					Total
		0-20	21-40	41-60	61-80	80-100	
Port McNeill	Pure Deciduous	5.1	69.9	106.2	63.9	42.4	287.6
	Deciduous-Leading	0.0	59.4	22.6	4.7	0.9	87.6
Holberg	Pure Deciduous	3.0	7.4	316.8	179.9	0.0	507.1
	Deciduous-Leading	0.3	77.1	179.5	52.0	60.9	369.7
Jeune Landing	Pure Deciduous	2.3	97.1	269.6	133.8	6.7	509.5
	Deciduous-Leading	3.4	8.8	195.0	95.3	32.4	334.9
Mahatta	Pure Deciduous	0	82.7	64.9	9.4	0.0	157.0
	Deciduous-Leading	0	32.2	77.0	2.8	2.6	114.6
Total		**14.2**	**434.6**	**1231.7**	**541.8**	**145.8**	**2368.0**

Although this section focuses on problem deciduous forest types it should be noted that some low quality conifer sites were removed during the operability classification. A more detailed explanation of the criteria used during the operability classification can be found in Appendix III.

6.16 Roads, Trails and Landings

6.16.1 Classified Roads, Trails and Landings

Classified roads, trails and landings are those that are mapped as polygons distinctly separate from adjacent polygons. Only the mainline roads have been identified as separate polygons on the forest cover maps. TABLE 15 summarizes the areas of classified roads for each Operating Area of the TFL.

TABLE 15

CLASSIFIED ROADS, TRAILS AND LANDINGS

Operating Area	Total Area (ha)		
	Roads	Landings	Trails
Port McNeill	146.3	0.0	0.0
Holberg	198.8	0.0	0.0
Jeune Landing	86.1	0.0	0.0
Mahatta	41.5	0.0	0.0
Total	**472.7**	**0.0**	**0.0**

6.16.2 Unclassified Roads, Trails and Landings

Unclassified roads on the TFL have been mapped as lineal features. For the purposes of determining the total area of unclassified roads, all are assumed to occupy a 10 metre wide unproductive width. As with classified trails and landings, all trails and the majority of the landings are rehabilitated and restocked immediately following logging and consequently there is no associated area reduction. TABLE 16 indicates the area of unclassified roads in the TFL that is excluded from the timber harvesting land base.

TABLE 16

UNCLASSIFIED ROADS, TRAILS AND LANDINGS

Operating Area	Total Road Length (km) in Productive Forest Land	Total Area Reduction (ha)		
		Roads	Landings	Trails
Port McNeill	1288.0	1249.9.	0.0	0.0
Holberg	673.7	664.2	0.0	0.0
Jeune Landing	395.8	393.7	0.0	0.0
Mahatta	219.2	215.7	0.0	0.0
Total	**2576.7**	**2523.5**	**0.0**	**0.0**

6.16.3 Future Roads, Trails and Landings

A projected road system was developed as part of the operability classification for TFL 6. This road system was digitized into the GIS in conjunction with operability loading which allowed for the same approach used with unclassified roads to predict area summaries. TABLE 17 indicates the area of future roads in the TFL that have yet to be developed.

TABLE 17

FUTURE ROADS, TRAILS AND LANDINGS

Operating Area	Total Road Length (km) in Productive Forest Land	Total Area (ha)		
		Roads	Landings	Trails
Port McNeill	1014.4	890.0	0.0	0.0
Holberg	1377.3	1310.6	0.0	0.0
Jeune Landing	573.8	524.2	0.0	0.0
Mahatta	123.1	112.9	0.0	0.0
Total	**3088.6**	**2837.7**	**0.0**	**0.0**

6.17 Exclusion of Specific, Geographically Defined Areas

The four recreational reserves noted above have already been excluded. Known scenic areas will be handled through forest cover constraints in the timber supply analysis. Parks designated by Order-In-Council total 24,384 ha of land and water within or adjacent to TFL 6 lands and include 1) Cape Scott Provincial Park (21,849 ha) 2) Marble River Park (1421.7 ha) 3) Misty Lake Ecological Reserve (59.6 ha) 4) Quatsino Park (633.7 ha) and 5) Raft Cove Park (423.0 ha). Some additional area of TFL 6 has also been identified as part of the Cluxewe Salt Marsh Reserve (35.2 ha), which was established in conjunction with The Nature Trust of British Columbia. Although this area has not been formally removed from the TFL a request to do so was made on November 18, 1992. All areas listed have already been accounted for in the inventory file (i.e. have been excluded from the THLB prior to the net down exercise). Consequently, no further reductions to the productive forest land base are required.

6.18 Any Other Land Base Exclusions

There are no other land base exclusions to be accounted for at this time.

6.19 Area Additions

There are no additions to the timber harvesting land base at this time.

7.0 INVENTORY AGGREGATION

In keeping with the approach used in Management Plan 8, the current inventory, dated January 1, 1998, has been aggregated on the basis of leading species, site productivity and age class. Of principal interest are the coniferous-leading stands. These stands are the only ones which have been aggregated into species-specific groups; the remainder of the deciduous stock is aggregated either as "pure deciduous" or "deciduous leading." Ecological classification completed for the TFL is used as the indicator of site quality.

7.1 Management Zones

For the purposes of this Timber Supply Analysis, the net operable and available land base is subject to normal management where the focus is timber production. Unique forest cover objectives can be easily handled in the models using cover constraints so aggregation of areas of similar objectives is unnecessary. Landscape Units, Special Management Zone (SMZ) and Resource Management Zone (RMZ) are delineated in the data set and will be used to report seral stage distributions and other ecological parameters for selected sensitivity analyses. TABLE 18 presents the Landscape Units, SMZ and RMZ found on the TFL.

TABLE 18

MANAGEMENT ZONES AND LANDSCAPE UNITS

Management Zone	Landscape Unit	Operating Area	THLB[1] Area (ha)	Management Considerations
RMZ 4	San Josef	Holberg Jeune Landing **Total**	32,017.2 535.3 **32,552.5**	Timber harvesting in this zone is considered to be relatively free of constraints except for P, PR and R VQO polygons which have maximum alteration limits of 1%, 5% and 15% as explained in the accompanying text. Sensitivity analyses will evaluate the impacts associated with the Interim BEO rating of Intermediate assigned to this zone.
RMZ 5	Holberg	Holberg Port McNeill **Total**	11473.4 11983.2 **23,456.6**	Timber harvesting in this zone is considered to be relatively free of constraints except for PR and R VQO polygons that have maximum alteration limits as noted above. Sensitivity analyses will evaluate the impacts associated with the Interim BEO rating of Intermediate assigned to this zone.
RMZ 6	Keogh	Port McNeill	24215.1	Timber harvesting in this zone is considered to be relatively free of constraints except for P, PR and R VQO polygons that have maximum alteration limits as noted above. Sensitivity analyses will evaluate the impacts associated with the Interim BEO rating of Intermediate assigned to this zone.
RMZ 7	Marble	Port McNeill Jeune Landing **Total**	11840.0 2864.0 **14,704.0**	Timber harvesting in this zone is considered to be relatively free of constraints except for PR and R VQO polygons that have maximum alteration limits as noted above. Sensitivity analyses will evaluate the impacts associated with the Interim BEO rating of Intermediate assigned to much of this zone.
RMZ 8	Mahatta Neroutsos	Mahatta Jeune Landing Jeune Landing Port McNeill **Total**	12038.6 6971.8 18171.7 1422.8 **38,604.9**	Timber harvesting in this zone is considered to be relatively free of constraints except for P, PR and R VQO polygons that have maximum alteration limits as noted above. Sensitivity analyses will evaluate the impacts associated with the Interim BEO ratings of Intermediate and High assigned to portions of this zone.
SMZ 2	San Josef	Holberg	11227.8	In the current management option, constraints imposed by the SMZ designation is removed; therefore, timber harvesting in these zones is considered relatively free of constraints except for P, PR and R VQO polygons which have maximum alteration limits as noted above. Sensitivity analyses will evaluate the impacts associated with the imposition of special management practices required under the SMZ designation. Sensitivity analyses will also evaluate the impacts associated with the Interim BEO rating of Intermediate assigned to these zones.
SMZ 4	San Josef	Holberg	4983.2	
Total			**149,747.0**	

1. Timber Harvesting Landbase as defined in Table 3.

TABLE 19

AREA BY LANDSCAPE UNIT AND BEC VARIANT

Landscape Unit	BEC	Seral Stage	Productive Forest (ha)	Non Contributing Area		THLB Area	
				ha	%	ha	%
Holberg	CWH vh 1	Early	1889.2	184.9	4.3%	1704.3	39.6%
		Mid	126.3	11.7	0.3%	114.6	2.7%
		Mature	125.9	3.4	0.1%	122.5	2.8%
		Old	2167.8	1222.9	28.4%	944.8	21.9%
	CWH vh 1 Total		***4309.1***	***1422.9***		***2886.3***	
	CWH vm 1	Early	9682.2	933.6	3.6%	8748.7	33.6%
		Mid	5772.0	361.5	1.4%	5410.5	20.8%
		Mature	471.0	22.0	0.1%	449.1	1.7%
		Old	10098.5	5195.4	20.0%	4903.1	18.8%
	CWH vm 1 Total		***26023.8***	***6512.5***		***19511.3***	
	CWH vm 2	Early	464.9	17.7	1.3%	447.2	31.8%
		Mid	152.6	0.3	0.0%	152.3	10.8%
		Mature	17.4	0.0	0.0%	17.4	1.2%
		Old	772.8	336.4	23.9%	436.5	31.0%
	CWH vm 2 Total		***1407.8***	***354.4***		***1053.4***	
	MH mm 1	Early	13.9	9.1	7.7%	4.8	4.0%
		Old	104.5	103.7	87.6%	0.8	0.7%
	MH mm 1 Total		***118.4***	***112.8***		***5.6***	
Holberg Total			**31859.1**	**8402.5**		**23456.6**	
Keogh	CWH vm 1	Early	11209.6	1338.7	5.0%	9870.9	36.6%
		Mid	6290.4	554.8	2.1%	5735.6	21.3%
		Mature	2522.4	170.4	0.6%	2352.0	8.7%
		Old	6915.6	2694.9	10.0%	4220.7	15.7%
	CWH vm 1 Total		***26938.0***	***4758.8***		***22179.2***	
	CWH vm 2	Early	1412.8	114.1	5.1%	1298.7	57.8%
		Mid	182.1	5.2	0.2%	176.9	7.9%
		Old	651.5	240.0	10.7%	411.5	18.3%
	CWH vm 2 Total		***2246.5***	***359.3***		***1887.2***	
	MH mm 1	Early	77.1	26.2	3.9%	50.9	7.6%
		Old	594.6	496.7	74.0%	97.8	14.6%
	MH mm 1 Total		***671.7***	***522.9***		***148.7***	
Keogh Total			**29856.2**	**5641.0**		**24215.1**	
Mahatta	AT p	Old	16.6	13.0	78.4%	3.6	21.6%
	AT p Total		***16.6***	***13.0***		***3.6***	
	CWH vh 1	Early	62.9	0.7	0.4%	62.2	39.7%
		Mid	19.2	0.0	0.0%	19.2	12.3%
		Old	74.6	38.1	24.3%	36.5	23.3%
	CWH vh 1 Total		***156.7***	***38.8***		***117.9***	
	CWH vm 1	Early	9919.2	668.5	3.7%	9250.7	51.1%
		Mid	3032.7	315.2	1.7%	2717.5	15.0%
		Mature	663.0	79.6	0.4%	583.4	3.2%
		Old	4489.1	1489.0	8.2%	3000.1	16.6%
	CWH vm 1 Total		***18104.0***	***2552.3***		***15551.7***	
	CWH vm 2	Early	1396.8	68.8	1.4%	1327.9	27.4%
		Mid	32.1	4.0	0.1%	28.1	0.6%
		Mature	44.2	4.3	0.1%	39.9	0.8%
		Old	3381.3	1495.6	30.8%	1885.7	38.8%
	CWH vm 2 Total		***4854.4***	***1572.7***		***3281.7***	
	MH mm 1	Early	0.4	0.0	0.0%	0.4	0.1%
		Old	311.8	256.6	82.2%	55.2	17.7%
	MH mm 1 Total		***312.2***	***256.6***		***55.6***	
Mahatta Total			**23443.9**	**4433.4**		**19010.5**	

Landscape Unit	BEC	Seral Stage	Productive Forest (ha)	Non Contributing Area		THLB Area	
				ha	%	ha	%
Marble	CWH vm 1	Early	6811.3	614.4	4.1%	6196.9	41.2%
		Mid	3013.8	181.5	1.2%	2832.3	18.8%
		Mature	2099.8	243.9	1.6%	1855.9	12.3%
		Old	3118.0	1077.0	7.2%	2041.0	13.6%
	CWH vm 1 Total		***15042.8***	***2116.8***		***12926.0***	
	CWH vm 2	Early	584.5	25.7	0.9%	558.8	20.6%
		Mid	237.3	12.6	0.5%	224.7	8.3%
		Mature	5.7	0.6	0.0%	5.0	0.2%
		Old	1881.2	1053.0	38.9%	828.2	30.6%
	CWH vm 2 Total		***2708.6***	***1091.9***		***1616.7***	
	MH mm 1	Early	12.0	0.2	0.0%	11.7	2.0%
		Old	583.4	433.9	72.9%	149.5	25.1%
	MH mm 1 Total		***595.3***	***434.1***		***161.2***	
Marble Total			**18346.7**	**3642.8**		**14704.0**	
Neroutsos	CWH vm 1	Early	4063.8	315.5	1.8%	3748.3	21.0%
		Mid	4615.3	244.4	1.4%	4370.9	24.5%
		Mature	2395.8	100.5	0.6%	2295.3	12.9%
		Old	6735.0	1326.4	7.4%	5408.6	30.4%
	CWH vm 1 Total		***17809.8***	***1986.8***		***15823.0***	
	CWH vm 2	Early	944.0	44.2	0.8%	899.8	16.2%
		Mid	180.2	6.5	0.1%	173.6	3.1%
		Mature	146.9	9.3	0.2%	137.6	2.5%
		Old	4280.2	1733.9	31.2%	2546.2	45.9%
	CWH vm 2 Total		***5551.3***	***1794.0***		***3757.3***	
	MH mm 1	Mid	2.6	2.0	1.9%	0.6	0.6%
		Old	99.2	85.6	84.1%	13.6	13.3%
	MH mm 1 Total		***101.7***	***87.6***		***14.2***	
Neroutsos Total			**23462.8**	**3868.4**		**19594.4**	
San Josef	CWH vh 1	Early	5171.2	547.3	3.1%	4623.9	26.5%
		Mid	493.9	51.5	0.3%	442.4	2.5%
		Mature	1504.1	91.2	0.5%	1412.8	8.1%
		Old	10270.4	3378.1	19.4%	6892.3	39.5%
	CWH vh 1 Total		***17439.5***	***4068.1***		***13371.5***	
	CWH vm 1	Early	14104.8	1195.3	3.2%	12909.5	34.1%
		Mid	4977.2	351.0	0.9%	4626.2	12.2%
		Mature	3525.7	225.4	0.6%	3300.3	8.7%
		Old	15277.7	3083.1	8.1%	12194.5	32.2%
	CWH vm 1 Total		***37885.4***	***4854.9***		***33030.5***	
	CWH vm 2	Early	607.0	19.1	0.6%	587.9	18.7%
		Mid	105.6	17.1	0.5%	88.5	2.8%
		Mature	133.5	5.9	0.2%	127.5	4.1%
		Old	2296.9	755.3	24.0%	1541.6	49.0%
	CWH vm 2 Total		***3143.0***	***797.6***		***2345.4***	
	MH mm 1	Old	54.4	35.4	65.0%	19.0	35.0%
	MH mm 1 Total		***54.4***	***35.4***		***19.0***	
San Josef Total			**58522.3**	**9755.8**		**48766.4**	
Grand Total			**185490.9**	**35743.9**		**149747.0**	

7.2 Analysis Units

Aggregation of species groups is necessary to facilitate forest level modelling. Species of similar biological, management and silvicultural attributes are grouped to reduce complexity. This must be balanced with creating groups narrow enough to allow accurate modelling of stand yields.

Ideally, every forest polygon should have its own yield table. While conceptually simple, this scenario would be very cumbersome to implement computationally. Instead, forest polygons were aggregated by age class, leading species, and site productivity group (1, 2 [which is a grouping of 2 and 3], and 4) derived from ecological classification. Aggregation based on the above criteria still resulted in the creation of a fair number of analysis units; however, this should not pose any problem in the timber supply model process.

TABLE 20 provides a breakdown of the analysis units that are to be used within the timber supply analysis for TFL 6 and TABLE 21 outlines the timber harvesting land base by analysis unit.

TABLE 20

ANALYSIS UNITS

Analysis Unit	Stand Type	Productivity Group[1]	Leading Species	Age Class	SI Origin	Treatment	Model
1	Managed Stands	1	C	1-2	Prod. Grp	Fertilized	Tipsy
2	Managed Stands	1	C	1-2	Prod. Grp	No Treatment	Tipsy
3	Managed Stands	2	C	1-2	Prod. Grp	Fertilized	Tipsy
4	Managed Stands	2	C	1-2	Prod. Grp	Spaced	Tipsy
5	Managed Stands	2	C	1-2	Prod. Grp	No Treatment	Tipsy
6	Managed Stands	4	C	1-2	Prod. Grp	Fertilized	Tipsy
7	Managed Stands	4	C	1-2	Prod. Grp	No Treatment	Tipsy
8	Managed Stands	1	H	1-2	Prod. Grp	Fertilized	Tipsy
9	Managed Stands	1	H	1-2	Prod. Grp	Spaced	Tipsy
10	Managed Stands	1	H	1-2	Prod. Grp	Fert Spaced	Tipsy
11	Managed Stands	1	H	1-2	Prod. Grp	No Treatment	Tipsy
12	Managed Stands	2	H	1-2	Prod. Grp	Fertilized	Tipsy
13	Managed Stands	2	H	1-2	Prod. Grp	Spaced	Tipsy
14	Managed Stands	2	H	1-2	Prod. Grp	Fert Spaced	Tipsy
15	Managed Stands	2	H	1-2	Prod. Grp	No Treatment	Tipsy
16	Managed Stands	4	H	1-2	Prod. Grp	Fertilized	Tipsy
17	Managed Stands	4	H	1-2	Prod. Grp	Spaced	Tipsy
18	Managed Stands	4	H	1-2	Prod. Grp	Fert Spaced	Tipsy
19	Managed Stands	4	H	1-2	Prod. Grp	No Treatment	Tipsy
20	Managed Stands	1	B	1-2	Prod. Grp	No Treatment	Tipsy
21	Managed Stands	2	B	1-2	Prod. Grp	Fertilized	Tipsy
22	Managed Stands	2	B	1-2	Prod. Grp	Spaced	Tipsy
23	Managed Stands	2	B	1-2	Prod. Grp	No Treatment	Tipsy
24	Managed Stands	4	B	1-2	Prod. Grp	No Treatment	Tipsy
25	Managed Stands	1	S	1-2	Prod. Grp	Fertilized	Tipsy
26	Managed Stands	1	S	1-2	Prod. Grp	Spaced	Tipsy
27	Managed Stands	1	S	1-2	Prod. Grp	No Treatment	Tipsy
28	Managed Stands	2	S	1-2	Prod. Grp	Spaced	Tipsy
29	Managed Stands	2	S	1-2	Prod. Grp	No Treatment	Tipsy
30	Managed Stands	4	S	1-2	Prod. Grp	Fertilized	Tipsy
31	Managed Stands	4	S	1-2	Prod. Grp	Spaced	Tipsy
32	Managed Stands	4	S	1-2	Prod. Grp	No Treatment	Tipsy
33	Managed Stands	1	D	1-2	Prod. Grp	No Treatment	Tipsy
34	Managed Stands	2	D	1-2	Prod. Grp	No Treatment	Tipsy
35	Managed Stands	4	D	1-2	Prod. Grp	No Treatment	Tipsy
36	Managed Stands	1	D mixed	1-2	Prod. Grp	No Treatment	Tipsy
37	Managed Stands	2	D mixed	1-2	Prod. Grp	No Treatment	Tipsy
38	Managed Stands	4	D mixed	1-2	Prod. Grp	No Treatment	Tipsy
39	Managed Stands	2	Cy	1-2	Prod. Grp	No Treatment	Tipsy
40	Managed Stands	4	Cy	1-2	Prod. Grp	No Treatment	Tipsy
41	Managed Stands	1	F	1-2	Prod. Grp	No Treatment	Tipsy
42	Managed Stands	2	F	1-2	Prod. Grp	Spaced	Tipsy
43	Managed Stands	2	F	1-2	Prod. Grp	No Treatment	Tipsy
44	Managed Stands	4	F	1-2	Prod. Grp	No Treatment	Tipsy
45	Managed Stands	2	Pl, Pw	1-2	Prod. Grp	No Treatment	Tipsy
46	Managed Stands	4	Pl, Pw	1-2	Prod. Grp	No Treatment	Tipsy

Analysis Unit	Stand Type	Productivity Group[1]	Leading Species	Age Class	SI Origin	Treatment	Model
47	Natural Immature Stands	1	C	3-7	Prod. Grp	No Treatment	VDYP
48	Natural Immature Stands	2	C	3-7	Prod. Grp	No Treatment	VDYP
49	Natural Immature Stands	4	C	3-7	Prod. Grp	Fertilized	VDYP
50	Natural Immature Stands	4	C	3-7	Prod. Grp	No Treatment	VDYP
51	Natural Immature Stands	1	H	3-7	Prod. Grp	No Treatment	WFP Model
52	Natural Immature Stands	2	H	3-7	Prod. Grp	No Treatment	WFP Model
53	Natural Immature Stands	4	H	3-7	Prod. Grp	No Treatment	WFP Model
54	Natural Immature Stands	1	B	3-7	Prod. Grp	No Treatment	VDYP
55	Natural Immature Stands	2	B	3-7	Prod. Grp	No Treatment	VDYP
56	Natural Immature Stands	4	B	3-7	Prod. Grp	No Treatment	VDYP
57	Natural Immature Stands	1	S	3-7	Prod. Grp	No Treatment	VDYP
58	Natural Immature Stands	2	S	3-7	Prod. Grp	No Treatment	VDYP
59	Natural Immature Stands	1	D	3-7	Prod. Grp	No Treatment	VDYP
60	Natural Immature Stands	2	D	3-7	Prod. Grp	No Treatment	VDYP
61	Natural Immature Stands	4	D	3-7	Prod. Grp	No Treatment	VDYP
62	Natural Immature Stands	1	D mixed	3-7	Prod. Grp	No Treatment	VDYP
63	Natural Immature Stands	2	D mixed	3-7	Prod. Grp	No Treatment	VDYP
64	Natural Immature Stands	4	D mixed	3-7	Prod. Grp	No Treatment	VDYP
65	Natural Immature Stands	2	Cy	3-7	Prod. Grp	No Treatment	VDYP
66	Natural Immature Stands	2	F	3-7	Prod. Grp	No Treatment	VDYP
67	Natural Immature Stands	4	F	3-7	Prod. Grp	No Treatment	VDYP
68	Natural Immature Stands	4	Pl, Pw	3-7	Prod. Grp	No Treatment	VDYP
69	Mature Forest	1	C	8-9	None	No Treatment	Flat Line
70	Mature Forest	2	C	8-9	None	No Treatment	Flat Line
71	Mature Forest	4	C	8-9	None	No Treatment	Flat Line
72	Mature Forest	1	H	8-9	None	No Treatment	Flat Line
73	Mature Forest	2	H	8-9	None	No Treatment	Flat Line
74	Mature Forest	4	H	8-9	None	No Treatment	Flat Line
75	Mature Forest	1	B	8-9	None	No Treatment	Flat Line
76	Mature Forest	2	B	8-9	None	No Treatment	Flat Line
77	Mature Forest	4	B	8-9	None	No Treatment	Flat Line
78	Mature Forest	1	S	8-9	None	No Treatment	Flat Line
79	Mature Forest	2	S	8-9	None	No Treatment	Flat Line
80	Mature Forest	4	S	8-9	None	No Treatment	Flat Line
81	Mature Forest	2	D	8-9	None	No Treatment	Flat Line
82	Mature Forest	1	Cy	8-9	None	No Treatment	Flat Line
83	Mature Forest	2	Cy	8-9	None	No Treatment	Flat Line
84	Mature Forest	4	Cy	8-9	None	No Treatment	Flat Line
85	Mature Forest	2	F	8-9	None	No Treatment	Flat Line
86	Mature Forest	4	F	8-9	None	No Treatment	Flat Line

[1] Productivity Groups 2 and 3 have been combined and referred to a “2”.

TABLE 21

TIMBER HARVESTING LANDBASE BY ANALYSIS UNIT AND OPERATION

Analysis Unit	Port McNeill	Holberg	Jeune Landing	Mahatta	Total	% of Total THLB
NSR	885.4	849.0	607.9	246.1	2588.3	1.73%
1	20.7	17.6			38.3	0.03%
2	110.9	153.2	26.7	4.5	295.4	0.20%
3	399.9	395.4		5.5	800.9	0.53%
4	17.4	7.5		3.4	28.3	0.02%
5	1445.8	1917.1	379.6	244.5	3987.0	2.66%
6	1818.2	326.3		9.8	2154.3	1.44%
7	3974.1	995.8	180.7	125.6	5276.1	3.52%
8	7.5	16.1			23.6	0.02%
9	83.8	53.7	38.9	1.1	177.5	0.12%
10		15.7			15.7	0.01%
11	306.6	456.4	192.0	319.7	1274.7	0.85%
12	395.7	1174.0		108.1	1677.8	1.12%
13	2495.2	1318.5	990.0	188.4	4992.1	3.33%
14	10.8	75.6			86.5	0.06%
15	7509.9	9531.0	4635.9	4633.9	26310.6	17.57%
16	174.5	748.2		24.0	946.6	0.63%
17	84.5	26.9	43.7	3.1	158.2	0.11%
18	0.7	4.5			5.2	0.00%
19	962.1	1725.4	333.2	651.0	3671.6	2.45%
20	32.5	32.7	66.0	8.3	139.5	0.09%
21	34.5	14.6			49.1	0.03%
22	14.0				14.0	0.01%
23	575.5	530.8	866.0	134.1	2106.5	1.41%
24	18.5	19.9	18.9	11.1	68.4	0.05%
25	1.2	13.9			15.1	0.01%
26	53.8	72.9			126.7	0.08%
27	256.9	363.3	294.9	173.9	1089.1	0.73%
28	11.0	41.1	1.1	17.5	70.7	0.05%
29	249.1	586.2	294.8	224.1	1354.2	0.90%
30	12.4	32.7			45.0	0.03%
31	2.7	34.5		1.2	38.4	0.03%
32	51.4	201.9	9.7	27.7	290.7	0.19%
33	24.6	6.0	2.3	21.6	54.6	0.04%
34	43.7	4.4	96.3	57.7	202.1	0.13%
35	6.7		0.8	3.3	10.8	0.01%
36	29.2	29.9	0.4	24.5	84.1	0.06%
37	27.0	37.7	11.9	7.2	83.8	0.06%
38	3.1	9.9		0.4	13.4	0.01%
39	158.3	65.8	17.4	16.6	258.1	0.17%
40	94.4	13.1	14.0	3.9	125.4	0.08%
41	11.7	0.5	0.9		13.1	0.01%
42	63.6	1.0	88.4		153.0	0.10%
43	743.9	160.2	203.8	24.1	1132.1	0.76%
44	175.5	70.1	0.3	0.4	246.4	0.16%
45	2.6	4.0	2.8		9.4	0.01%
46	33.8	0.4			34.2	0.02%
47	7.8	3.5			11.3	0.01%

Analysis Unit	Port McNeill	Holberg	Jeune Landing	Mahatta	Total	% of Total THLB
48	223.6	41.7	9.7	16.8	291.8	0.19%
49	262.9				262.9	0.18%
50	1244.2	6.7	1.8	21.2	1273.9	0.85%
51	407.2	523.9	301.8	170.6	1403.6	0.94%
52	11941.8	11106.4	5563.4	1885.8	30497.4	20.37%
53	1062.0	743.9	226.5	163.7	2196.2	1.47%
54	4.5	16.7			21.2	0.01%
55	381.7	249.3	75.5	7.2	713.7	0.48%
56	23.7	3.3		4.5	31.5	0.02%
57	3.9	3.1	7.3	1.5	15.8	0.01%
58	12.9	10.7	11.9		35.6	0.02%
59	91.4	201.1	153.1	48.4	494.0	0.33%
60	114.7	285.5	248.5	25.3	674.0	0.45%
61	6.4	10.1	6.6	0.7	23.8	0.02%
62	20.0	71.7	55.2	44.6	191.6	0.13%
63	7.4	216.1	259.9	29.7	513.2	0.34%
64	0.8	4.5	7.5	8.1	20.8	0.01%
65		9.0			9.0	0.01%
66	183.9		188.0		371.9	0.25%
67	40.3				40.3	0.03%
69	39.3	252.6	5.3	4.2	301.4	0.20%
70	1244.4	6087.9	1324.9	354.7	9011.8	6.02%
71	3483.9	3397.8	366.1	182.9	7430.7	4.96%
72	95.5	443.9	211.7	29.0	780.2	0.52%
73	3800.2	11532.0	8611.3	1440.9	25384.3	16.95%
74	341.0	895.9	388.2	107.5	1732.6	1.16%
75	8.6	6.2		0.7	15.6	0.01%
76	252.0	217.9	585.4	80.0	1135.3	0.76%
77	13.8	2.8	21.2	1.2	39.0	0.03%
78	27.4	220.1	14.1	9.9	271.5	0.18%
79	17.5	233.1	40.0	6.6	297.1	0.20%
80	0.5	56.5	10.4	0.4	67.8	0.05%
81			1.9		1.9	0.00%
82	9.8	3.1	1.1		14.0	0.01%
83	338.9	394.2	230.1	56.5	1019.7	0.68%
84	238.5	288.3	40.6	9.7	577.1	0.39%
85	65.1	9.3	154.2		228.6	0.15%
86	8.1				8.1	0.01%
Total	*49461.1*	*59704.6*	*28542.7*	*12038.6*	*149747.0*	*100 %*

8.0 GROWTH AND YIELD

The yield tables for this data package were provided by J.S. Thrower and Associates and were divided into two broad groups: existing stands and future stands. Yield tables for existing stands were further divided into:

1. managed stands (stands that are 40 years of age or less);
2. natural immature stands (stands between 41 and 140 years of age), and
3. mature stands.

Input information for producing yield tables for the existing stands were based on the summaries of the inventory database. Input information for the future stands was based on silviculture strategies that are currently being implemented by ecological units.

No growth was assumed for old-growth stands, the current volumes are assumed to remain constant.

8.1 Site Index Assignments

Site indices for the leading species were as recommended by J.S. Thrower & Associates Ltd. (1997)[5]. The productivity groups were a refinement of those used in the last Management Plan and the rationale for amalgamation is given by J.S. Thrower & Associates Ltd. Although Block 2 was not then part of the TFL, we expect the results to apply as Block 2 is in the same BEC variant, has been classified using the same ecological units, and is in the same geographic area.

TABLE 22

SITE INDEX VALUES USED TO COMPUTE CURRENT INVENTORY

Productivity Group	Ecosystems	Leading Species				
		H	C	B	S	F (Dec & Pl, Pw)
1	S3	33.7	27	30.1	35.2	37.5
2/3	S4, S5, S13, S1ha, M1, M3, M5, S12	28.0	23.8	26.2	31.0	31.4
4	S1ch, S2, S6, S10, M2, M4	20.1	19.2	20.7	25.0	22.6

8.2 Utilization Levels

The utilization level was 12.5 cm for all existing stands less than 41 years old and for future stands. Stump height for these stands was 30 cm and top diameter inside bark (DIB) was 10 cm. Utilization level for immature and mature conifer stands was 17.5 cm, with stump height of 30 cm and top DIB of 15 cm. TABLE 23 summarizes the utilization levels currently in effect for existing stands and proposed for utilization of future stands.

[5] J.S Thrower &Associates Ltd. 1997. Second Growth Site Index on Western Forest Products TFL 6. Unpublished Report, Contract No WPV-091-002. 31 p. Refer to Appendix II.

TABLE 23

UTILIZATION LEVELS

Species Group	Utilization			Firmwood Standard (%)
	Minimum DBH (cm)	Stump Height (cm)	Top DIB (cm)	
Managed Conifers (0 to <=40 years)	12.5	30.0	10.0	50
Immature Conifers (41 to <140 years)	17.5	30.0	15.0	50
Mature Conifer (≥141 years)	17.5	30.0	15.0	50
Older Mature Deciduous > 41 years old	17.5	30.0	15.0	50
Younger Mature Deciduous > 24 years old but < 40 years old	12.5	30.0	10.0	50

8.3 Decay, Waste and Breakage

The default decay, waste, and breakage factors for TFL 6 within VDYP 6.4 were used for existing natural stands.

8.4 Operational Adjustment Factors

For existing managed stands and for all future stands, an OAF1 of 10% and OAF2 of 5% were used for these MSYTs. The MOF recommends using 15% for OAF1 where local information is not available, however, information from the TFL suggests that the proportion of unmapped NP in second-growth stands is very low. Results from a random sample of 68 plots throughout second-growth stands on the TFL showed that less than 5% of the sampled areas was classified as NP. These results were supported by J.S. Thrower and Associates' field staff that noted generally excellent stocking in these stands and very little NP area that was not mapped. Therefore, an OAF1 of 10% was used in these yield tables to better reflect the excellent stocking and low proportion of NP area on the TFL 6 landbase.

8.5 Volume Deductions

Deciduous volumes existing in pure or mixed stands have been deducted from the productive forest as a problem forest type. No additional deductions are therefore required for the deciduous component.

8.6 Yield Tables For Unmanaged Stands

8.6.1 Natural Immature Stand Volumes

For existing natural immature stands, an analysis unit was assigned to every forest cover polygon based on criteria defined in Section 7.2. Area-based weighted averages of species composition, site index, and crown closure were calculated. The stocking class with the most area within an analysis unit was selected as the stocking class for the whole analysis unit. This information was then used in VDYP 6.4 to generate yield tables. The VDYP input information for existing, natural stands is given in TABLE 24.

TABLE 24

VDYP INPUTS FOR EXISTING NATURAL IMMATURE STANDS

Analysis Unit	Site Index	Spc 1	%	Spc 2	%	Spc 3	%	Spc 4	%	Spc 5	%	Spc 6	%	Utiliz.	PSYU	FIZ	Stocking Class
47	27.0	C	69	H	31									17.5	306	A	1
48	23.8	C	75	H	25									17.5	306	A	1
49	24.0	C	100											17.5	306	A	1
50	19.2	C	90	H	10									17.5	306	A	1
51	33.7	H	78	B	10	S	8	D	2	C	2			17.5	306	A	1
52	28.0	H	76	B	15	S	5	F	2	C	1	D	1	17.5	306	A	1
53	20.1	H	77	B	12	C	5	S	3	F	2	D	1	17.5	306	A	1
54	30.1	B	50	H	50									17.5	306	A	1
55	26.2	B	53	H	47									17.5	306	A	1
56	20.7	B	59	H	41									17.5	306	A	1
57	35.2	S	63	H	28	D	9							17.5	306	A	1
58	31.0	S	64	H	24	D	12							17.5	306	A	1
59	37.5	D	100											17.5	306	A	1
60	31.4	D	100											17.5	306	A	1
61	22.6	D	100											17.5	306	A	1
62	37.5	D	57	H	25	S	8	B	8	C	2			17.5	306	A	1
63	31.4	D	51	H	36	B	4	S	4	F	3	C	2	17.5	306	A	1
64	22.6	D	50	H	38	B	8	S	2	F	2			17.5	306	A	1
65	23.8	Cy	50	H	50									17.5	306	A	1
66	31.4	F	59	H	37	B	3	C	1					17.5	306	A	1
67	22.6	F	54	H	46									17.5	306	A	1
68	22.6	Pl	100											17.5	306	A	1

After this work was completed, local data from an in progress Vegetation Resources Inventory became available for hemlock leading stands. This data was used to calibrate a spreadsheet basal area-based yield model developed internally and based on Forest Renewal funded work carried out by Olympic Resource Management. The provincial permanent sample plot database was used to create a basal area-height function that was used to generate volume-height curves. These were fitted to preliminary VRI data by transforming and normalizing deviations around maximum MAI to ensure that the curve's best fit was in the vicinity of probable harvest. Site index was then changed to generate a curve for each analysis unit. After discussions with Timber Supply Branch, yield curves 51, 52, and 53 were replaced with the revised yield estimates.

To ensure that the volume yields by analysis units did not over or under estimate yields calculated by individual inventory polygons a comparison exercise was carried out. TABLE 25 provides this comparison of the natural immature timber harvesting land base volume for individual polygons by the volume calculated by analysis units. All analysis units fell within acceptable tolerances with the exception of Analysis Unit 49. This particular analysis unit represents pure cedar stands that have been fertilized and adjustments to the analysis unit yield table were made to reflect this. Unfortunately, the adjustment was not repeated when the individual polygon volumes were calculated using VDYP.

TABLE 25

NATURAL IMMATURE VOLUME VERIFICATION

Analysis Unit	THLB Area	Polygon Volume	Analysis Unit Volume	% Difference of Polygon vs. Analysis Unit Volume
47	11.3	5081.4	5206.7	2%
48	291.8	84652.3	86812.2	2%
49	262.9	17377.2	32861.6	47%
50	1273.9	135419.3	142812.1	5%
51	1403.6	915666.6	916629.6	0%
52	30497.4	18294527.0	18494355.4	1%
53	2196.2	613356.5	618145.5	1%
54	21.2	18085.1	20034.9	10%
55	713.7	507342.5	509261.1	0%
56	31.5	14542.1	14681.4	1%
57	15.8	13239.4	13801.9	4%
58	35.6	22419.9	24013.4	7%
59	494.0	174002.8	174053.1	0%
60	674.0	212384.1	212548.0	0%
61	23.8	4848.6	4850.5	0%
62	191.6	100966.6	104919.4	4%
63	513.2	280185.6	280309.8	0%
64	20.8	5587.3	5613.2	0%
65	9.0	5211.8	5330.8	2%
66	371.9	231852.4	240966.3	4%
67	40.3	16300.9	16610.9	2%
68	0.0	0.0	0.0	0%
Total	39093.6	21673047.4	21923817.9	1%

8.6.2 Existing Mature Stand Volumes

The timber volume in existing mature stands (those ≥ 140 years) was determined for each analysis unit noted in TABLE 20 by using area weighted average volumes from the inventory plots located in these stands.

TABLE 26 provides a comparison of the mature timber harvesting land base volume based on the inventory plots and by weighted average volumes used for the Analysis Units. TABLE 27 provides a breakdown of current mature timber harvesting landbase volumes by operations.

TABLE 26

EXISTING MATURE VOLUME VERIFICATION

Analysis Unit	THLB Area	Weighted Average Volume	Weighted Avg. Total Volume	Inventory Volume	% Difference of Inventory vs. Analysis. Unit
69	301.4	991.8	298928.5	298893.4	0.012%
70	9011.8	944.9	8515250.0	8514986.0	0.003%
71	7430.7	826.7	6142960.0	6142603.0	0.006%
72	780.2	954.1	744388.8	744324.9	0.009%
73	25380.5	945.0	23984573.0	23983236.0	0.006%
74	1732.1	919.3	1592320.0	1592376.0	-0.004%
75	15.6	1048.2	16351.9	16319.7	0.197%
76	1135.3	1022.9	1161298.0	1161284.0	0.001%
77	39.0	982.0	38298.0	38301.4	-0.009%
78	271.5	1067.8	289907.7	289960.9	-0.018%
79	297.1	1136.8	337743.3	337760.5	-0.005%
80	67.8	979.8	66430.4	66404.0	0.040%
81	1.9	571.2	1085.3	1079.6	0.523%
82	14.0	892.8	12499.2	12506.1	-0.055%
83	1019.7	799.0	814740.3	814740.4	0.000%
84	577.1	836.8	482917.3	482881.5	0.007%
85	228.6	955.5	218427.3	218405.2	0.010%
86	8.1	617.5	5001.8	4971.3	0.609%

TABLE 27

TIMBER HARVESTING LAND BASE MATURE VOLUMES (M^3) BY OPERATION

Analysis Unit	Leading Species and Site Productivity[1]	Port McNeill	Holberg	Jeune Landing	Mahatta	Total
69	Cedar (PG 1)	38,258	251,409	4,863	4,363	298,893
70	Cedar (PG 2)	1,182,915	5,789,763	1,242,655	299,654	8,514,986
71	Cedar (PG 4)	2,630,322	3,020,560	344,777	146,944	6,142,603
72	Hemlock (PG 1)	91,857	429,187	192,681	30,600	744,325
73	Hemlock (PG 2)	3,564,450	11,052,256	7,963,951	1,405,120	23,985,778
74	Hemlock (PG 4)	317,156	821,164	352,374	102,000	1,592,694
75	Balsam (PG 1)	8,849	6,654		817	16,320
76	Balsam (PG 2)	264,166	233,852	573,213	90,055	1,161,284
77	Balsam (PG 4)	13,726	2,049	21,451	1,075	38,301
78	Spruce (PG 1)	28,401	238,127	13,599	9,834	289,961
79	Spruce (PG 2)	21,767	269,697	36,060	10,236	337,761
80	Spruce (PG 4)	523	56,233	9,065	583	66,404
81	Deciduous (PG 2)			1,080		1,080
82	Cypress (PG 1)	9,293	2,481	732		12,506
83	Cypress (PG 2)	302,880	321,890	151,381	38,590	814,740
84	Cypress (PG 4)	211,678	237,800	26,773	6,630	482,882
85	Douglas-fir (PG 2)	52,444	6,547	159,414		218,405
86	Douglas-fir / Pine (PG 4)	4,971				4,971
Total		***8,743,656***	***22,739,667***	***11,094,070***	***2,146,501***	***44,723,894***

[1] PG = Productivity Group.

8.7 Yield Tables for Managed Stands

8.7.1 Existing Managed Stand Volumes

For existing managed stands, area-based weighted average of species composition and productivity groups were estimated. Fertilized stands were assumed to move to the next highest group. Establishment densities were adjusted to simulate free growing densities expected from application of stocking standards and typical planting densities for age class 1 stands. All age class 1 stands were assumed planted with one-year-old stock. Age class 2 stands were assumed to have been established through natural regeneration, which is reflected in the growth model by using higher establishment densities. For spaced stands curves were generated at higher initial densities to simulate natural infill. OAF1 was 10%, OAF2 was 5%. Regeneration delay was assumed to be 0 within the growth model as the timber supply model addresses the application of regeneration delay. The existing, managed stand yield tables were generated using Batch TIPSY version 2.1. Input information is given in TABLE 28.

TABLE 28

TIPSY INPUTS FOR EXISTING MANAGED STANDS – AGE CLASS 1

Analysis Unit	Site Index	Spc 1	%	Spc 2	%	Spc 3	%	Spc 4	%	Spc 5	%	Planted/ Natural	Initial Density	PCT Density
1	27.0	C	87	H	11	Cy	2					P	1375	
2	27.0	C	85	H	8	S	7					P	1375	
3	23.8	C	87	H	12	S	1					P	1375	
4	23.8	C	80	H	13	F	7					P	1375	850
5	23.8	C	85	H	12	F	1	S	1	B	1	P	1375	
6	24.0	C	92	H	8							P	1375	
7	19.2	C	92	H	6	Cy	1	Pl	1			P	1375	
8	33.7	H	71	C	13	S	11	B	5			P	1100	
9	33.7	H	79	S	10	B	6	C	4	F	1	P	1400	850
10	33.7	H	61	S	38	C	1					P	1400	850
11	33.7	H	71	S	14	B	7	C	7	F	1	P	1100	
12	28.0	H	71	C	18	B	6	S	5			P	1100	
13	28.0	H	77	B	10	C	5	S	4	F	4	P	1400	850
14	28.0	H	73	S	11	B	9	C	7			P	1400	850
15	28.0	H	76	B	13	C	6	S	4	F	1	P	1100	
16	28.0	H	67	C	19	S	10	B	3	F	1	P	1100	
17	20.1	H	76	F	7	B	7	C	6	S	4	P	1400	850
18	28.0	H	73	S	17	C	7	B	3			P	1400	850
19	20.1	H	71	C	12	B	8	S	7	F	2	P	1100	
20	30.1	B	73	S	16	H	8	Bg	2	C	1	P	1100	
21	26.2	B	69	H	21	C	10					P	1100	
22	26.2	B	70	H	30							P	1400	850
23	26.2	B	67	H	26	S	5	C	1	Cy	1	P	1100	
24	20.7	B	60	H	29	S	6	C	3	Cy	2	P	1100	
25	35.2	S	49	H	35	C	16					P	1100	
26	35.2	S	56	H	34	F	9	C	1			P	1400	850
27	35.2	S	71	H	22	C	5	B	2			P	1100	
28	31.0	S	59	H	35	F	4	C	2			P	1400	850
29	31.0	S	62	H	33	C	4	B	1			P	1100	
30	31.0	S	70	H	19	C	11					P	1100	
31	25.0	S	51	F	25	H	24					P	1400	850
32	25.0	S	57	H	29	C	12	B	2			P	1100	
33	37.5	D	96	Ac	4							P	1100	
34	31.4	D	99	Ac	1							P	1100	
35	22.6	D	100									P	1100	
36	37.5	D	65	H	15	S	15	C	4	B	1	P	1100	
37	31.4	D	59	H	24	S	9	C	4	B	4	P	1100	
38	22.6	D	63	H	25	C	10	B	2			P	1100	
39	23.8	Cy	66	H	14	C	12	B	8			P	1100	
40	19.2	Cy	85	C	11	H	4					P	1100	
41	37.5	F	67	H	15	C	9	B	7	S	2	P	1100	
42	31.4	F	54	H	46							P	1400	850
43	31.4	F	65	H	28	C	3	S	2	D	2	P	1100	
44	22.6	F	55	C	20	H	20	S	5			P	1100	
45	31.4	Pl	76	C	14	H	5	Cy	5			P	1100	
46	22.6	Pl	50	C	37	Pw	8	F	5			P	1100	

TABLE 29

TIPSY INPUTS FOR EXISTING MANAGED STANDS – AGE CLASS 2

Analysis Unit	Site Index	Spc 1	%	Spc 2	%	Spc 3	%	Spc 4	%	Spc 5	%	Planted/ Natural	Initial Density	PCT Density
1	27.0	C	87	H	11	Cy	2					N	3000	
2	27.0	C	85	H	8	S	7					N	3000	
3	23.8	C	87	H	12	S	1					N	3000	
4	23.8	C	80	H	13	F	7					N	3000	850
5	23.8	C	85	H	12	F	1	S	1	B	1	N	3000	
6	24.0	C	92	H	8							N	3000	
7	19.2	C	92	H	6	Cy	1	Pl	1			N	3000	
8	33.7	H	71	C	13	S	11	B	5			N	2200	
9	33.7	H	79	S	10	B	6	C	4	F	1	N	2800	850
10	33.7	H	61	S	38	C	1					N	2800	850
11	33.7	H	71	S	14	B	7	C	7	F	1	N	2200	
12	28.0	H	71	C	18	B	6	S	5			N	2200	
13	28.0	H	77	B	10	C	5	S	4	F	4	N	2800	850
14	28.0	H	73	S	11	B	9	C	7			N	2800	850
15	28.0	H	76	B	13	C	6	S	4	F	1	N	2200	
16	28.0	H	67	C	19	S	10	B	3	F	1	N	2200	
17	20.1	H	76	F	7	B	7	C	6	S	4	N	2800	850
18	28.0	H	73	S	17	C	7	B	3			N	2800	850
19	20.1	H	71	C	12	B	8	S	7	F	2	N	2200	
20	30.1	B	73	S	16	H	8	Bg	2	C	1	N	2200	
21	26.2	B	69	H	21	C	10					N	2200	
22	26.2	B	70	H	30							N	2800	850
23	26.2	B	67	H	26	S	5	C	1	Cy	1	N	2200	
24	20.7	B	60	H	29	S	6	C	3	Cy	2	N	2200	
25	35.2	S	49	H	35	C	16					N	2200	
26	35.2	S	56	H	34	F	9	C	1			N	2800	850
27	35.2	S	71	H	22	C	5	B	2			N	2200	
28	31.0	S	59	H	35	F	4	C	2			N	2800	850
29	31.0	S	62	H	33	C	4	B	1			N	2200	
30	31.0	S	70	H	19	C	11					N	2200	
31	25.0	S	51	F	25	H	24					N	2800	850
32	25.0	S	57	H	29	C	12	B	2			N	2200	
33	37.5	D	96	Ac	4							N	2200	
34	31.4	D	99	Ac	1							N	2200	
35	22.6	D	100									N	2200	
36	37.5	D	65	H	15	S	15	C	4	B	1	N	2200	
37	31.4	D	59	H	24	S	9	C	4	B	4	N	2200	
38	22.6	D	63	H	25	C	10	B	2			N	2200	
39	23.8	Cy	66	H	14	C	12	B	8			N	2200	
40	19.2	Cy	85	C	11	H	4					N	2200	
41	37.5	F	67	H	15	C	9	B	7	S	2	N	2200	
42	31.4	F	54	H	46							N	2800	850
43	31.4	F	65	H	28	C	3	S	2	D	2	N	2200	
44	22.6	F	55	C	20	H	20	S	5			N	2200	
45	31.4	Pl	76	C	14	H	5	Cy	5			N	2200	
46	22.6	Pl	50	C	37	Pw	8	F	5			N	2200	

8.7.2 Future Stand Volumes

For future stands, a series of silvicultural strategies were derived based on what is currently being done on the TFL and what Western Forest Products intends to do in the future. These silvicultural strategies were based on ecological units. Input information is given in TABLE 30. For existing managed stands, OAF1 was 10%, OAF2 was 5%, utilization limit was 12.5 cm, and there was no regeneration delay. Genetic gains were incorporated into the TIPSY model for all planted regeneration curves. A genetic gain of 5% was used for Western Hemlock and Cypress in the first decade of future planting and 15% for plantations established after that. A genetic gain of 3% was used for all future Western Red Cedar and Sitka spruce plantations.

TABLE 30

SILVICULTURE STRATEGIES FOR FUTURE STANDS

Analysis Unit	Productivity Group	Ecosystem Class.	Treatment	Estab. Density	Resid. Density	Spc 1	%	Spc 2	%	Spc 3	%	Site Index[1]	Planted/ Natural
120	2	S1HA	Spaced	2500	850	H	80	B	20			28	N
121	2	S1HA	No treatment	1050		H	80	B	20			28	P
122	4	S1CH	No treatment	1050		C	80	H	20			19	P
123	4	S1CH	Fertilized	1050		C	80	H	20			24	P
124	4	S2	No treatment	1050		C	90	H	10			19	P
125	4	S2	Fertilized	1050		C	90	H	10			24	P
126	1	S3	No treatment	1150		B	40	S	40	H	20	30	P
127	2	S4	No treatment	1150		B	40	S	40	H	20	26	P
128	2	S5	No treatment	1050		C	50	S	50			24	P
129	4	S6	No treatment	1050		C	100					19	P
130	2	S12HA	No treatment	1150		H	80	B	20			28	P
131	2	S12HA	Spaced	2500	850	H	80	B	20			28	N
132	2	S12CH	No treatment	1050		C	80	H	20			24	P
133	2	S13	No treatment	1150		H	50	B	40	S	10	28	P
134	2	M1	No treatment	1050		H	70	B	20	C	10	28	P
135	4	M2	No treatment	1050		H	60	C	30	B	10	20	P
136	2	M3	No treatment	1050		H	60	C	20	B	20	28	P
137	4	M4	No treatment	1050		H	70	C	30			20	P
138	2	M5	No treatment	1050		H	60	C	20	B	20	28	P

[1]Note that site index assignment by species is described in Section 8.1.

Current Site Degradation

Western Forest Products' standard operating practices include the ripping and restocking of trails once logging is completed. Highlead landings are typically small and of no measurable consequence. Helicopter landings are rehabilitated. No allowance for current site degradation has been made in TABLE 15 or TABLE 16.

Future Site Degradation

Future road systems have been projected within the TFL and area reductions will be applied once the model harvests the polygon. Section 6.16.3 outlines the amount of future road to be built in the TFL over the long term.

8.7.3 Regeneration Delay

The regeneration delay refers to the average time elapsed between harvesting and establishment of new plantations on the TFL. TABLE 31 indicates the regeneration delay period used to shift the yield curve for each regenerated analysis unit. Regeneration delay will be applied in the timber supply model, not in the TIPSY yield model. Time-of-planting fertilization is current management practice on all salal sites (S1ch, S12ch, S6). In cases where fertilizer is applied during planting field results and research trials indicate that an “effective” one-year reduction of regeneration delay is appropriate and conservative.

TABLE 31

REGENERATION DELAY PERIOD

Analysis Unit	Regeneration Delay Years[1]
120	1.5
121	1.5
122	1.5 (0.5)
123	1.5 (0.5)
124	2.0 (1.0)
125	2.0 (1.0)
126	1.0
127	1.0
128	1.0 (0.0)
129	1.5 (0.5)
130	1.5
131	1.5
132	1.5 (0.5)
133	1.0
134	4.0
135	4.0
136	4.0
137	4.0
138	4.0

[1] Years in ( ) indicate regeneration delay period for stands planted with fertilizer.

8.7.4 Regeneration Assumptions

The timber supply analysis for the TFL will use the regeneration assumptions outlined in TABLE 32. Assigning a regeneration curve considers two elements 1) the ecosystem classification and 2) the site productivity for each stand. Due to the ecosystem classification factor a one to one relationship between the existing analysis unit and the regeneration analysis unit cannot be made. For regeneration analysis units that indicate spacing or fertilization the management option will influence which existing analysis unit gets assigned.

TABLE 32

REGENERATION AFTER HARVESTING

Existing Analysis Unit	Ecosystem Classification	Regenerated Analysis Unit(s)
3,4,5.12,13,14,15,21,22,23,28,29, 33,34,37,39,42,43,45,48,52,55,58,60, 63,65,66,70,73,76,79,81,83,85	S1ha	120, 121
6,7,16,17,19,24,30,31,32,35,38,40,44,46,49,50,53,56,61,64,71,74,77,80, 84,86	S1ch	122, 123
6,7,16,17,19,24,30,31,32,35,38,40,44,46,49,50,53,56,61,64,71,74,77,80, 84,86	S2	124, 125
1,2,8,9,10,11,20,25,26,27,33,36,39,41,45,47,51,54,57,59,62,66,69,72,75 ,78,82,85	S3	126
3,4,5.12,13,14,15,21,22,23,28,29, 33,34,37,39,42,43,45,48,52,55,58,60, 63,65,66,70,73,76,79,81,83,85	S4	127
3,4,5.12,13,14,15,21,22,23,28,29, 33,34,37,39,42,43,45,48,52,55,58,60, 63,65,66,70,73,76,79,81,83,85	S5	128
6,7,16,17,19,24,30,31,32,35,38,40,44,46,49,50,53,56,61,64,71,74,77,80, 84,86	S6	129
6,7,16,17,19,24,30,31,32,35,38,40,44,46,49,50,53,56,61,64,71,74,77,80, 84,86	S10	122, 123
3,4,5.12,13,14,15,21,22,23,28,29, 33,34,37,39,42,43,45,48,52,55,58,60, 63,65,66,70,73,76,79,81,83,85	S12ha	130, 131
3,4,5.12,13,14,15,21,22,23,28,29, 33,34,37,39,42,43,45,48,52,55,58,60, 63,65,66,70,73,76,79,81,83,85	S12ch	132
3,4,5.12,13,14,15,21,22,23,28,29, 33,34,37,39,42,43,45,48,52,55,58,60, 63,65,66,70,73,76,79,81,83,85	S13	133
3,4,5.12,13,14,15,21,22,23,28,29, 33,34,37,39,42,43,45,48,52,55,58,60, 63,65,66,70,73,76,79,81,83,85	M1	134
6,7,16,17,19,24,30,31,32,35,38,40,44,46,49,50,53,56,61,64,71,74,77,80, 84,86	M2	135
3,4,5.12,13,14,15,21,22,23,28,29, 33,34,37,39,42,43,45,48,52,55,58,60, 63,65,66,70,73,76,79,81,83,85	M3	136
6,7,16,17,19,24,30,31,32,35,38,40,44,46,49,50,53,56,61,64,71,74,77,80, 84,86	M4	137
3,4,5.12,13,14,15,21,22,23,28,29, 33,34,37,39,42,43,45,48,52,55,58,60, 63,65,66,70,73,76,79,81,83,85	M5	138

8.7.5 Species Conversion

A small amount of non-productive brush type (NP BR) is converted on a yearly basis within the TFL. This type occurs in small patches and is usually contiguous to productive forestland. This area is site prepared in conjunction with the harvested area and planted. The area converted on a yearly basis is difficult to quantify and insignificant, it will not be modelled.

Conversion of alder leading stands has been discussed previously and will be modelled as a sensitivity analysis.

8.8 Silviculture History

8.8.1 Existing Managed Immature

TABLE 33 provides a breakdown of the extent of immature managed stands in the TFL by analysis group, silviculture treatment and age class.

TABLE 33

IMMATURE MANAGEMENT HISTORY BY THLB AREA AND AGE CLASS

Analysis Unit	Natural								Planted								Total
	No Treatment		Spaced		Fertilized		Fertilized and Spaced		No Treatment		Spaced		Fertilized		Fertilized and Spaced		
	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	
1						7.7							30.6				38.3
2	6.7	54.1							228.0	6.6							295.4
3					34.2	88.8							612.3	65.6			800.9
4			3.2	11.4							10.2	3.5					28.3
5	129.7	410.3							3,339.4	107.5							3,987.0
6					58.2	626.0							1,417.9	52.1			2,154.3
7	220.9	1,222.1							3,641.2	192.0							5,276.1
8					9.5	7.7							5.6	0.8			23.6
9			5.7	119.2							21.2	31.4					177.5
10								14.7								1.0	15.7
11	127.4	531.0							254.2	362.0							1,274.7
12					211.1	683.1							531.1	252.6			1,677.8
13			841.8	2,127.9							1,086.5	936.0					4,992.1
14							9.1	26.4							40.9	10.0	86.5
15	5,173.5	10,320.3							7,667.5	3,149.3							26,310.6
16					77.4	414.0							126.5	328.8			946.6
17			18.3	64.7							34.2	41.0					158.2
18							0.5	0.7							3.0	1.1	5.2
19	490.2	1,676.5							551.8	953.2							3,671.6
20									139.5								139.5
21													49.1				49.1
22			10.5								3.5						14.0
23	202.1	236.7							1,566.4	101.2							2,106.5
24	6.1	2.5							59.8								68.4
25						0.7							14.4				15.1
26				1.7							5.0	120.0					126.7
27	18.3	5.1							602.3	463.3							1,089.1
28				2.8							18.7	49.2					70.7
29	14.3	25.4				6.0			452.0	796.3			14.0	46.2			1,354.2
30						12.4							2.7	30.0			45.0
31				1.2							0.8	36.4					38.4
32		6.0							21.4	263.3							290.7
33	0.2	43.0							2.3	9.0							54.6
34	5.4	184.1							2.5	10.2							202.1
35		10.2								0.7							10.8
36		47.1							0.4	36.5							84.1
37		64.6							3.3	15.8							83.8
38		12.3								1.1							13.4
39	3.9	101.3							152.9								258.1
40	20.1	1.0							104.3								125.4
41									4.2	8.9							13.1
42											1.4	151.6					153.0
43	34.7	27.9							626.2	443.3							1,132.1
44		3.1							40.7	202.6							246.4
45									9.4								9.4
46									15.7	18.5							34.2
Total	***6,453.5***	***14,984.4***	***879.5***	***2,328.9***	***390.3***	***1,846.5***	***9.6***	***41.9***	***19,485.7***	***7,141.4***	***1,181.4***	***1,369.2***	***2,804.2***	***776.1***	***43.8***	***12.1***	***59,748.4***

8.8.2 Backlog and Current Non-Stocked Areas (NSR)

As of January 1, 1998 the total area of NSR amounted to 2864.7 ha. Of the NSR area within the TFL, 2588.3 ha are in the timber harvesting land base with the remainder in constrained areas. Currently, 27.4 ha of backlog areas are reported in the GIS; however, operational staff estimates indicate that most of these area are incorrectly classified and are in fact SR or NP. Natural NSR areas, blow-down and old slash fire escapes areas, are also reported in the GIS. These areas are also believed to be misclassified and are most likely fully stock stands. Western Forest Products target is committed to re-stocking denudated areas within three years of harvest. Since 1987 WFP has planted an average of approximately 1,700 ha/yr.

TABLE 34

NSR AREA

Operation	THLB Area (ha)			Non-THLB Area (ha)	
	Backlog NSR	Current NSR	Natural NSR	Current NSR	Natural NSR
Holberg	15.9	745.9	87.2	56.7	5.7
Jeune Landing	6.3	576.9	24.7	42.2	1.5
Mahatta	0.0	244.2	1.9	13.8	0.0
Port McNeill	5.2	856.7	23.5	143.8	12.6
Total	***27.4***	***2423.7***	***137.3***	***256.5***	***19.8***

Timber supply analysis assumption for dealing with reported NSR is as followed:

- Backlog NSR and Natural NSR areas are assumed to be fully stocked and will be given an age of ten years.
- Current NSR will be regenerated to their appropriate Analysis Unit within the specified regeneration delay period.

9.0 PROTECTION

9.1 Unsalvaged Losses

The risk of loss of timber in the TFL is relatively slight. The TFL has a wet climate characterized by cool, wet summers and fire suppression has been efficient; hence the likelihood of losses to forest fire is small. Historically there have been no losses due to pests and disease recorded and negligible losses to wildfire. As road networks expand throughout the TFL, the likelihood of significant losses in the timber harvesting land base being left to waste continues to decrease.

The only potential unsalvageable loss of significance in the TFL is windthrow. Loss of single trees or small groups is accounted for in the plot sampling used to calibrate VDYP and TIPSY with operational adjustment factors accounting for larger gaps. Large scale events, such as occurred circa 1904-6, would undoubtedly be the subject of a major salvage program if such an event were to recur. Under current District and company policy smaller windthrow patches would be salvaged if near a road and otherwise operable. Completion of the Vegetation Resource Inventory in 2000 should confirm that unsalvaged patches are infrequent and very minor. If found, their size and approximate year of origin may be used to establish a trend line and estimate their impact and frequency of occurrence.

Windthrow losses unlikely to be salvaged or accounted for are small patches in the order of tenths of a hectare. Timber edges around clearcuts would make up the majority of the losses in this category.

A pilot project to develop a surveying methodology to estimate the amount of unsalvaged windthrow per meter of clearcut edge proved unworkable and had to be abandoned. Dr. Steve Mitchell at the University of B.C. is developing a windthrow hazard rating system and in 2000 we hope to use the occurrence data collected for that project on TFL 6 as a GIS overlay. This would estimate the area of edge damage for several TFL 6 mapsheets and could be extrapolated to the TFL inventory. If salvage recovery and constrained areas can be successfully netted out, the result would be the area containing unsalvaged losses due to windfall. If volume per hectare can be estimated, this would provide a check of the reasonableness of operational estimates.

Engineering personnel in TFL 6 estimated average annual unsalvaged losses of 5,750 m3 (estimates ranged from 500 to 3,000 m3 within each of three operations) on the unconstrained timber harvesting land base for the past decade.

In December 1997 a freak snow and windstorm in the vicinity of Port McNeill, caused noticeable damage of second growth. Other parts of the TFL were undamaged. A survey of this damage, suggested that up to 15,000 m3 of damaged second growth was present. However, as it was widely scattered and/or inaccessible, only a tiny fraction that was in close proximity to roads could be salvaged. As this kind of damage was previously unheard of, it was assumed to be an event with at least a twenty-year return interval. Therefore, the associated annual loss would be expected to average at most 750 m3/yr. Consequently, we estimate that unsalvageable losses from the timber harvesting land base would be no more than 7,000 m3/yr within TFL 6 or equivalent to about 10 ha/year.

10.0 INTEGRATED RESOURCE MANAGEMENT

10.1 Forest Resource Inventory

TABLE 35 summarizes the forest resource inventories currently be maintained for the TFL.

TABLE 35

FOREST RESOURCE INVENTORY STATUS

Forest Resource Inventory	Standard	Date Completed	Date Approved	Comments
Timber Inventory	MoF	1970	Used in MP 5, 6, 7, 8	Second growth examined in 1982 for Block 2 Site productivity work – JS Thrower 1997 VRI Photo Interpretation phase started
Ecosystem and Terrain	WFP	1982 - 1985	Used in MP 8	Completed by T. Lewis. Minor classification revision in 1998.
Terrain Stability Classification	MoF	Block 1 1995 Block 2 1992	TFL 6 Block 2 used in MP 9 for TFL 25	Completed by T. Lewis Block 2 updated to MoF standards in 1998
Landscape Inventory	MoF	1994	1995	Completed by Neufeldt Reeve MacDougall Consultants Visual Mitigation Strategy conducted in 1998 Upgrading of Rec and Visual Inventory to new standards is currently be done by MoF as part of the Regional Roll-up.
Recreation Inventory	MoF	1990	1991	
Recreation Analysis (ROS)	MoF	1990	1991	
Karst Inventory	WFP	1979	-	Overview completed by P.Griffith Inventory not released as request by BC Speleological Society.
Wildlife Habitat Inventory	MOELP	Ongoing	-	UWR surveys, Eagle Nest Surveys, Habitat Modeling for Marbled Murrelet, Deer, Elk, Bear
Stream Classification	MOELP	Ongoing	-	Reconnaissance Mapping in SMZs FPC Classification for current development
Operability Classification	WFP	1999	1999	

10.2 Forest Cover Requirements

10.2.1 Forest Cover Objectives - Rationale

The rationale for each forest cover objectives reported in the timber analysis is described below. The rationales are based on the unique attributes of the TFL.

10.2.1.1 Visual Quality Objectives

Visual quality is currently being managed in all *"known"* Scenic Areas having a VQO in the TFL inventory. Visual Quality Objectives to be modelled in the timber supply analysis are Preservation (P), Retention (R), Partial Retention (PR) and Modification (M). The amount of area that can be disturbed at any one-time (i.e. has not achieved visually effective green-up) is 1%, 5%, 15% and 25% respectively. These levels are set at the upper end of the % denudation range for use in timber supply analyses as WFP has been incorporating more visual landscape design (i.e. irregular boundaries) within cutblock layout.

A 5.7 m visually effective green-up (VEG) height is proposed for TFL 6 Block 1, while a 4.1 m VEG height will be used for TFL 6 Block 2. VEG heights were calculated based on slope analysis and followed MOF *Procedures for Managing Visual Resources to Mitigate Impacts on Timber Supply, 1998 Section 5.2.2*. As both Complan and Atlas use volume over age curves for yield tables, an age surrogate will be established to represent VEG height for each analysis unit.

10.2.1.2 Recreation Sites

Significant recreation features have been included within the delineated Recreational Reserve Areas (See Section 6.12) and appropriate reductions to the productive land base have been applied to account for the significance of the resource. Recreation opportunities will also be protected through harvesting constraints designed to maintain visual landscape quality.

10.2.1.3 Winter Range

Ungulate winter ranges have been identified in wildlife habitat inventories and delineated. These areas are deducted from the timber harvesting land base.

10.2.1.4 Adjacent Cutblock Green-up

A 3 metre green-up height is proposed for areas without visual quality objectives. As described in Section 10.2.1.4, an age surrogate for each analysis unit will be used within the model to represent height.

Adjacency is expected to pose some significant challenges within the spatial timber supply modelling. Past spatial analyses completed by WFP, UBC, MOF and various consulting firms in the Interior have shown that timber supply is significantly impacted when modelled using traditional non-spatial timber constraints. WFP will work with

MOF during the timber supply analysis to determine an appropriate methodology to deal with these challenges.

10.2.1.5 Landscape Level Biodiversity

As Biodiversity Emphasis Options assigned to Landscape Units remain in draft form, the current management option will have forest cover constraints imposed based on government policy. According to the policy, approximately 45 percent of each LU will be in the lower BEO, 45 percent in the intermediate BEO and 10 percent in the high BEO. As a result, in the current management option the area-weighted average (i.e. 45/45/10) biodiversity constraints (old seral only) for the three BEOs will be applied for each variant in each draft LU.

Sensitivity analyses will evaluate the impacts of managing for biodiversity as specified by the interim BEO ratings. Modelling of the management of Landscape Units assigned Low, Intermediate and High BEO ratings will be guided by the Biodiversity Guidebook. As indicated to date by government only old seral targets will be modelled during the sensitivity.

10.2.1.6 Reductions to Reflect Volume Retention in Cutblocks

As previously mentioned WTP are mainly in constrained areas such as riparian reserves, unmerchantable stands or unstable slopes. In order to capture WTP in otherwise unconstrained areas a volume reduction will be implemented in the timber supply model. Current management practice is that at least 13% WTP retention is being managed for (MoF Regional directive until Landscape Unit Planning is implemented). Assuming 75% of the WTP retention is in constrained areas (based on the *Forest Practices Code Timber Supply Analysis*) a volume reduction of 3.25% (4% will be used due to model limitation and to be conservative) is used within the timber supply analysis. It is expected that this retention level will also address gully management areas left around non-fish bearing streams.

10.2.1.7 Managing Identified Wildlife

Significant wildlife habitat areas have been excluded from the operable land base as highlighted in Sections 6.11 and 10.1.3. However, no Wildlife Habitat Areas (WHA) have been designated within the TFL. Any future WHA that are designated in the TFL will be managed in accordance to the guidelines specified in the Identified Wildlife Management Strategy. Timber supply implications that may result from the future designations will be addressed in subsequent Management Plans.

10.2.1.8 Community Watersheds

Within the TFL there is one designated, but unused, community watershed and one non-designated watershed from which water is drawn for community use. Due to the small size of these watersheds issues surrounding water quality are dealt with at an operational level. Implementing forest cover constraints in the timber supply model for these areas is not expected to be necessary.

10.2.1.9 Higher Level Plans

Currently, sections of the Vancouver Island Land Use Plan await designation as higher-level plan objectives. WFP is conducting operations within the SMZs to meet the spirit and intent of the draft management objectives for these zones. For modelling purposes, current management constraints such as UWRs, VQOs, and FPC requirements and sensitivity analyses for BEOs will be adequate to address most SMZ objectives, hence no additional forest cover constraints are being modelled specifically for Special Management Zones

10.3 Timber Harvesting

10.3.1 Minimum Harvestable Age

Minimum harvestable ages are simply minimum criteria. While harvesting may occur in stands at the minimum requirements in order to meet forest level objectives (i.e. maintaining overall timber flows) many stands will not be harvested until well past the minimum timber production ages because consideration of other resource values may take precedence.

In the previous analysis, minimum harvest age was selected to be the age closest to the time when average stand DBH reached a threshold diameter. In keeping with this criterion, rotation length will again be set at the age closest to the time when average stand DBHq reaches 42cm for coniferous species on Productivity Group 1 sites, 37 cm average DBHq on Productivity Group 2/3 sites and 30 cm average DBHq on Productivity Group 4 sites. These minimums have been modestly reduced from the 45,40,35 cm criteria used in the previous analysis. For deciduous and deciduous/conifer mixtures a rotation that matched the conifer component of the stand was used.

The original DBHq criteria, which define maturity in terms of a minimum tree size, were selected on the basis of values and cost structures associated with current logging and manufacturing systems. Universally adopting a maximum physical wood production criterion (maximum Mean Annual Increment [MAI]), as is commonly used elsewhere on the coast, was thought unsuitable on operational grounds because of the smaller piece sizes obtained at the indicated rotation length.

10.3.2 Operability

The criteria used to determine operability for use in the timber supply analysis are highlighted in Section 6.7. A *Terms of Reference* document outlining the operability classification process was submitted to Ministry of Forests in August 1998 and contains detailed information regarding the assumptions and criteria used. This document has been included as Appendix III.

10.3.3 Initial Harvest Rate

The initial rate for the timber supply analysis will be set at the currently approved harvest level of 1,476,758 m^3. Rates will be varied to meet the objectives stated in Section 10.3.7. Once a suitable flow is established sensitivity analyses will be performed. Should these analyses suggest an alternative flow pattern is warranted, additional runs will be initiated. All pertinent rate of harvest information will be included in the Timber Supply Analysis report.

10.3.4 Harvest Rules

Since the timber supply model is spatially based, a couple of options are available to implement harvest rules. Like typical timber supply models, harvesting stands on an oldest first basis is available as a harvest rule. However, an additional rule of closest to the log dump can be used. This rule allows the model to harvest in a pattern typical of actual operations. Additional rules can be placed on the model to control the harvest levels by operating area. A number of options are expected to be run to determine which rule set gives the most realistic approximation of current and future harvest scheduling.

10.3.5 Harvest Profile

Harvesting to the inventory profile in TFL 6 has been achieved and will continue. No constraints will be imposed in the model to target certain species or product grades.

10.3.6 Silvicultural Systems

The majority of the TFL is currently harvested using clearcut with reserve or retention harvest methods. There is no significant dispersed partial cutting occurring at this time.

For the purposes of modelling clumped retention, volume reductions as discussed in Section 10.2.1.6 in combination with even-aged growth and yield projections for the remaining harvested area should be adequate, albeit imperfect.

To date the Licensee has focussed management strategies for conservation of biodiversity at the landscape level. Riparian reserves, wildlife tree patches and other exclusions from the timber harvesting land base are examples of areas being managed for conservation. Strategies for stand level retention within the TFL are now being investigated to augment higher-level conservation plans. A committee is active within WFP to explore the use of a variety of silviculture systems to retain more within-stand structure during harvesting.

As pressures to adopt non-traditional cutting methods and uneven-aged silviculture systems mount, growth and yield models need to be developed and calibrated for predicting the long term outcome of partial cutting in coastal old-growth and second-growth stands. As there is little experience on the coast and few, if any, stands to sample for partial cutting response, models will have to deviate significantly from the usual strategy of permanent sample plot analyses. Due to the lack of growth and yield data and predictive tools, the licensee will not attempt to model partial cutting for this timber supply analysis. However the Licensee is, and will be, supportive of any initiatives of the Ministry of Forests to meet the challenge of developing uneven-aged models for the Coastal Western Hemlock Zone.

10.3.7 Harvest Flow Objectives

The objective of the volume-based analysis in the TFL is to maintain current harvest levels for as long as reasonable. As the transition from current harvest levels to the long-term harvest level occurs efforts will be made to restrict the rate of volume change per decade to <10%.

Harvest levels within TFL 6 currently include a partitioned AAC for problem (low sites) forest types (52,000m^3/yr) and helicopter types (10,200 m^3/yr). Since implementation, performance within the problem forest type partition has been exceeded suggesting that a partition component to the AAC is no longer needed. Approximately 40,000 m^3 or 80% of partitioned helicopter wood has been logged over the past five years.

Appendix I

DETAILED AREA AND VOLUME SUMMARIES

AREA AND CURRENT VOLUME BY LEADING SPECIES GROUP, AGE CLASS AND OPERATING AREA

Operation	Leading Species	Age Class 0	1	2	3	4	5	6	7	8	Immature Area	Mature Area	Total Area	Immature Volume	Mature Volume	Total Volume
Holberg	H		6,864.5	8,281.5	4,724.6	3,576.7	3,824.3	67.4	181.3	12,871.8	27,520.2	12,871.8	40,392.0	8,389,564	12,302,607	20,692,171
	B		598.0		2.2	47.9	163.5	32.5	23.2	226.9	867.4	226.9	1,094.3	193,210	242,554	435,764
	C		3,550.7	262.4	5.9	15.5	21.5	4.6	4.5	9,738.3	3,865.0	9,738.3	13,603.3	36,910	9,061,732	9,098,642
	Cy		69.2	9.6			9.0			685.6	87.9	685.6	773.5	5,384	562,171	567,555
	S		516.6	829.8	4.7	9.1				509.6	1,360.3	509.6	1,870.0	171,027	564,057	735,084
	F		85.9	145.9						9.3	231.8	9.3	241.1	19,335	6,547	25,882
	Pl		4.4								4.4	0.0	4.4	216		216
	Dec		3.3	84.5	496.3	231.8	60.9				876.9	0.0	876.9	362,887		362,887
	NSR	849.0									849.0	0.0	849.0	0		0
Holberg Total		*849.0*	*11,692.7*	*9,613.7*	*5,233.7*	*3,881.0*	*4,079.2*	*104.6*	*209.0*	*24,039.3*	*35,662.9*	*24,041.6*	*59,704.6*	*9,178,533*	*22,739,667*	*31,918,201*
Port McNeill	H		4,963.7	7,067.6	5,977.7	2,730.2	4,286.4	171.8	244.9	4,236.7	25,442.3	4,236.7	29,678.9	8,737,662	3,973,463	12,711,125
	B		541.0	134.0	19.4	28.9	203.3	84.0	74.2	274.4	1,084.9	274.4	1,359.3	296,676	286,740	583,416
	C		5,202.2	2,584.7	1,703.4	8.9	7.2	10.7	8.4	4,767.5	9,525.5	4,767.5	14,293.0	401,360	3,851,495	4,252,855
	Cy	0.0	160.1	92.6						587.1	252.7	587.1	839.8	4,731	523,851	528,582
	S		180.8	457.8	6.5	1.7	5.6	0.0	3.1	45.4	655.5	45.4	700.9	111,517	50,691	162,208
	F		581.3	413.4	10.2	206.6	7.0			72.8	1,218.5	72.8	1,291.3	188,244	57,230	245,474
	Pl		17.8	18.5		0.5				0.4	36.9	0.4	37.2	634	186	820
	Dec		5.1	129.3	128.9	68.6	43.3				375.2	0.0	375.2	140,577		140,577
	NSR	885.4									885.4	0.0	885.4	0		0
Port McNeill Total		*885.4*	*11,652.1*	*10,898.0*	*7,846.1*	*3,045.3*	*4,552.8*	*266.5*	*330.5*	*9,984.3*	*39,476.8*	*9,984.3*	*49,461.1*	*9,881,401*	*8,743,656*	*18,625,057*
Jeune Landing	H		4,444.3	1,789.5	1,993.7	2,343.8	1,349.6	51.8	352.8	9,211.2	12,325.5	9,211.2	21,536.7	4,134,681	8,509,006	12,643,687
	B		766.8	184.1		13.2	33.5	18.2	10.5	606.6	1,026.4	606.6	1,633.0	67,143	594,664	661,807
	C		586.2	0.8		8.8	2.7			1,696.3	598.5	1,696.3	2,294.8	5,356	1,592,295	1,597,651
	Cy		31.4							271.8	31.4	271.8	303.3	0	178,887	178,887
	S		410.4	190.0	2.8	1.1	10.4		4.9	64.5	619.6	64.5	684.2	42,307	58,724	101,031
	F		39.3	254.3	95.6	30.1	62.3			154.2	481.5	154.2	635.7	139,025	159,414	298,439
	Pl		2.8								2.8	0.0	2.8	1		1
	Dec		5.7	105.9	464.6	229.1	37.2			1.9	842.5	1.9	844.4	356,499	1,080	357,579
	NSR	607.9									607.9	0.0	607.9	0		0
Jeune Landing Total		*607.9*	*6,287.0*	*2,524.5*	*2,556.6*	*2,626.1*	*1,495.8*	*70.0*	*368.3*	*12,006.5*	*16,536.2*	*12,006.5*	*28,542.7*	*4,745,012*	*11,094,070*	*15,839,082*
Mahatta	H		1,014.5	4,914.8	917.6	776.7	520.0		5.9	1,577.4	8,149.5	1,577.4	9,726.9	1,924,749	1,537,719	3,462,468
	B		131.3	22.3	4.7	7.0				82.0	165.2	82.0	247.2	5,611	91,947	97,558
	C		393.2			34.2	3.7			541.8	431.2	541.8	973.0	14,472	450,962	465,434
	Cy		20.5							66.2	20.5	66.2	86.7	0	45,220	45,220
	S		56.0	388.4			1.5			16.8	445.9	16.8	462.7	91,684	20,653	112,337
	F		0.8	23.8							24.5	0.0	24.5	560		560
	Dec			114.9	141.9	12.2	2.6				271.6	0.0	271.6	116,356		116,356
	NSR	246.1									246.1	0.0	246.1	0		0
Mahatta Total		*246.1*	*1,616.2*	*5,464.1*	*1,064.1*	*830.1*	*527.8*		*5.9*	*2,284.2*	*9,754.4*	*2,284.2*	*12,038.6*	*2,153,432*	*2,146,501*	*4,299,933*
Total		**2,588.3**	**31,248.0**	**28,500.4**	**16,700.6**	**10,382.5**	**10,655.7**	**441.1**	**913.7**	**48,314.3**	**101,430.3**	**48,316.6**	**149,747.0**	**25,958,378**	**44,723,894**	**70,682,272**

Appendix II

SECOND GROWTH SITE INDEX OF TFL 6

Prepared by:
J.S. Thrower and Associates
March 19, 1997

Project: WPV-091-002 (96-08-WPa)

Executive Summary

The primary objective of this study was to develop statistically based estimates of average site index for the major commercial tree species and ecosystems on Western Forest Products Limited Quatsino Tree Farm (TFL 6). This was achieved by sampling four Productivity Groups created by grouping Lewis ecological classification units having similar site potential. The average site index for hemlock was estimated from 61 randomly located plots within these four groups. Field sampling was restricted to hemlock-leading polygons aged 21-140 years to ensure good estimates of site index. Estimates for the average site index for balsam, cedar, and Sitka spruce were derived from equations relating the average site index of hemlock to the averages of these other species. The equations were derived from samples taken on the random plots and thus provide a representative sample of the TFL. An equation developed by the Ministry of Forests was used to estimate the average site index for Douglas-fir.

The average site indices computed in this study differ only slightly from the estimates derived previously by Western Forest Products when compared by Productivity Groups. The overall average site index for the TFL is virtually identical to the previous estimate. The results present a matrix of average site indices recommended for use in timber supply analyses for TFL 6.

Table of Contents

1.0 Introduction

1.1 Background

Western Forest Products (WFP) was one of the first companies in BC to use ecosystem classification for estimating site index in timber supply analyses. This was done in the last Management Plans for TFL 6 on northern Vancouver Island, and TFL 24 on the Queen Charlotte Islands. The average site indices in these projects were computed from data collected primarily from subjectively located permanent sample plots. This project aims to fulfill WFP's commitment to the Ministry of Forests (MOF) to update and refine the ecosystem-based estimates of site index using statistically-based sampling for the TFL 6 Management Plan 9 due in December 1999.

1.2 Objectives

The primary objective of this project was to:

Develop statistically-based estimates of average site index for the major commercial tree species and ecosystems on TFL 6 for use in timber supply analyses.

A secondary objective was to collect basal area information from a cluster of point-sample plots around randomly located site index plots to supplement forest inventory information.

This report was prepared for Dr. John Barker, *R.P.F.* at WFP to describe and document the methods and procedures used to develop the site index estimates.

2.0 TFL 6 Landbase

2.1 Geographic Location and TFL Area

The WFP Quatsino Tree Farm (TFL 6) is located on northern Vancouver Island. The TFL encompasses the area from Port Hardy on the east coast, to south of Cape Scott Park on the west coast, and south to Quatsino Sound, Neroutsos Inlet, and Victoria Lake. The total area of the TFL is 170,194 ha; over 93% of which is productive forest land (Table 36).

Table 36. Landbase summary for TFL 6.

Land Type	Area (ha)	
Non-Productive Land	11,483	6.7 %
Productive Land	158,711	93.3 %
NSR Polygons	4,448	2.6 %
Roads	2,164	1.3 %
Forested Polygons	152,099	89.4 %
Total TFL 6 Area	170,194	100 %

2.2 Forest Cover

About two-thirds of the 152,099 ha of forested area in the TFL is leading western hemlock (Hw) (102,664 ha) (Figure 2). Red cedar is the second largest component (18.8%) followed by yellow cedar (5.4%), Sitka spruce (2.8%), and balsam (2.3%). Nearly one-half of the forested area in the TFL is age class 8 (141 yrs+) (Figure 3, Appendix II.1) (WFP Age Class 8 includes MOF Classes 8 and 9). Other major components are age class 1 (1-20 yrs - 19%) and age class 2 (21-40 yrs - 15%).

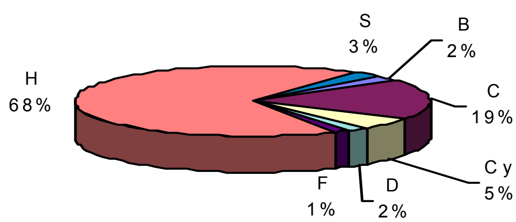


Figure 2. Species distribution of TFL 6.

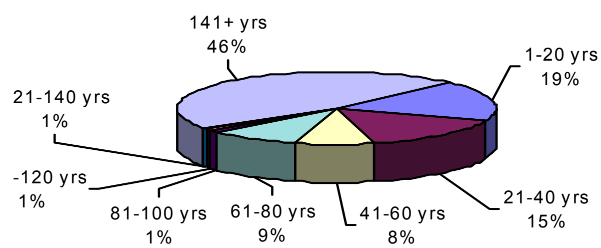


Figure 3. Age distribution of forested land in TFL 6.

2.3 Ecological Classification

2.3.1 Lewis Ecosystem Units

An ecological inventory and terrain classification was completed on the TFL in 1985 by Dr. Terry Lewis.[6] These eco-units are not named the same as BEC units; however, there is a very close relation between the two systems and for most units, boundaries are identical.[7] The Lewis system was designed for TFL 6 and thus recognizes local ecosystems not included in BEC.

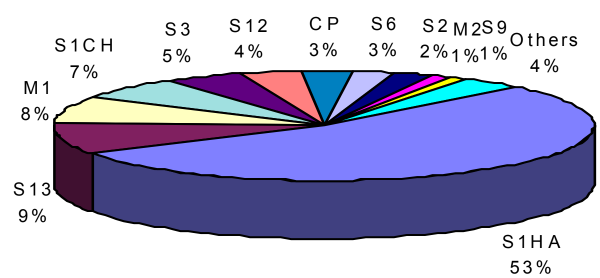


Figure 4. Distribution of Lewis ecosystem units on TFL 6

Approximately one-half of the forested area in the TFL is the S1HA ecosystem (Figure 4, Appendix II.2). This equates to the zonal CWHvm1/01 (Appendix III.3). Other major ecosystems are the S13 (composite of CWHvm1/05 and 07), the M1 (CWHvm2/01), and the S1CH (salal phase of CWHvm1/01).

2.3.2 BEC Conversion

The Lewis eco-units were converted to BEC units by Bob Green, *R.P.F.* of B.A. Blackwell and Associates Ltd. Figure 5, Appendix II.3). This was not necessary to complete this

[6] Lewis, T. 1985. Ecosystems of Quatsino Tree-Farm Licence (TFL 6). Contract Rep. to WFP. 77 pp.
[7] Personal communication with Bob Green, R.P.F., March 1996.

project; however, it provided a common link to other site studies allowing us to refine our Productivity Groups. This conversion was based on Mr. Green's extensive knowledge of the area, familiarity with Dr. Lewis' classification system, and documentation of the classification on the TFL including detailed descriptions of the ecosystems.

The Lewis classification system identifies some sites unique to TFL 6. These special sites are mostly related to non-forested and wetland ecosystems and did not affect the sampling plan for this project. The S13 eco-unit is a composite of the CWHvm1/05 and 07; these two BEC site series are very close in productivity and would be grouped into the same sampling unit for any forest-level sampling program.

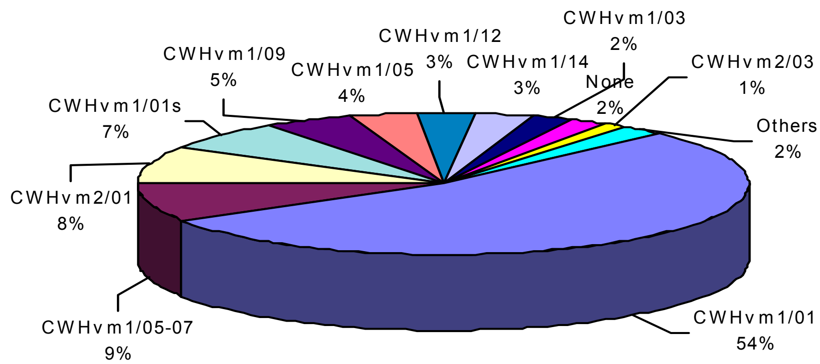


Figure 5. Distribution of BEC site series on TFL 6 converted from Lewis ecosystem map-units.

3.0 Sampling Plan

3.1 Sampling Approach

The primary objective of this project was to develop estimates of average site index for the major commercially important species on the TFL, including Hw, Ba, Cw, Ss, and Fdc. However, lack of sufficient area of species other than Hw across the various ecosystems on the TFL precluded observing site index for these species in the random plots. Therefore, the objective of developing estimates of average site index for all species on all major ecosystems on the TFL was achieved in a two-phase process:

- Random Sampling for Hemlock Site Index – where statistically-based estimates of average site index were developed for Hw from plots randomly located across the TFL in stands aged 21-140 years.

- Sampling for Site Index Conversion Equations – where pairs of site index estimates were measured from the random plots and nearby areas to develop equations to predict the average site index of Ba, Cw, and Ss from the average site index of Hw from the random samples.

3.2 Productivity Groups

3.2.1 Overview

The field sampling and analyses focused on four Productivity Groups (PGs). These groups provided an ecologically-based stratification of the TFL aggregating areas of similar productivity. This provided the stratification necessary to reduce within-group variation in site index to increase sampling efficiency. This stratification also provides a link to current management and planning systems on the TFL based on the Lewis ecosystem classification. These PGs formed the basis for sampling and will form the basis for applying the results.

3.2.2 Original Sampling Plan

The previous estimates of average site index for the TFL were developed by Dr. John Barker using Lewis ecosystem units to create seven PGs (Table 37). These Barker PGs were used in the previous Management Plan 8 for yield prediction. The original sampling plan[8] for this project combined six of the Barker PGs into three PGs for our project. This original plan aimed to reduce sampling costs by reducing the number of sampling strata. However, the original plan was changed to add another PG to recognize the higher productivity of the S3 eco-unit. This was achieved by allocating plots proportionally to the two PGs and confirming the difference in interim analysis.

Table 37. The four Productivity Groups (PGs) related to Barker/Lewis groups and BEC ecological units.

	Barker PG	Lewis Unit	BEC Site Series	ha in TFL	% of TFL	% of PG
PG 1	EE	S3	CWHvm1/09	7,407	5.0	100
Subtotal				*7,407*	*5.0*	
PG 2	EG+	S4	CWHvm1/10	113	0.1	0.8
		S5	CWHvm1/14	693	0.5	4.7
		S13	CWHvm1/05+07	13,862	9.3	94.5
Subtotal				*14,668*	*9.9*	*100*
PG 3	EG	S1-ha	CWHvm1/01	81,730	54.8	80.8
		S12	CWHvm1/05	5,651	3.8	5.6
	EG-	M1	CWHvm2/01	12,561	8.4	12.4
		M3	CWHvm2/05	106	0.1	0.1
		M5	CWHvm2/08	1,127	0.8	1.1
Subtotal				*130,511*	*67.9*	*100*
PG 4	EM	S1-ch	CWHvm1/01s	9,838	6.6	48.4
	EL	S2	CWHvm1/03	4,364	2.9	21.5
		S6	CWHvm1/14	3,913	2.6	19.3
		M2	CWHvm2/03	1,878	1.2	9.2
		M4	CWHvm2/11	332	0.3	1.6
Subtotal				*281,347*	*13.6*	*100*
Total				143,485	94.3	
PG 5	EVL	S7	CWHvm1/13	61	0.0	1.1
		CP	CWHvm1/12	4,622	3.1	83.6
		MH	MH/01	845	0.6	15.3
Subtotal (not sampled)				*711,707*	*3.7*	*100*

[8] J.S. Thrower and Associates Ltd. 1996. Sampling Plan: Second Growth Site Index Inventory in Western Forest Products TFL 6. Confidential Contract Report to Dr. John Barker, R.P.F. and Western Forest Products Ltd. May 21, 1996. 26 pp.

3.2.3 Final Productivity Groups

The four PGs used in this project are aggregations of the existing Barker groups (Table 2). Three of the former seven Barker PGs were retained in the original form. The lowest productivity group (PG 5) represented only 3.7% of the landbase and thus was not sampled. The other two PGs in this project were combinations of two of the original Barker groups. About 3,000 ha (2.0%) of the forested polygons (avalanche, wetlands, bog, and park areas) were not assigned to a Barker PG and were not included in our PGs before creating the additional group.

3.3 Sample Size

The target number of plots detailed in the sampling plan was achieved in the field sampling (Table 38). The small sample size in PG 1 (8 plots) was the result of splitting the original highest productivity PG in the preliminary sample plan into two separate groups as noted above. The ecosystems in this PG were very homogeneous and thus a small standard error was achieved even with the small sample size.

Table 38. Preliminary target and actual sample size for the randomly located site index plots.

Prod. Group	Lewis Eco-Unit	Target Initial Sample Size	Actual Sample Size
1	S3	7	8
2	S4, S5, S13	13	13
3	S1HA, S12, M1, M3, M5	25	25
4	S1CH, S2, S6, M2, M4	15	15
5	S7, CP, MH	0	0
Total		60	61

3.4 Random Sampling for Hemlock Site Index

3.4.1 Target and Sample Populations

The target population for application of the results of this project is the total land area within each PG. The sample population is the land area in each PG suitable for sampling. The sample population is always smaller than the target population in forest-level site productivity sampling. This is because stands of all species suitable for estimating site index do not occur everywhere across the landscape. The sample population was defined as polygons with leading Hw forest cover aged 21-140 years (age class 2-7) (Table 39).

Table 39. Area of target population, sample population, and sampling intensity.

Prod. Group	Plots	Target Population			Sample Population		
		Area (ha)		Sample Intensity (ha/plot)	Area (ha)		Sample Intensity (ha/plot)
1	8	7,407	5.0 %	926	2,155	29.1 %	269
2	13	14,668	9.9 %	1,128	2,052	14.0 %	158
3	25	101,175	67.9 %	4,047	37,280	36.8 %	1,491
4	15	20,325	13.6 %	1,355	3,873	19.1 %	258
5	0	5,528	3.7 %	NA	44	0.8 %	NA
Total	*61*	*149,103*	*100 %*		*45,404*	*100 %*	

3.4.2 Statistical Inference

The sampling plan was designed to select random points on the landbase to measure Hw site index within each of the four PGs. The only restriction to the randomization and application of statistical theory in this design is that Hw stands suitable for estimating site index do not occur on all areas in the TFL. Theoretically, this imposes a bias in the sampling design. However, most of the potential for bias from this restriction is removed through the ecological classification system. The potential effect of this randomization restriction is addressed through the assumption that the productive potential of the land within a given PG is the same for areas that could be sampled as for those that could not be sampled. This is a reasonable assumption and should not be of major concern in the application of these estimates.

3.4.3 Polygon and Plot Selection

Random sample points within each PG were selected in a two-phase process. First, polygons were selected randomly with probability proportional to area (with replacement) from a list of all polygons in each PG. Points were then randomly located on maps within the selected polygons and transferred to air photos for field sampling.

3.5 Sampling for Site Index Conversion Equations

Equations to predict the site index of Ba, Cw, and Ss from the site index of Hw were developed from field measurements. The intent was to develop these Hw-Ba, Hw-Cw, and Hw-Ss relationships from trees that adequately represent the range of conditions on the TFL. To achieve this representative sample of the land base, site index measurements were taken from three sources:

1. Trees measured in the randomly located Hw sample plots.
2. Trees measured in the basal area plots systematically located around each random plot.
3. Trees measured from areas adjacent to the random plots.

3.6 Basal Area Sampling

A cluster of five point-sample plots was sampled in conjunction with each randomly located site index plot (Figure 6). The cluster contained a central plot located at the random site index plot center, and four satellite plots located at 50 m in the cardinal directions. The objective was to obtain additional information about the basal area and volume of these randomly sampled second-growth stands to supplement WFP's existing forest cover inventory and to aid in the development of growth and yield models.

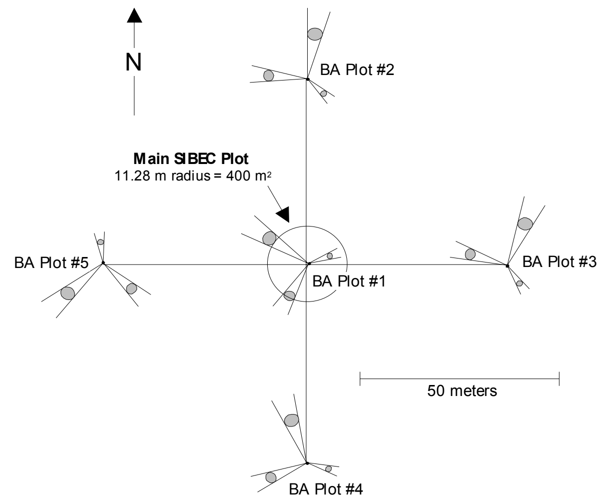


Figure 6. Sample plot cluster showing main SIBEC plot and the five basal area plots.

4.0 Field Methods

4.1 Plot Establishment

4.1.1 Plot Location and Suitability Assessment

The random site index plots were established as closely as possible to the point indicated on the photos and maps by compassing and chaining from the nearest photo tie point. The first step upon reaching a plot was to ensure the plot was in the target forest-cover and ecosystem polygon. The second step was to determine if the stand condition was suitable for estimating site index. When stand conditions were not suitable for estimating site index, plots were systematically offset in a pre-determined 25 m grid pattern. Suitable plots contained at least three healthy, dominant or co-dominant, unsuppressed Hw site trees. Approximately one-third of the plots (21 of 61) were offset because the original location did not contain suitable site trees.

4.1.2 Tie Points and Monuments

Tie points to sample plots were located using metal tags indicating the plot number, date, crew initials, bearing, and distance to the plot. A double band of blue and yellow ribbon was located above and below each metal tag. Compass lines to the plots were flagged using blue ribbon. A bearing tree was located close to each plot center identified in the same manner as the tie points, including bearing and distance to plot center written on a metal tag. Plot centers were marked with a 24" wooden survey stake. Site trees on the plot were marked with orange flagging tape. Trees measured for site index pairs were marked with yellow flagging tape. Original maps and air photos showing plot locations were submitted to Dr. John Barker at WFP head office.

4.2 Random Plots For Hemlock Site Index

Three types of information were taken on each of the randomly located sample plots: site tree, mensurational, and ecological.

4.2.1 Site Tree Measurements

The Hw site trees were selected as the largest diameter, dominant or co-dominant, healthy, and unsuppressed trees within the 400 m^2 sample plots. Four Hw site trees were measured where possible, however, three trees were considered an adequate sample if one tree was not suitable for estimating site index. Total heights were measured and ages at breast height determined from increment cores (all samples included the 1996 growing season). Site index for the Hw site trees were computed in the field and later confirmed with computer analyses using Wiley's site index equation.[9] The bearing and distance from plot center to each site tree was recorded to facilitate possible future relocation of the samples.

Site index was estimated using an age correction to account for measurements taken during the growing season. The growing season for height was assumed to occur between May 15 and July 15 (62 days). The age of a tree for estimating site index was thus the ring count minus the portion of the growing season for height missing when the tree was measured.

4.2.2 Stand Mensurational Data

Measurements were taken from all trees in the sample plots to describe the general stand conditions from which site indices were estimated. These measurements included quadrant (starting north and progressing clockwise), diameter, species, and live/dead code for all trees. In dense stands where it was estimated that plots contained more than 50 trees and the stand was relatively uniform, randomly selected half plots were measured.

4.2.3 Ecological Data

Each 400 m^2 sample plot was classified to the BEC site series level. Classification was based on detailed descriptions of the vegetation, soil, and environmental conditions.

[9] Wiley, K.N. 1978. Site index tables for western hemlock in the Pacific Northwest. Weyerhaeuser Company, For. Res. Center For. Paper. 28 pp.

4.3 Site Index Conversion Measurements

Site index pairs of Hw with the other three target species were taken using standard measurement and site tree selection procedures. To reduce site variation between pairs, the sample trees were located within 10 m of each other, and on the same BEC site series. Site indices for all trees were computed in the field and later confirmed with computer analyses. Site indices were estimated using the curves and equations for coastal species presented by Thrower and Nussbaum.[10]

4.4 Basal Area Measurements

The basal area factor (BAF) for each plot cluster was selected at the main plot and used in all plots in the cluster (Figure 6). Randomly selected half sweeps were used when an initial sweep indicated the plot tally would exceeded 12 trees. Measurement of the selected trees included diameter, species, and the live/dead code. Three additional Hw trees were measured from across the range of diameters in each cluster to provide data to construct height-diameter equations for volume estimation.

5.0 Results and Discussion

5.1 Hemlock Site Index

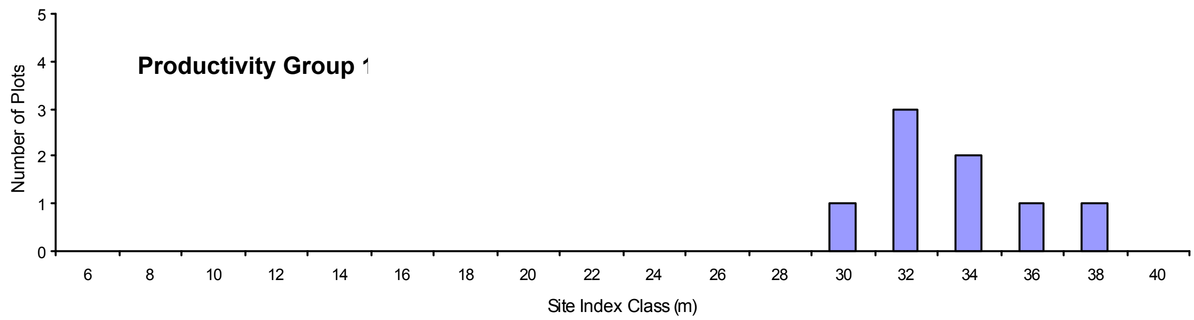
5.1.1 Average Site Index and Ecological Trends

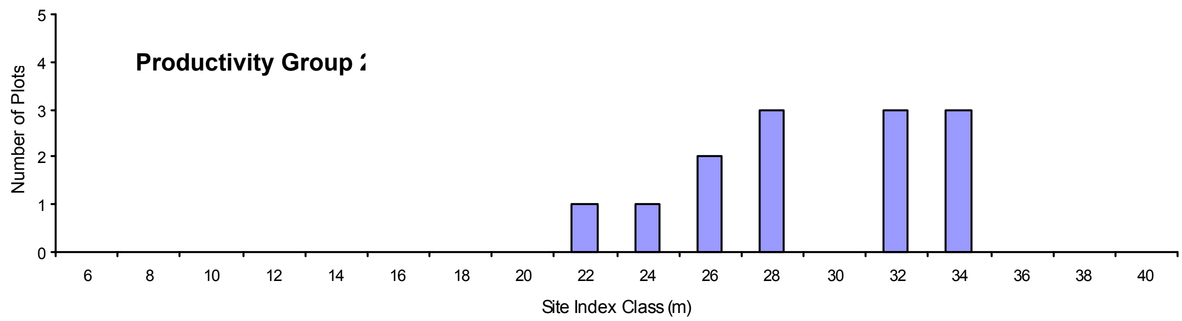
The results showed the expected trend where the average Hw site indices increased from PG 4 to PG 1 (Table 40, Figure 7, Appendix II.4). Productivity Group 1 showed the highest site indices and the lowest standard error. This was expected as the alluvial ecosystems in this PG are relatively homogeneous and are among the highest in productivity on northern Vancouver Island. Productivity Group 2 also showed a relatively narrow range of observed site index compared to PGs 3 and 4 (Figure 7). The wide range of site indices in these two broad groups reflects the wide ecological amplitude of Hw and also the variation inherent in these PGs. Further work could be done to narrow the range of productivity in these groups by more closely examining ecological conditions and trends for use in future studies.

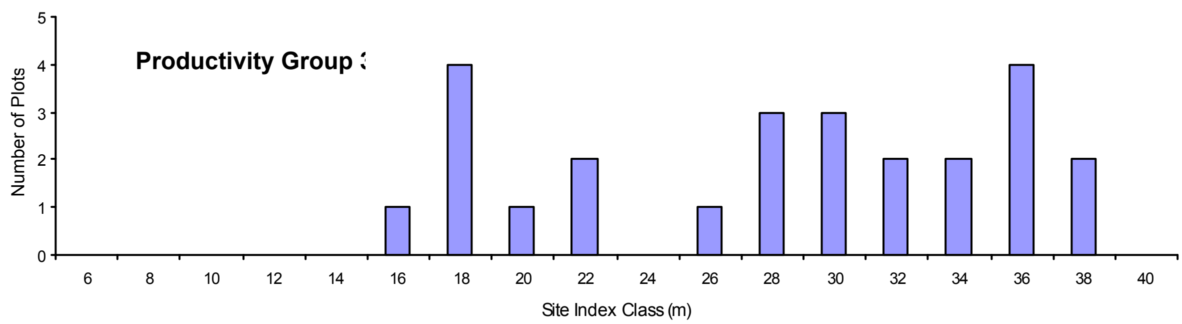
Table 40. Site index statistics (using corrected ages) by Productivity Group.

Prod. Group	No. Plots	Avg. SI (m)	Min SI (m)	Max SI (m)	SD SI (m)	SE SI (m)
1	8	33.7	30.9	38.5	2.73	0.96
2	13	29.3	22.0	34.5	4.15	1.15
3	25	27.9	16.3	37.2	7.12	1.42
4	15	20.1	6.0	30.6	8.22	2.12

[10] Thrower, J.S. and Nussbaum, A.F. 1991. Site index curves and tables for British Columbia. BC Min. For. Res. Br. Lnd. Mgmt. Hdbk. Field Guide Insert 3.







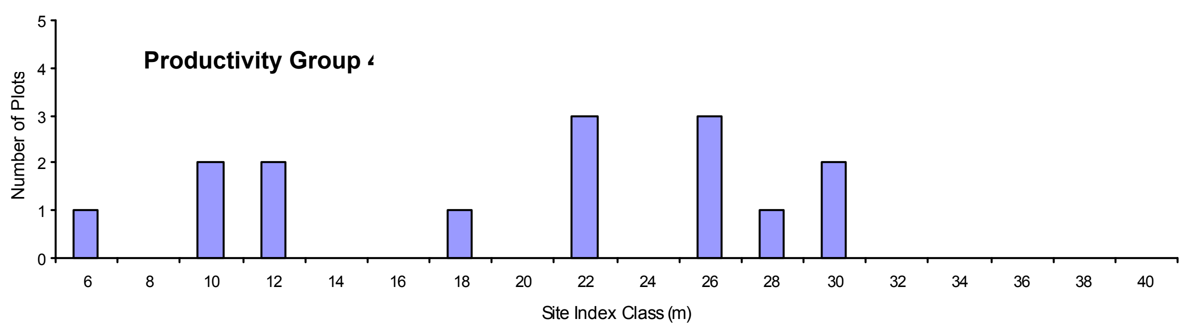


Figure 7. Distribution of sample plots by site index and Productivity Group.

5.1.2 Combining Productivity Groups 2 and 3

Considerable variation in site index was observed in the PGs, and increased from PG 1 through PG 4. This wide range of site index within relatively homogeneous ecological units is a characteristic of Hw not shared by other coastal species such as Douglas-fir. This suggests that some of the PGs could be combined. For example, PG 2 and PG 3 differ in site index by only 1.4 m. This difference is probably not significant, and combining the two PGs would simplify timber supply analysis procedures. The combined PG represents about 78% of the TFL. Maintaining these two groups as separate units probably will not provide significant advantages given the small difference in average site index. We thus recommend combining these two groups.

5.1.3 Western Hemlock Age Correction

The site index of each plot was calculated with and without the age correction accounting for the measurement of trees during the period of active height growth. Overestimating age by half a growing season on average resulted in site index underestimation of 0.4 m on average. As the final results are rounded to the nearest meter of site index, this additional adjustment resulted in the rounding up of site index for only Hw in PG 1.

5.2 Site Index Conversion Equations

A linear equation was used to express the relationship between the site index of Hw and the other target species:

$$SI = B_0 + B_1 SI_{Hw}$$

The intercept and slope were estimated using the least-squares technique.

The relationship between the site index of the Hw and neighboring Ba and Cw site trees was relatively strong (Table 41, Figure 8, Figure 9), and the relationship with Ss was somewhat weaker (Table 41, Figure 10).

Table 41. Regression statistics for the site index conversion equations.

Equation	No. Site Index Pairs	Equation Intercept (m)	Equation Slope	R^2	Standard Error (m)
Hw-Ba	49	6.8	0.691	77.2 %	2.58
Hw-Cw	29	7.7	0.572	70.4 %	3.48
Hw-Ss	28	9.8	0.756	44.8 %	3.47

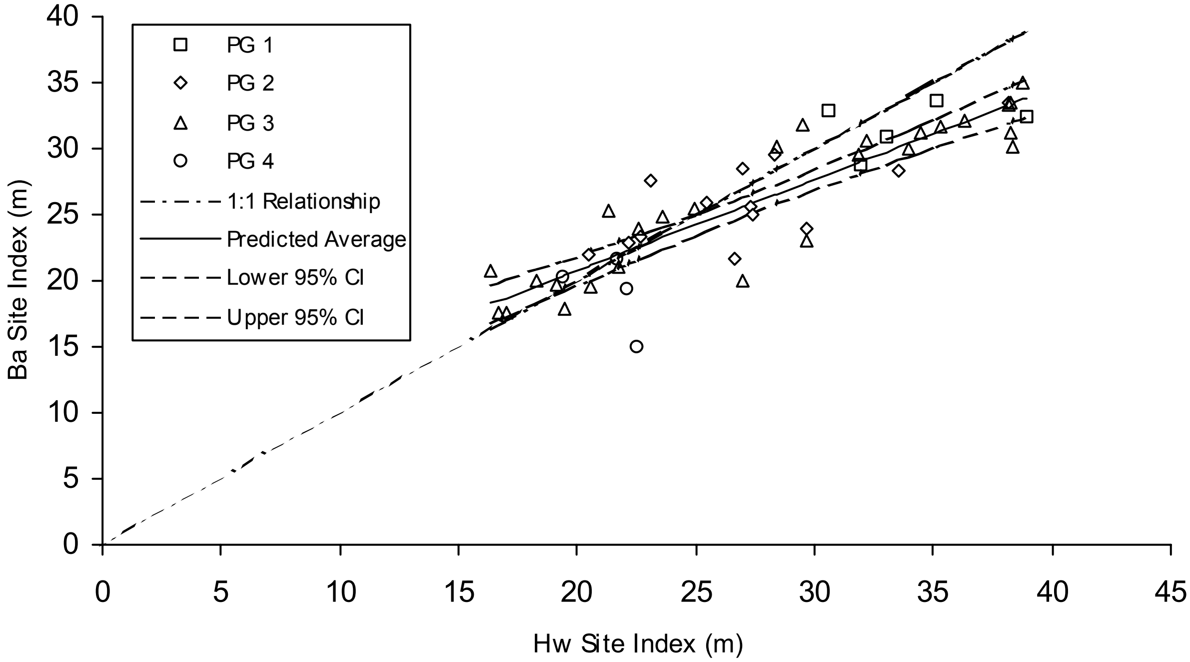


Figure 8. Relationship between Hw and Ba site index.

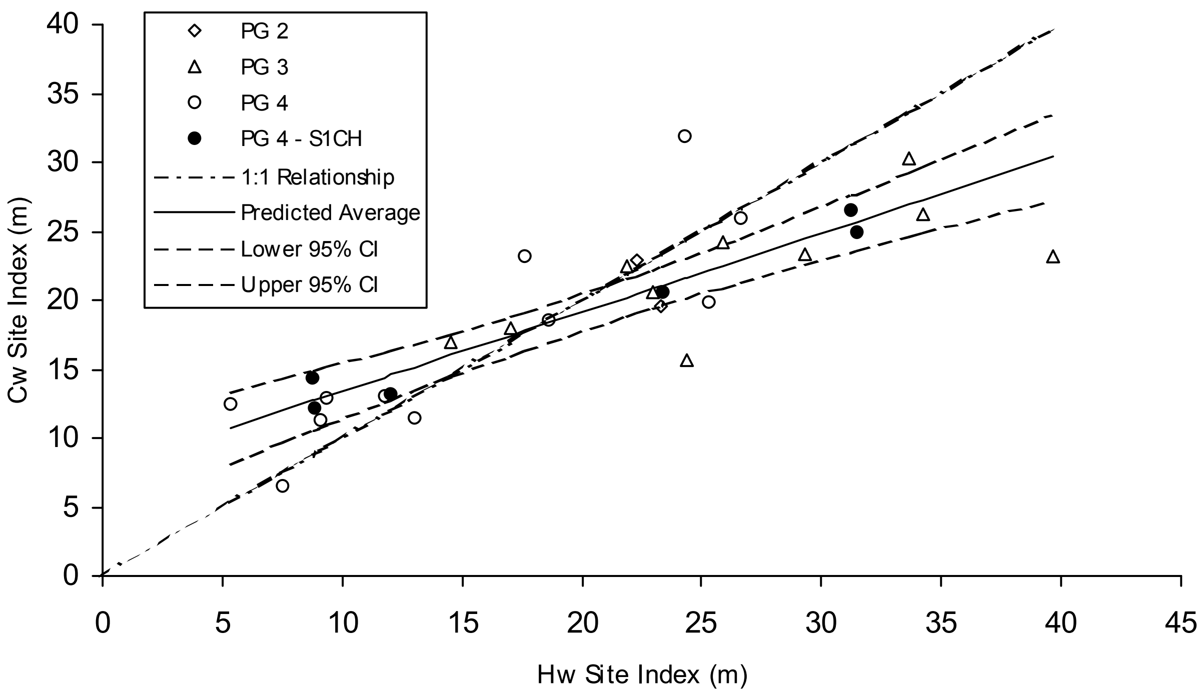


Figure 9. Relationship between Hw and Cw site index.

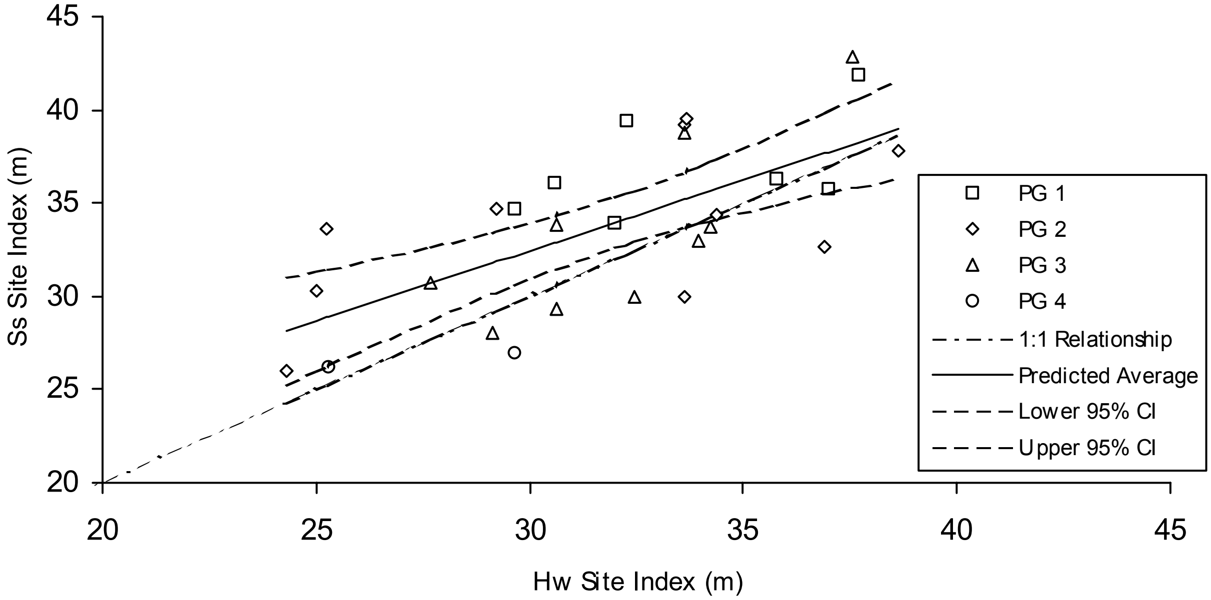


Figure 10. Relationship between Hw and Ss site index.

5.3 Site Index Estimates for All Target Species

The average site indices for Ba, Cw, and Ss were estimated with the conversion equations developed in this project (Table 41) and for Fdc using the equations developed by the MOF.[11] Together with the estimated averages from the random Hw plots, this provides estimates for the four major commercial species and ecosystems on the TFL (Table 42). These estimates apply to over 96% of the forest landbase.

Table 42. Average site index (m) for the five target species by Productivity Group.

Prod. Group	Hw Random Sampling Estimate	Ba TFL 6 Conversion Equation	Cw TFL 6 Conversion Equation	Ss TFL 6 Conversion Equation	Fdc Nigh's Conversion Equation
1	33.7	30.1	27.0	35.2	37.5
2	29.3	27.1	24.5	31.9	32.7
3	27.9	26.1	23.7	30.9	31.2
2+3	28.0	26.2	23.8	31.0	31.4
4	20.1	20.7	19.2	25.0	22.6

[11] Site Index (Fdc) = 0.48 + 1.11 Site Index (Hw). Nigh, G.D. 1995. Site index conversion equations for mixed species stands. BC Min. For. Res. Rep. No. 1. 20 pp.

5.4 Comparison with Previous Estimates

The estimated average Hw site indices in this study are very similar to the previous estimates derived by Dr. Barker (Table 43). The largest proportion of the landbase is in PG3 (68%) where the new estimated average Hw site index is higher than the previous estimate by only 0.4 m. The largest difference in Hw site index is in PG2 (10% of the landbase) where the new estimate is lower by 1.7 m. The average site index weighted by the proportion of area in each PG is only 0.3 m higher than the previous estimates (Table 44).

Table 43. Comparison of Hw site indices.

Prod. Group	Barker Group	Area in TFL (ha)	Prop. of TFL (%)	Barker Site Index (m)	New Site Index (m)	Diff. from Barker (m)
1	EE	7,405	5.0 %	33.0	33.7	+ 0.7
2	EG+	14,668	9.9 %	31.0	29.3	- 1.7
3	EG	87,381		28.0		
	EG-	13,795		24.0		
	Total	*101,175*	*67.9 %*	*27.5*	*27.9*	*+ 0.4*
4	EM	9,838		20.0		
	EL	10,485		18.0		
	Total	*20,323*	*13.6 %*	*19.0*	*20.1*	*+ 1.1*

The average site index for Ba, Cw, Ss, and Fdc predicted by the conversion equations differed from the previous estimates more than Hw. Overall, the new estimates are about 1.3 m lower for Ba and about 0.4 m lower for Cw. The new estimates for Fdc and Ss were higher overall. However, the averages shown in Table 44 are weighted by the proportion of area in each PG. Thus, the overall averages are not meaningful for Fdc and Ss, which do not occur on all the PGs.

Table 44. Comparison of new estimates of average site index (m) with estimates from Management Plan 8.

	Hw Random Sampling			Ba TFL 6 Conversion Equations			Cw TFL 6 Conversion Equations			Ss TFL 6 Conversion Equations			Fdc Nigh Conversion Equations		
Prod. Group	New SI	MP 8 SI	Change	New SI	MP 8 SI	Change	New SI	MP 8 SI	Change	New SI	MP 8 SI	Change	New SI	MP 8 SI	Change
1	33.7	33.0	+0.7	30.1	33.0	-2.9	27.0	30.0	-0.2	35.2	36.0	-0.8	37.5	36.0	+1.5
2	29.3	31.0	-1.7	27.1	31.0	-3.9	24.5	27.0	-0.2	31.9	33.0	-1.1	32.7	33.0	-0.3
3	27.9	27.5	+0.4	26.1	27.5	-1.4	23.7	24.5	-0.5	30.9	29.3	+1.6	31.2	29.3	+1.9
2+3	28.0	27.9	+0.1	26.2			23.8			31.0			31.4		
4	20.1	19.0	+1.1	20.7	19.0	1.7	19.2	15.0	+0.6	25.0	18.5	+6.5	22.6	18.5	+4.1
Wtd Avg	26.3	26.0	+0.3	24.7	26.0	-1.3	22.5	22.8	-0.4	29.3	27.5	+1.8	29.4	27.5	+1.9

5.5 Basal Area

Average basal area in PG 2 was significantly higher than the other three PGs (Table 45). However, because stand density is generally lower on higher sites, the quadratic mean diameter was larger in PG 1. These results show excellent stocking in all PGs, which was also noted during fieldwork. Statistics for the 61 plots are given in II.5 and Appendix I.6.

The height-diameter relationships for Hw and Ba were very strong (Table 46, Appendix II 7). Individual tree volume was predicted using the MOF logarithmic volume equations[12] using diameter at breast height and the predicted heights. For minor species, the Hw equation was used to estimate volume. The volume/basal area ratio (VBAR) was about 8 m^3/m^2 for PGs 1, 2, and 3, and 5.5 m^3/m^2 for PG 4. This reflects the generally shorter trees in the lower productivity PG 4.

Table 45. Average inventory attributes from the random sample plots.

Prod. Group	Average Top Ht (m)	Average BH Age (yrs)	Average Site Index (m)	Average Density (no/ha)	Average Basal Area (m^2/ha)	Average DBHq (cm)	Total Gross Volume (m^3/ha)	Average VBAR (m^3/m^2)
1	21.6	27.6	33.7	2,431	46.6	15.6	374.3	8.0
2	22.5	37.0	29.3	4,363	58.2	13.0	490.5	8.4
3	21.0	35.8	27.9	4,383	47.4	11.7	373.9	7.9
4	12.3	28.0	20.1	3,498	27.1	9.9	150.8	5.6

Table 46. Height-diameter equations.

Spp	No. Trees	HT-DBH Equation	R^2	SEE (m)
Ba	60	$Ht = 1.3 + 83.347(1 - e^{-0.004516DBH^{1.19949}})$	90.9%	2.58
Cw	44	$Ht = 1.3 + 25.045(1 - e^{-0.036DBH^{0.915}})$	80.1%	1.86
Hw	371	$Ht = 1.3 + 95.045(1 - e^{-0.005548DBH^{1.121847}})$	85.5%	3.54
Ss	38	$Ht = 1.3 + 0.6543DBH - 0.00179DBH^2$	68.0%	4.42

[12] Ministry of Forests. 1976. Whole Stem Cubic Metre Volume Equations.

6.0 Recommendations

This study provides statistically-based estimates of average site index for the major commercial tree species and ecosystems on TFL 6. We suggest the following recommendations for application of these estimates:

- Productivity Groups 2 and 3 should be combined for estimating average site index for timber supply analyses (Table 47). The increased simplicity in analyses from the reduced number of PGs is more significant than the very small difference in the average site index between these two groups (Table 42).

Table 47. Site indices recommended for timber supply analyses for TFL 6.

	Hw	Ba	Cw	Ss	Fdc
Prod. Group	Random Sampling (m)	TFL 6 Conversion Equation (m)	TFL 6 Conversion Equation (m)	TFL 6 Conversion Equation (m)	Nigh's Conversion Equation (m)
1	34	30	27	35	38
2+3	28	26	24	31	31
4	20	21	19	25	23

- These average site indices are applicable for growth and yield estimation in timber supply analyses. The estimates were developed independently for each PGs and should be applied to the landbase in the same manner.

- These PGs should be saved as a separate layer in the GIS. This will provide a framework for application of the estimates that is consistent with the sampling used in their derivation. This is needed in the event that future modifications or adjustments are made to map boundaries for the ecological classification.

- These estimates are not applicable for predicting the growth and yield of old-growth stands. The appropriate method of yield prediction should reflect actual growth in these stands, and be linked to the stand-level inventory attributes.

- These site indices should be applied to existing post-harvest regenerated stands where the current inventory site index was derived from a pre-harvest (old-growth) stand.

- The site indices for timber supply for areas not sampled in this project should be assigned the values used in the previous analyses for Management Plan 8.

Appendix II.1 - TFL 6 Area by species and age class

Table 48. Area (ha) by species and WFP age class.

	Age Class									
Species	1	2	3	4	5	6	7	8	Total	% Total
A	1	0	0	0	0	0	0	0	1	0.0 %
Ac	4	0	0	0	0	0	0	0	4	0.0 %
B	1,201	147	105	175	88	80	26	1,619	3,442	2.3 %
Bg	3	0	0	0	0	0	0	0	3	0.0 %
Bp	1	0	0	0	0	0	0	0	1	0.0 %
C	5,232	230	69	44	9	14	9	23,000	28,608	18.8 %
Cy	201	1	0	9	0	0	0	8,001	8,212	5.4 %
D	96	901	1,155	476	106	0	1	0	2,735	1.8 %
Dc	0	0	0	0	0	0	0	0	0	0.0 %
Ds	1	0	0	0	0	0	0	0	1	0.0 %
F	891	598	300	67	2	0	68	225	2,151	1.4 %
H	18,969	19,458	10,424	13,139	1,465	928	765	37,516	102,664	67.5 %
HS	2	0	0	0	0	0	0	0	2	0.0 %
Pl	9	0	2	0	0	0	0	0	12	0.0 %
S	1,702	1,610	25	10	10	7	5	890	4,261	2.8 %
Total	28,319	22,945	12,080	13,920	1,680	1,030	875	71,251	152,099	100.0 %
% Total	18.6 %	15.1 %	7.9 %	9.2 %	1.1 %	0.7 %	0.6 %	46.8 %	100.0 %	

Appendix II.2 - TFL 6 Ecological Classification

Table 49. Area (ha) by species and Lewis Ecosystem unit.

Spp	A	Ac	B	Bg	Bp	C	Cy	D	Dc	Ds	F	H	HS	Pl	S	Total	% Total
A						4	29	2				88				123	0.1 %
CP						2,951	1,581					90				4,622	3.0 %
M1			1,194		0	1,407	1,099	3			0	8,844			14	12,561	8.3 %
M2			37			471	712					657				1,878	1.2 %
M3			7			4	3					93				106	0.1 %
M4			3			74	144					111				332	0.2 %
M5			18			278	136	3				686			6	1,127	0.7 %
M6						1										1	0.0 %
MH			57			115	205					468				845	0.6 %
NV								1				8			7	17	0.0 %
P			18			125	501					510				1,154	0.8 %
PM							1									1	0.0 %
S11			5			31	3	10			3	49			13	114	0.1 %
S12			23			417	13	162			539	4,491			6	5,651	3.7 %
S13			318			1,790	217	212			100	10,982		0	243	13,862	9.1 %
S1CH			47			5,187	646	34			68	3,629		4	224	9,838	6.5 %
S1HA		1	1,574		1	10,804	897	1,134	0	1	1,301	64,626		4	1,386	81,730	53.7 %
S2			20			1,371	282	2			11	1,733			57	3,477	2.3 %
S2F			2			185	3	17			86	574			13	880	0.6 %
S2P						1					1	2		3		7	0.0 %
S3		2	102	3		564	11	1,035			18	3,702	2	1	1,776	7,216	4.7 %
S3B			2			6		10				58			113	189	0.1 %
S4			1			1	1	29				63			18	113	0.1 %
S5	1	1	7			193		74			1	181			235	693	0.5 %
S6			7			2,381	360	6			24	988			147	3,913	2.6 %
S7						33	26					2				61	0.0 %
S9						215	1,342					19				1,575	1.0 %
W						0		3				6			2	11	0.0 %
None			0			0	0					2			0	3	0.0 %
Total	1	4	3,442	3	1	28,608	8,212	2,735	0	1	2,151	102,664	2	12	4,261	152,099	100.0 %
% Total	0.0 %	0.0 %	2.3 %	0.0 %	0.0 %	18.8 %	5.4 %	1.8 %	0.0 %	0.0 %	1.4 %	67.5 %	0.0 %	0.0 %	2.8 %	100.0 %	

Appendix II.3 - BEC-Lewis Conversions

Table 50. BEC-Lewis conversions.

Lewis Unit	BEC Unit	Name	Comments
	CWHvm1		
S1-ha	01	HwBa – Blueberry	
S1-ch	01s	HwBa – Blueberry: Salal phase	
S2	03	HwCw- Salal	
$S2^F$ and $S2^P$	02	HwPl – Cladina	
S3	09	Ss – Salmonberry (high bench)	Inactive floodplains may be 07 BaCw –Salmonberry
$S3^B$	04	CwHw – Swordfern	May include 05 BaCw - Foamflower if SMR is fresh
S4	10	Act -Red osier dogwood	
S5	14	CwSs - Skunk cabbage	Rich alluvial Skunk cabbage
S6	14	CwSs- Skunk cabbage	Poor non-alluvial Skunk cabbage
S7	13	Pl – Sphagnum	
S8	13	Pl – Sphagnum	
S9	na	no site unit	Non-forested bog
S10	06s	HwBa - Deer fern: salal phase	Moist, poor CwHw salal site
S11	na	no site unit	Non-forested fen
S12	05	BaCw – foamflower	Limestone influence evident
$S12^{CH}$	01	HwBa –Blueberry	Limestone influence masked by thick for. floor
$S12^F$	05	BaCw –Foamflower	Limestone influence evident
S13	05 & 07	BaCw –Foamflower & BaCw – Salmonberry	
S15	06	HwBa - Deer Fern	Not described in TFL 6 report
CP	12	CwYc – Goldthread	
	CWHvm2		
M1	01	HwBa – Blueberry	
M2	03	HwCw – Salal	
M3	05	BaCw – Foamflower	
M4	11	CwSs _ Skunk cabbage	
M5	08	BaSs -Devil's club	
	MH & AT		
MH or MH1	01	HmBa – Blueberry	No information provided in reports
MH2	03	HmBa – Mountain heather	No information provided in reports
MH4	05	HmYc - Deer cabbage	No information provided in reports
P	na	Parkland	
A	na	Snow avalanche	
W	na	Wetlands	
AT	na	Alpine Tundra	

Appendix II.4 - Site index statistics by Barker Productivity Groups, Lewis Ecosystem Units, and BEC Site Series

Table 51. Site Index statistics by Barker Productivity Groups.

Prod. Group	Random Sample Plots	Average Site Index (m)	Minimum Site Index (m)	Maximum Site Index (m)	Standard Deviation (m)	Standard Error (m)
EE	8	33.7	30.9	38.5	2.73	0.96
EG+	13	29.3	22.0	34.5	4.15	1.15
EG	17	31.5	17.1	37.2	5.29	1.28
EG-	8	20.3	16.3	27.3	3.56	1.26
EL	8	19.6	6.0	30.6	9.31	3.29
EM	7	20.7	9.8	28.4	7.46	2.82

Table 52. Site index statistics by Lewis Ecosystem Units.

Lewis Eco. Unit	Random Sample Plots	Average Site Index (m)	Minimum Site Index (m)	Maximum Site Index (m)	Standard Deviation (m)	Standard Error (m)
S1CH	7	20.7	9.8	28.4	7.46	2.82
S1HA	15	32.1	25.4	37.2	3.87	1.00
S2	1	17.3	17.3	17.3		
S3	8	33.7	30.9	38.5	2.73	0.96
S5	1	33.5	33.5	33.5		
S6	6	19.5	6.0	30.6	10.90	4.45
S12	2	26.8	17.1	36.5	13.74	9.71
S13	12	29.0	22.0	34.5	4.13	1.19
M1	8	20.3	16.3	27.3	3.56	1.26
M2	1	22.3	22.3	22.3		

Table 53. Site index statistics by Lewis units converted to BEC site series.

Converted BEC Site Series	Random Sample Plots	Average Site Index (m)	Minimum Site Index (m)	Maximum Site Index (m)	Standard Deviation (m)	Standard Error (m)
CWHvm1/01	15	32.1	25.4	37.2	3.87	1.00
CWHvm1/01s	7	20.7	9.8	28.4	7.46	2.82
CWHvm1/03	1	17.3	17.3	17.3		
CWHvm1/05	2	26.8	17.1	36.5	13.74	9.71
CWHvm1/05-07	12	29.0	22.0	34.5	4.13	1.19
CWHvm1/09	8	33.7	30.9	38.5	2.73	0.96
CWHvm1/14	7	21.5	6.0	33.5	11.27	4.26
CWHvm2/01	8	20.3	16.3	27.3	3.56	1.26
CWHvm2/03	1	22.3	22.3	22.3		

Table 54. Site index statistics by field classified BEC site series.

Field Class. BEC Site Series	Random Sample Plots	Average Site Index (m)	Minimum Site Index (m)	Maximum Site Index (m)	Standard Deviation (m)	Standard Error (m)
CWHvm1/01	21	29.5	17.3	37.2	5.22	1.14
CWHvm1/01s	2	18.2	11.0	25.4	10.15	7.17
CWHvm1/05	9	33.2	26.6	38.5	3.68	1.23
CWHvm1/06	5	25.7	11.0	32.1	8.60	3.85
CWHvm1/06s	2	15.5	9.8	21.3	8.11	5.73
CWHvm1/07	9	32.0	27.4	37.2	3.79	1.26
CWHvm1/13	2	9.2	6.0	12.4	4.49	3.17
CWHvm1/14	1	26.9	26.9	26.9		
CWHvm2/01	6	20.4	17.1	22.6	2.32	0.95
CWHvm2/05	1	18.6	18.6	18.6		
CWHvm2/06	1	17.4	17.4	17.4		
CWHvm2/07	1	27.3	27.3	27.3		
CWHvm2/11	1	16.3	16.3	16.3		

Appendix II.5 - Plot Statistics

Table 55. Inventory statistics for the 61 sample plots.

PG	Plot No.	Lewis Unit	Map BEC Site Series	Barker PG	No. BA Plots	Ht (m)	BH Age (yrs)	Site Index (m)	Density All Stems (no/ha)	BA (m^2/ha)	Total Gross Vol. (m^3/ha)	DBHq (cm)
1	1-01	S3	CWHvm1/09	EE	3	23.0	29.0	34.4	245	32.0	388.2	40.8
1	1-02	S3	CWHvm1/09	EE	4	28.3	34.3	36.8	632	54.0	608.6	33.0
1	1-04	S3	CWHvm1/09	EE	2	19.1	24.5	33.1	7,310	85.0	444.9	12.2
1	1-05	S3	CWHvm1/09	EE	4	25.1	35.0	32.5	2,002	56.0	484.7	18.9
1	1-06	S3	CWHvm1/09	EE	5	16.6	21.7	32.3	2,214	48.0	308.6	16.6
1	1-07	S3	CWHvm1/09	EE	4	14.7	19.9	31.1	3,351	40.5	220.0	12.4
1	1-08	S3	CWHvm1/09	EE	5	27.2	30.6	38.5	747	41.6	388.6	26.6
1	1-09	S3	CWHvm1/09	EE	5	18.4	25.7	30.9	4,740	34.8	236.8	9.7
2	2-01	S13	CWHvm1/05-07	EG+	5	41.3	67.3	34.5	2,108	92.8	933.3	23.7
2	2-03	S13	CWHvm1/05-07	EG+	5	43.9	83.2	32.5	604	80.0	1,189.6	41.1
2	2-04	S13	CWHvm1/05-07	EG+	4	19.2	32.9	26.6	2,172	45.0	290.4	16.2
2	2-05	S13	CWHvm1/05-07	EG+	2	29.3	44.1	31.9	2,000	60.0	541.2	19.5
2	2-07	S13	CWHvm1/05-07	EG+	3	14.0	18.0	32.5	6,441	58.7	270.6	10.8
2	2-08	S13	CWHvm1/05-07	EG+	2	12.6	19.7	27.4	1,828	8.0	37.3	7.5
2	2-10	S13	CWHvm1/05-07	EG+	3	16.5	26.5	27.4	1,041	34.7	295.9	20.6
2	2-11	S13	CWHvm1/05-07	EG+	5	35.7	53.6	34.2	835	68.0	711.7	32.2
2	2-12	S13	CWHvm1/05-07	EG+	5	16.2	31.3	23.5	13,931	80.0	491.9	8.6
2	2-13	S13	CWHvm1/05-07	EG+	4	13.6	20.4	28.6	4,342	22.0	93.9	8.0
2	2-14	S5	CWHvm1/14	EG+	4	19.1	24.1	33.5	615	44.0	379.3	30.2
2	2-15	S13	CWHvm1/05-07	EG+	5	17.7	38.0	22.0	5,532	70.8	379.9	12.8
2	2-16	S13	CWHvm1/05-07	EG+	4	13.6	22.4	26.4	12,210	38.0	143.1	6.3
3	3-01	S1HA	CWHvm1/01	EG	4	29.1	49.8	29.4	881	33.0	300.7	21.8
3	3-02	S1HA	CWHvm1/01	EG	5	15.0	24.1	27.2	6,220	45.6	195.9	9.7
3	3-03	S12	CWHvm1/05	EG	5	27.8	33.9	36.5	740	52.8	595.7	30.1
3	3-04	S1HA	CWHvm1/01	EG	3	29.8	37.6	36.2	698	56.0	661.4	32.0
3	3-06	S1HA	CWHvm1/01	EG	4	14.2	18.5	32.1	14,207	54.0	267.4	7.0
3	3-07	S1HA	CWHvm1/01	EG	4	26.4	37.8	32.3	751	38.0	365.3	25.4
3	3-09	S1HA	CWHvm1/01	EG	4	20.7	34.7	27.4	899	48.0	449.2	26.1
3	3-10	S1HA	CWHvm1/01	EG	5	18.0	19.7	37.2	2,365	52.4	349.7	16.8
3	3-11	M1	CWHvm2/01	EG-	4	16.5	34.1	22.3	1,632	58.0	465.6	21.3
3	3-12	S1HA	CWHvm1/01	EG	5	26.5	34.1	34.8	2,566	27.6	235.7	11.7
3	3-16	S12	CWHvm1/05	EG	5	10.9	29.2	17.1	4,098	23.2	89.9	8.5
3	3-19	S1HA	CWHvm1/01	EG	5	10.9	15.8	29.5	1,677	8.8	31.7	8.2
3	3-20	S1HA	CWHvm1/01	EG	5	17.9	32.2	25.4	5,546	41.6	198.4	9.8
3	3-26	S1HA	CWHvm1/01	EG	5	22.6	25.5	37.2	2,635	51.6	359.9	15.8
3	3-27	S1HA	CWHvm1/01	EG	5	31.7	43.0	35.0	727	60.0	681.2	32.4
3	3-28	S1HA	CWHvm1/01	EG	5	39.1	59.9	35.0	464	48.0	580.0	36.3
3	3-30	S1HA	CWHvm1/01	EG	3	38.6	78.3	29.2	1,014	96.7	1,126.2	34.8
3	3-31	S1HA	CWHvm1/01	EG	5	30.8	43.5	33.8	1,688	70.4	630.5	23.0
3	3-53	M1	CWHvm2/01	EG-	4	19.2	31.9	27.3	14,899	49.0	278.0	6.5
3	3-64	M1	CWHvm2/01	EG-	3	8.3	16.2	22.6	6,949	21.3	80.4	6.3
3	3-71	M1	CWHvm2/01	EG-	2	8.0	18.9	18.6	5,609	33.0	113.9	8.7
3	3-72	M1	CWHvm2/01	EG-	4	12.6	31.6	18.5	19,258	77.5	331.2	7.2
3	3-75	M1	CWHvm2/01	EG-	4	27.7	83.9	19.6	2,222	67.5	642.4	19.7
3	3-76	M1	CWHvm2/01	EG-	4	11.5	30.6	17.4	11,315	41.0	222.5	6.8
3	3-79	M1	CWHvm2/01	EG-	4	11.2	31.3	16.3	4,881	40.0	192.4	10.2
4	4-02	S1CH	CWHvm1/01s	EM	5	5.4	24.0	9.8	523	6.4	34.4	12.5
4	4-03	S1CH	CWHvm1/01s	EM	2	20.2	44.0	22.3	1,986	39.0	233.9	15.8
4	4-04	S2	CWHvm1/03	EL	3	13.9	38.0	17.3	3,852	42.7	218.6	11.9
4	4-05	S6	CWHvm1/14	EL	5	14.0	20.2	30.1	1,708	26.4	151.5	14.0
4	4-08	S6	CWHvm1/14	EL	4	7.4	27.0	12.4	4,417	26.0	126.4	8.7
4	4-09	M2	CWHvm2/03	EL	4	11.8	23.5	22.3	12,448	34.0	119.3	5.9

PG	Plot No.	Lewis Unit	Map BEC Site Series	Barker PG	No. BA Plots	Ht (m)	BH Age (yrs)	Site Index (m)	Density All Stems (no/ha)	BA (m^2/ha)	Total Gross Vol. (m^3/ha)	DBHq (cm)
4	4-11	S6	CWHvm1/14	EL	4	22.9	40.5	26.9	3,888	58.0	445.2	13.8
4	4-12	S1CH	CWHvm1/01s	EM	3	16.9	26.1	28.4	3,832	18.0	90.9	7.7
4	4-15	S1CH	CWHvm1/01s	EM	4	6.6	26.5	11.0	1,942	12.0	49.7	8.9
4	4-18	S1CH	CWHvm1/01s	EM	1	9.8	20.3	21.3	430	8.0	46.0	15.4
4	4-28	S6	CWHvm1/14	EL	5	4.2	31.9	6.0	909	15.2	83.5	14.6
4	4-31	S1CH	CWHvm1/01s	EM	4	18.7	31.4	26.9	6,169	57.5	294.3	10.9
4	4-32	S1CH	CWHvm1/01s	EM	3	7.9	13.4	25.4	8,258	32.0	152.1	7.0
4	4-54	S6	CWHvm1/14	EL	5	6.3	26.0	11.0	2,636	16.8	69.5	9.0
4	4-55	S6	CWHvm1/14	EL	5	19.0	26.8	30.6	565	21.6	172.5	22.1

APPENDIX II.6 – Main Plot Attributes in Relation to Top Height

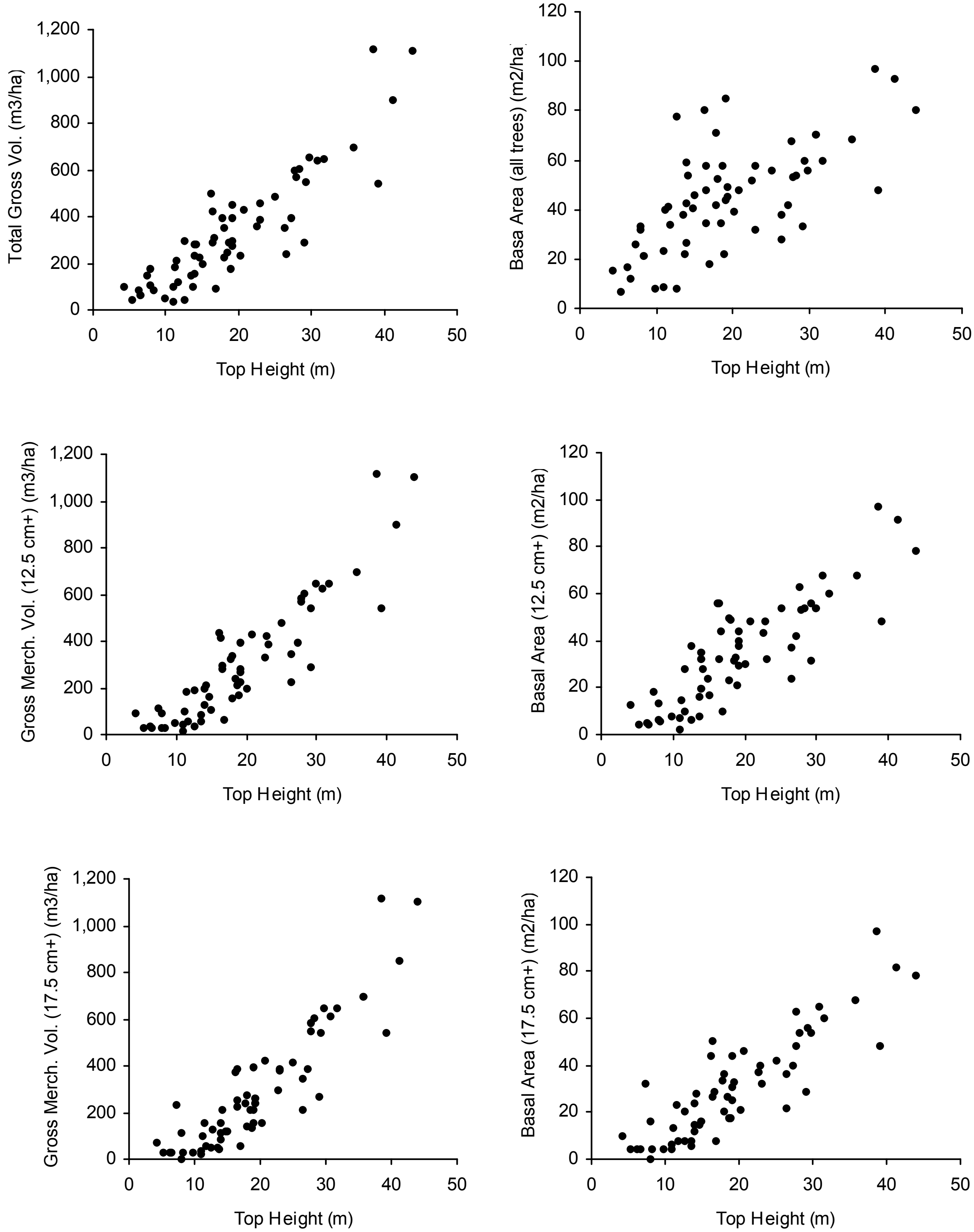


Figure 11. Plot volume and basal area in relation to top height (total, 12.5 cm+, and 17.5 cm+).

Appendix II.7 - Height-Diameter Curves

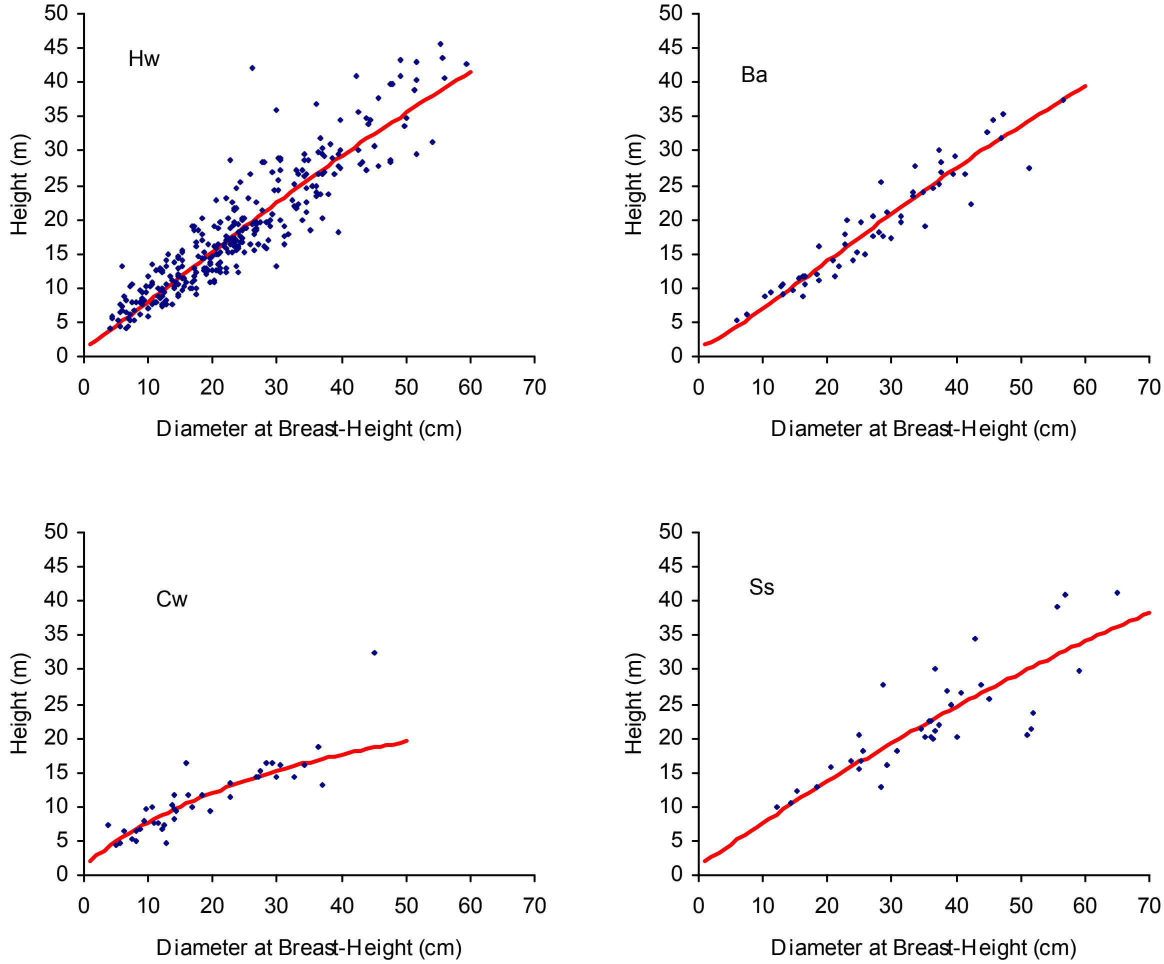


Figure 12. Height-diameter curves for Hw, Ba, Cw, and Ss.

Appendix III

OPERABILITY CLASSIFICATION

Terms of Reference

August 19, 1998

1) SOURCES OF INFORMATION

a) Resource Inventories

i) Operational mapping
   - Forest cover and topography – K.C. Hoel; 1:5,000 scale; 7.6 meter and 10 meter contour intervals

ii) Forest cover mapping – WFP Pamap G.I.S. inventory base; 1:10,000 scale

iii) Terrain stability overview mapping (TSIL “C”) – T. Lewis; MoF 5 Class System; 1:20,000 scale; TFL 25 Blk 4 is currently Lewis 5 Class System – this will be converted to MoF 5 Class

iv) Stream classification mapping – WFP; known fish streams; 1:20,000 scale

v) Landscape Inventory and Analysis – Neufeldt, Reeves, MacDougall & Associates

vi) Recreation Inventory and Analysis – Neufeldt, Reeves, MacDougall & Associates

vii) Ecosystem Mapping – T. Lewis; 1:10,000 scale and Low Ecosite Mapping; 1:20000 scale

b) Reconnaissance

i) Aerial

ii) Ground

c) Photography

i) 1:15,000 scale aerial photography (1995)

2) ASSUMPTIONS AND PLANNING CONSIDERATIONS

Terrain Stability Note:

The level “C” terrain stability overview mapping is by definition, a relatively coarse filter. Local knowledge and historical evidence show that at a more refined level, Class 4 and Class 5 terrain as identified on the overview may include terrain of more stable classifications. There will, therefore, be small areas identified as operable, which will be in apparent conflict with the overview mapping.

Ultimately, the area excluded from the operable land base as Class 4 and Class 5 terrain, will be that identified by the overview mapping net of those areas deemed to be of a more stable classification. Prior to any development activity, field terrain stability assessments will be conducted on all areas identified on the overview as having stability concerns, as required by the Forest Practices Code.

a) **Forest Road Specifications**
   i) Grades
      (1) Favourable
         (a) Maximum sustained grades of +18%
         (b) Switchbacks and short pitches up to +20%

      (2) Adverse
         (a) Maximum sustained grades of –8%
         (b) Short intervals up to –12%

   ii) Terrain

      (1) Roads are not proposed on Class 5 terrain

      (2) Roads may cross Class 4 inclusions to access timber (terrain field assessments will be conducted prior to development as per the Forest Road Regulation)

      (3) Roads can be constructed on Class 1 through 3 terrain

b) **Yarding Systems - Physical Constraints**
   i) Conventional Yarding Systems (**O_C or O_{CE}**)

      Conventional yarding is subdivided into two operable types based on forest cover: "**O**perable **C**onventional" (**O_C**) and "**O**perable **C**onventional with **E**conomic constraints based on forest cover" (**O_{CE}**). (Refer to section 2)c); Yarding Systems – Forest Cover Constraints). The physical constraints described hereafter hold true for both conventionally operable subtypes.

      (1) Highlead (includes 27.4 meter tower and grapple yarders)
         (a) Square Lead
            (i) 200 meters preferred maximum yarding distance
            (ii) 250 meters acceptable in occasional situations with adequate deflection

         (b) Corners
            (i) 300 meters preferred maximum yarding distance
            (ii) 350 meters acceptable in occasional situations with adequate deflection

         (c) Terrain
            (i) Logs are fully suspended on Class 4
            (ii) Not considered on Class 5

(2) Longline
  (a) Distance Constraints – Uphill Yarding (shotgun system preferable)
    (i) Maximum yarding - 800 meters
    (ii) Maximum tail hold - 1000 meters

  (b) Distance Constraints – Downhill Yarding
    (i) Maximum yarding and tail hold – 650m

  (c) Not considered on Class 5 terrain

  (d) Situations indicating consideration for use
    (i) Terrain stability concerns
      1. Largely continuous terrain Class 4 road development required to yard conventionally
      2. Improve deflection to minimize ground disturbance

    (ii) Portion of setting inaccessible by road due to terrain constraints
      1. Class 5
      2. Rock bluffs
      3. Canyons

    (iii) Minimize isolation of timber
    (iv) Preferable to heli-logging where useable
    (v) Economics dictates skyline over extra and expensive road

(3) Ground Based (hydraulic hoe forwarders) (note: this type may also include ground based systems used in alternative systems such as forwarders and skidders where suitable)
  (a) Distance Constraints
    (i) 150 meters maximum distance to road side
    (ii) May be used, where appropriate, to forward to highlead system (60m maximum)

  (b) Terrain
    (i) Class 1 and 2 terrain with minor inclusions of Class 3
    (ii) 30% maximum sustained slope
    (iii) Small inclusions of steeper ground acceptable

ii) Non-Conventional Yarding Systems

Non-conventional yarding is subdivided into two operable types, **O**perable **H**elicopter (**O_H**) or **O**perable **H**elicopter with **E**conomic constraints (**O_{HE}**). (Refer to section 2)c); Yarding Systems – Forest Cover Constraints). The physical constraints hereafter apply to both the economically constrained and non-economically constrained helicopter operable types.

(1) Helicopter (**O_H or O_{HE}**)
   (a) Flight Distance
      (i) 1.0 kilometer or less preferred
      (ii) Up to 2.0 kilometers acceptable where no alternative exists

   (b) Both water and land drops are considered

   (c) Uphill flight acceptable using same constraints as in 2)c)i)(1) above

   (d) Slope constraint determined by terrain class (i.e. not considered on Class 5 terrain; steep slopes on class 4 or less terrain are considered)

   (e) Situations indicating consideration for use
      (i) Timber inaccessible by road due to terrain constraints
         1. Class 5
         2. Rock bluffs
         3. Canyons

      (ii) Isolated location (i.e. conventional development uneconomic due to sheer distance from current development and insufficient merchantable timber in between)

      (iii) Terrain stability issues

c) Yarding Systems – Forest Cover Constraints

As previously mentioned, forest cover is broken into two operable types, one with economic constraints (denoted by the subscript “E” in the operability descriptor) and the other without economic constraints (and no modifier in the descriptor).

The economic constraint is indicative of timber which is on the margin of operability in terms of volume, quality and species. In good economic times, operability types with the “E” modifier will be operable. In poor economic times these same types may not be operable. These types are seen as opportunity timber and given the unpredictability of the economy, should have no associated requirement to harvest for cut control purposes.

i) Conventional yarding systems (**O_C**)
   (1) All height class 4 and above
   (2) All height class 3 with cedar, cypress, Douglas fir or spruce as primary species with the exception of stocking class 3 stands which are excluded.
   (3) Height class 3 stands with hemlock or balsam as primary species which are in close proximity to **Oc** types noted in points (1) and (2)

ii) Conventional Yarding Systems with Economic Constraints (**O_{CE}**)
   (1) Height class 3 stands with hemlock or balsam as primary species which are not in close proximity to **Oc** types described in points i) (1) and (2)
   (2) Stocking class 3 stands with Douglas fir, cedar, cypress or spruce as primary or secondary species
   (3) Deciduous stands – operability determination based upon local knowledge

iii) Non-Conventional Yarding Systems
   (1) Helicopter (**O_H**)
      (a) All height class 4 and above
      (b) All height class 3 stands with cedar, cypress, spruce or Douglas fir as primary species (excluding stocking class 3 stands) that are in close proximity to **O_H** (height class 4 and better) and/or **O_C** types.

   (2) Helicopter with Economic Constraints (**O_{HE}**)
      (a) Height class 3 stands with cedar, cypress, spruce or Douglas fir as primary species (excluding stocking class 3 stands) which are not in close proximity to **O_H** or **O_C** types
      (b) Height class 3 stands with cedar, cypress, spruce or Douglas fir as secondary species with the exception of all height class 3 stocking class 3 combinations which are excluded
      (c) Pure hemlock balsam height class 3 stands are excluded

d) Economically Inoperable Forest Cover (I_E)
   (1) All mature height class 1 and 2
   (2) Pure hemlock balsam height class 3 stocking class 3 open stands
   (3) Pine dominant stands

e) Physically Inoperable Lands (I_P)
   (1) All non-productive types (i.e. rock, brush, swamp, alpine, lakes, rivers, dryland sorts, camps, quarries, etc.)
   (2) Land feature limitations (eg. major gullies)
   (3) Areas rendered physically and/or economically inaccessible by extreme terrain and/or distance, from development, which is physically and/or economically possible. (This distinction pertains to those areas to which access is physically possible, but so physically onerous that it is economically prohibitive)

f) Other Inoperable

Areas that are inoperable for environmental or institutional reasons will be withdrawn through the G.I.S. (eg. terrain class 5, riparian, wetland or lake reserves, deer winter ranges, research plots, etc.)

Appendix III

Timber Supply Analysis

Western Forest Products Limited

TIMBER SUPPLY ANALYSIS

IN PREPARATION OF

MANAGEMENT PLAN 9

FOR

Tree Farm Licence 6

Submitted to the Timber Supply Branch,
Ministry of Forests

August, 2000

REGISTERED PROFESSIONAL
PROVINCE OF
R. PAUL BAVIS
BRITISH COLUMBIA
FORESTER

R. Paul Bavis, *R.P.F*
Manager, Timber Supply and Planning
Western Forest Products Limited

Executive Summary

This report proposes, after allowances for non-recoverable losses, that the harvest level for Tree Farm Licence 6 be reduced to 1,469,900 m^3/year through the period 2001-2005. This includes an optional, experimental commercial thinning partition of 3,000 m^3/year and a partition of 14,500 m^3/year for economically marginal forest to be harvested only if and when conditions permit. The basic harvest level of 1,452,400 m^3/year is consistent with a step down of harvest levels forecast five years ago during the preparation of Management Plan 8.

Complan 3.0, a spatial harvest model, was used to simulate timber supply based on current management practices in the licence area. The proposed harvest flow was derived after ensuring sustainability of environmental values. Simulation of current management practices maintained at least 30,000 ha of old forest and 83 million m^3 of growing stock throughout the 250-year simulation horizon. In response to increased emphasis on environmental values, the projected harvest levels have been and are continuing below the unrestrained timber production potential of the Tree Farm Licence.

Sensitivity analyses around the current management or “base case” simulation suggest that timber supply is sensitive to land base withdrawals, adjacency and block size policies, seral stage targets, minimum stem size at harvest, and inventory changes. Further constraints would reduce flexibility and make impacts most likely forty to fifty years into the future.

After 2060, gains from current and past silviculture efforts produce an abundance of growing stock and facilitate a predicted “fall up” in harvest levels to a long term level of 1,640,000 m^3/year.

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Acknowledgements

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Western Forest Products Limited

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1.0 Introduction

1.1 Purpose

This report summarizes the results of timber supply modelling for Western Forest Products' Quatsino Tree Farm Licence 6 (TFL 6) on northern Vancouver Island. The work was done in conjunction with the preparation of Management Plan 9 and in support of the Chief Forester of British Columbia's determination of the Allowable Annual Cut (AAC) for the period 2001 – 2005. The report also is intended to demonstrate the sustainability of resource management in TFL 6.

1.2 Objectives

1.2.1 Non-timber Resources

Ensure that non-timber resources in TFL 6 are appropriately accommodated through net downs of potential timber yields, net downs of forest area otherwise capable of producing timber, and/or by application of cover constraints. Forecast age class structure changes through time.

1.2.2 Timber Resource

After considering non-timber resources, estimate the residual timber sustainability of TFL 6 considering forest growth potential, treatments to enhance growth, potential losses, operational and legislative constraints, alternate timber flows, and the impacts and uncertainties around key assumptions.

1.2.3 Social Considerations

Declines in harvest levels greater than 10% per decade are to be avoided if feasible.

1.3 Timber Supply Model

Complan 3.0, a spatially-explicit supply model developed by Olympic Resource Management, was used to project timber supply 250 years into the future. The model is described in more detail in Timber Supply Information Package, subsection 4.1.

1.3.1 Model Set-up

Initial harvesting was "hard-wired" to follow the currently approved Forest Development Plan and to bring the model starting state forward from the inventory date (January 1, 1998) to correspond to the proposed management plan period (2001-2005). Blocks were harvested to match actual areas harvested in 1998-9 and projected harvest volume for 2000. An undercut of 303,000 m^3 from the 1995-1999 cut control period is expected to be carried forward as a series of timber sale licences for local First Nations and is to be harvested over the next 5-7 years. This was modelled as a one-time “non-recoverable loss” by adding an additional annual harvest of 60,600 m^3 to the 2001-2005 harvest period. Although this volume is included in all model runs, to avoid confusion this volume is not presented in the timber flows and graphs herein with the exception of Figure 4.

1.3.2 Analysis Units

Eighty-six analysis units were created to represent stands of different composition, age, and productivity. Further details are provided in Timber Supply Information Package, subsection 7.2.

1.3.3 Yield Curves

Yield curves and assumptions for different past or future management regimes for each of the analysis units were developed and are described in more detail in Timber Supple Information Package, section 8. A text file of the analysis unit yield curves is provided as *CM0 yield tables.txt* on the enclosed diskette.

1.3.4 Queuing Method

The model queues stands for harvest using any or all of the following priorities:

- *Yield Table Priority* is used to ensure that specific stands — old growth for example — are harvested ahead of others.
- *Minimize Growth Loss* instructs the model to consider yield curves with different growth rates and culmination points and first seek out stands past culmination and priorize those with the lowest ratio of periodic annual increment to standing volume.
- *Oldest First* priorizes stands by age.
- *Minimize Costs* selects harvesting blocks with the lowest access cost first[13].

1.4 Description of TFL

1.4.1 Land base summary

Effective October 1, 1998 Block 4 of TFL 25 was amalgamated into TFL 6 and became Block 2 thereof. The total area of TFL 6 is now 198,113 ha of which 149,747 ha is classified as within the timber harvesting land base (THLB). Further breakdowns of the land base are provided in Timber Supply Information Package, section 6.

1.4.2 Age Class Structure

Current 20-year age class[14] structure for TFL 6 is unbalanced and is illustrated in Figure 1.

[13] Was not used in this timber supply analysis except to produce a more operationally realistic 20-year harvest projection. Test runs confirmed that there was no impact on overall harvest levels.
[14] Class 8 is >140 years.

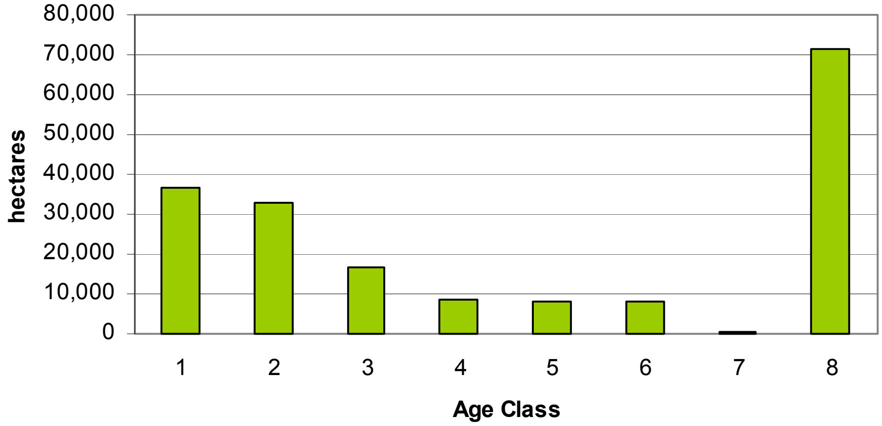

Figure 1 - Current forested age class structure (2001)

1.4.3 Prior Management Plans

Allowable annual cuts for prior periods are shown in Figure 2. In general terms harvesting levels for the past 30 years have been relatively stable. Note however that there have been additions and deletions to the Tree Farm Licence areas, timber supply analyses have improved, and inventory and yield estimates have improved considerably; hence more variability, although not apparent, might have been expected.

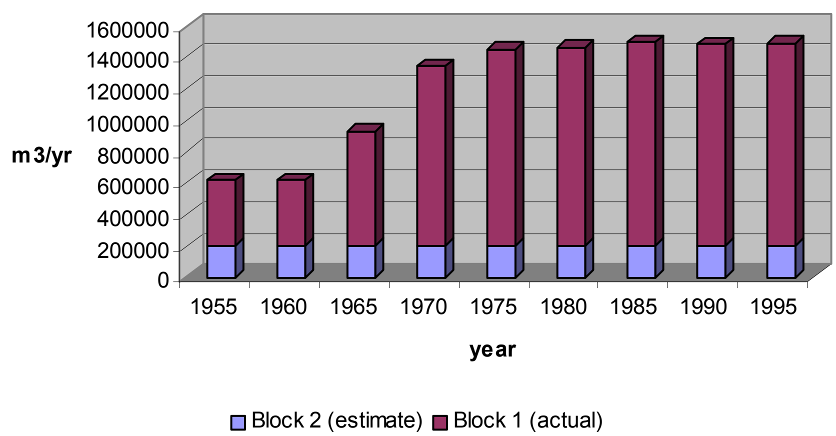

Figure 2 - Historical AAC

1.4.4 Timber Potential

In any given landscape, only a portion of the forested area actually contributes to timber supply. Several areas have been removed from TFL 6 for parks and other lands, such as inaccessible areas, riparian or other reserves, ungulate winter ranges, etc., have been deemed not operable even though they are otherwise capable of producing timber. Timber supply simulations attempt to produce the maximum annual cut that can be sustained, but only after setting aside timber potential to sustain other resource values. Often the timber potential of these set-asides is not well documented and can be substantial.

(CM0t -con) Unconstrained Timber Harvest

In this simulation there were no constraints imposed or land base withdrawals for non-timber values. All physically operable portions of the land base were included but inoperable areas were not. Figure 3 estimates the basic timber production potential of TFL 6 at 1,585,000 m^3/year for the next century and is presented for comparison purposes to the proposed current management run (CM0). The introduction of current management measures for non-timber resources reduces potential cut levels by up to 390,000 m^3/year or 25% (period 2041-2050).

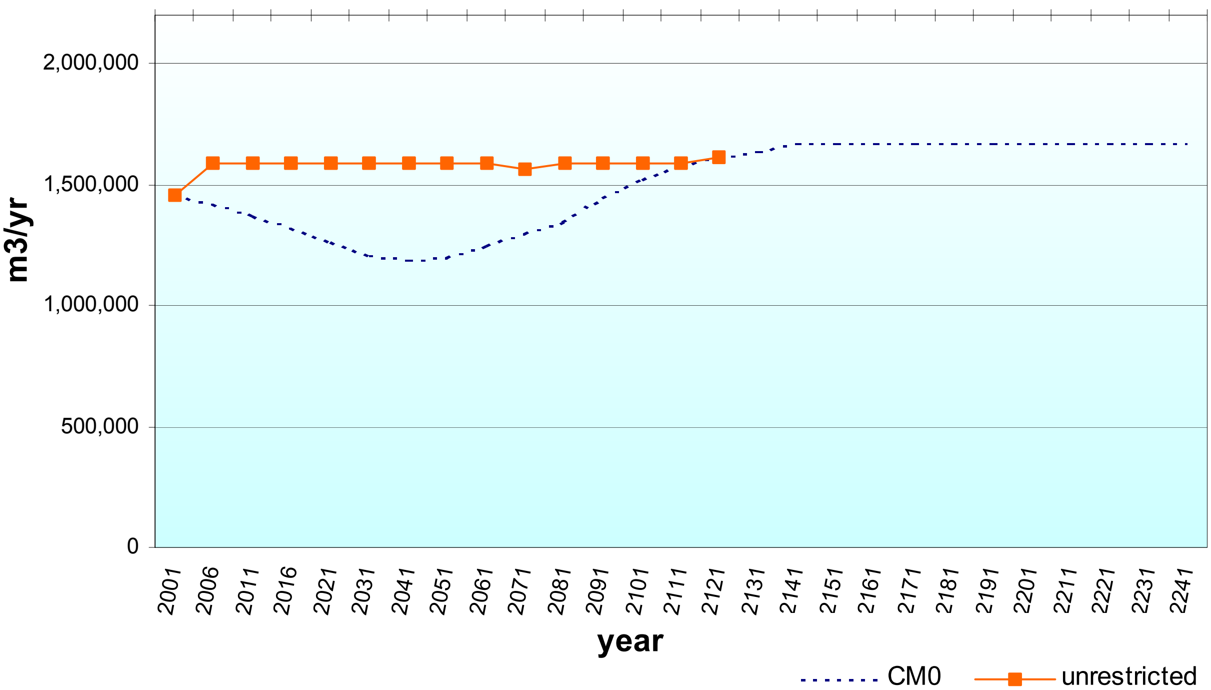

Figure 3 - Unconstrained Timber Harvesting

2.0 Current Management

The "current management" simulation assumes the following based on past performance with particular focus on the past five years:

1. Implementation of the Forest Practices Code and Vancouver Island Land Use Plan including:
    - removal of new parks from land base
    - 40 ha maximum opening size
    - Application of visual quality objectives (VQOs) to known scenic areas
    - Riparian management areas
    - Wildlife tree patch retention
    - A generalized old seral stage target based on expected 10% high and 45% intermediate and low designations is used to defer old forest in each landscape unit[15]
    - Retaining old (not grandparented) Deer Winter Ranges as modelling surrogates for as yet undefined Old Growth Management Areas and Wildlife Habitat Areas
2. Minimum harvest size for good, medium, and poor[16] sites set to 42 cm, 37 cm, and 35 cm quadratic mean stand diameter at breast height (DBHq) respectively.
3. Harvesting to the inventory and operability profile.
    - Based on a three-year average, volumes and area harvested from inventory polygons of lower site classes have exceeded partition requirements. Therefore this assumption may be conservative, but the assessment interval has been relatively short.
    - Helicopter logging has been ongoing for several years.
4. Gradual conversion of older alder leading stands to conifer leading stands through normal forest succession with no utilization of the alder component.
5. Planting of most harvested sites.
    - For the past five years an average of 1,790 ha/year has been planted which exceeds the average 1,610 ha/year logged.
6. Broadcast fertilization of cedar-salal sites to achieve crown closure and improve long term productivity.
    - Research results continue to support the hypothesis of long term shifts in productivity. For the past five years the average 1,769 ha/year fertilized exceeds the average annual area logged and far exceeds the current cedar-salal area harvested each year. The Licensee expects to continue aggressive fertilization as stands become available. Although operational restrictions will prevent fertilization of every regenerated cedar-salal site, the program is expanding into similar types; for the purposes of the model run, all cedar-salal sites are assumed to be fertilized.

[15] Subsequent to submission of the information package and model set-up, interim designations for each landscape unit were near approval. An additional sensitivity run has been added to test the guidebook application of old seral targets by the specific biodiversity emphasis applicable to each of the landscape units.

[16] Productivity Groups 1, 2/3, and 4 respectively.

7. Spacing of 300 ha of primarily hemlock-leading stands is assumed, subject to availability of suitable stands.
    - For the past five and thirty-three years respectively, on average 460 ha and 320 ha per year were juvenile spaced suggesting that the "current management" run is conservative with respect to the amount of juvenile spacing.
8. Tree improvement volume gains were modelled to be up to 5% for stock planted currently and rising to 15% for western hemlock, Douglas-fir, and yellow cypress after the next decade.
    - Recent research results suggest that these estimates are likely to prove conservative17.
9. Yield Table Priority, then Minimize Growth Loss
    - Stands were priorized by using Yield Table Priority to select old growth stands (with flat-lined inventory volumes) first. With second growth stands, Minimize Growth Loss was used to target stands where periodic annual increment was lowest. In the event of two or more stands being otherwise of equal priority, an Oldest First rule was invoked.
10. Compartment Balancing
    - Community stability has been long recognized by the Licensee as an important objective; hence simulated harvest was balanced between operating areas to ensure that the harvesting communities dependent on each would be sustained through the simulation.

For further information and specific details on current management assumptions, refer to Timber Information Package, section 3.1.

[17] Vandalism of our seed production facility remains a risk and events to date have had a slight short term impact, but replacement and salvage of damaged trees has been effective and security is improved.

3.0 Harvest Flow Options

3.1 Prior Harvest Flow

The previously determined Allowable Annual Cut for Blocks 1 and 2 (formerly TFL 25, block 4) combined is 1,489,000 m^3. The combined harvest flow as derived from the timber supply analyses is illustrated in Figure 5 and was projected to step down to a combined long-term harvest level of 1,148,772 m^3 after the year 2050.

(CM0)

3.2 Current Management, controlled step down

The proposed harvest flow is illustrated in Table 1 and graphically in Figure 4 and Figure 5. The one-time TSL of 303,000 m^3, expected to be issued to local First Nations over the next 5-7 years, was modelled by increasing the harvest request in the first 5-year period by 60,600 m^3/year over and above the proposed allowable cut (Figure 4 only). A slight deficit forecast in 2041-2050 indicates that optimum harvest levels are being achieved through the original to managed forest transition.

Table 1 - Proposed Harvest Flow

Period Start	Proposed annual cut (m^3/year)	Δ (%)	TSL (m^3/year)	Deficits
	1,489,000			
2001	1,459,400	-2.0	+60,600	
2006	1,420,000	-2.7		
2011	1,375,000	-3.2		
2016	1,322,000	-3.9		
2021	1,263,000	-4.5		
2031	1,206,000	-4.5		
2041	1,200,000	-0.5		-6,245
2051	1,200,000	0.0		
2061	1,250,000	+4.2		
2071	1,300,000	+4.0		
2081	1,350,000	+3.7		
2091	1,450,000	+7.4		
2101	1,520,000	+4.8		
2111	1,580,000	+4.0		
2121	1,610,000	+1.9		
2131	1,640,000	+1.9		
2141	1,670,000	+1.8		
2151	1,670,000	0.0		
2161+	1,670,000	0.0		

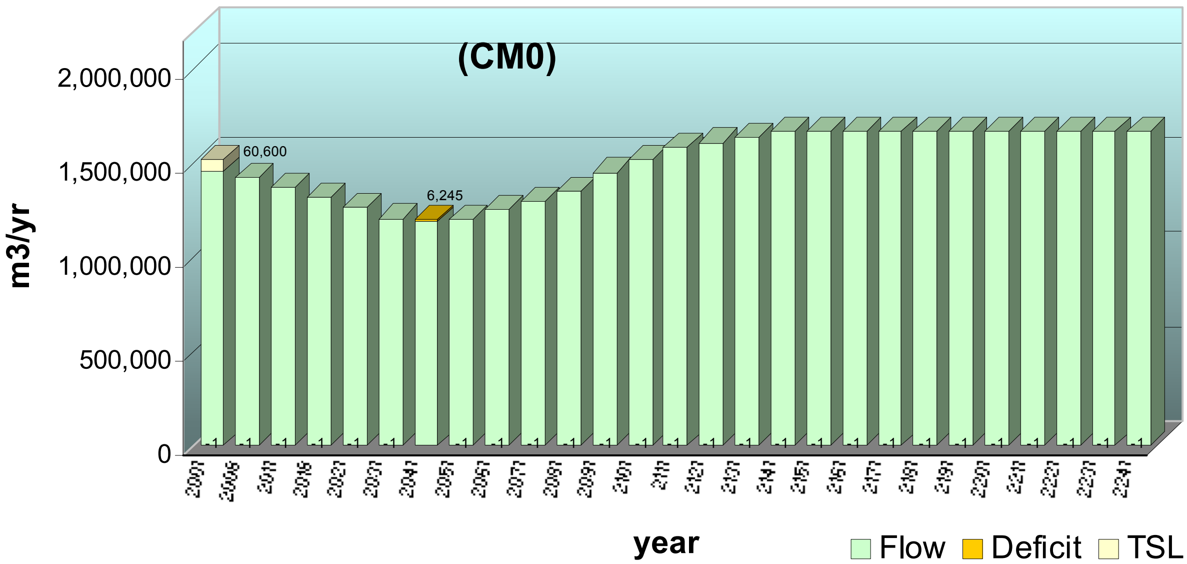

Figure 4 - Proposed Harvest Flow

Direct comparison of the current analysis with previous ones (Figure 5) is complicated by the amalgamation of Blocks 1 and 2, lack of exact time synchronization, and different assumptions and data inputs and therefore can be of a general nature only. The proposed current management flow is similar to the flow projected in the previous analyses of Blocks 1 and 2, but approaches the previously predicted long term harvest level earlier (2031 vs 2051). Beyond 2051 the curves diverge significantly due to several factors. The current management run, as requested by the Ministry of Forests, simulates yield gains due to tree improvement, fertilization and tree planting programs whereas the previous analysis considered modest tree planting yield gains only. Recent inventory projects suggested previous operational adjustments had been more conservative than necessary and have provided revised yield estimates for some maturing stands. As well, in the previous simulations the long term harvest level was frozen at the 2050 harvest level and no attempt was made to recover to higher levels after that point. A wave of available timber in the circa 2070-2130 period (Figure 11, "as available" run) suggests that higher harvest levels than projected in both the previous analyses and the current management run are conceivable in this period, irrespective of the future silviculture effect (Figure 12).

Harvest age (Figure 7) and stand diameter (Figure 6) decline abruptly through the 2021-2050 period corresponding to a shift in harvesting to second growth timber. The area harvested (Figure 6) declines in sync with reductions in harvest levels to 2040 and then recovers slowly through the simulation as increasing harvest levels are tempered by higher managed stand yields (Figure 7). In this TFL, second growth volumes at the projected harvest ages tend to be higher than currently harvested old growth volumes and lead to a "fall up" in harvest levels after the transition to second growth, rather than the often expected "fall down".

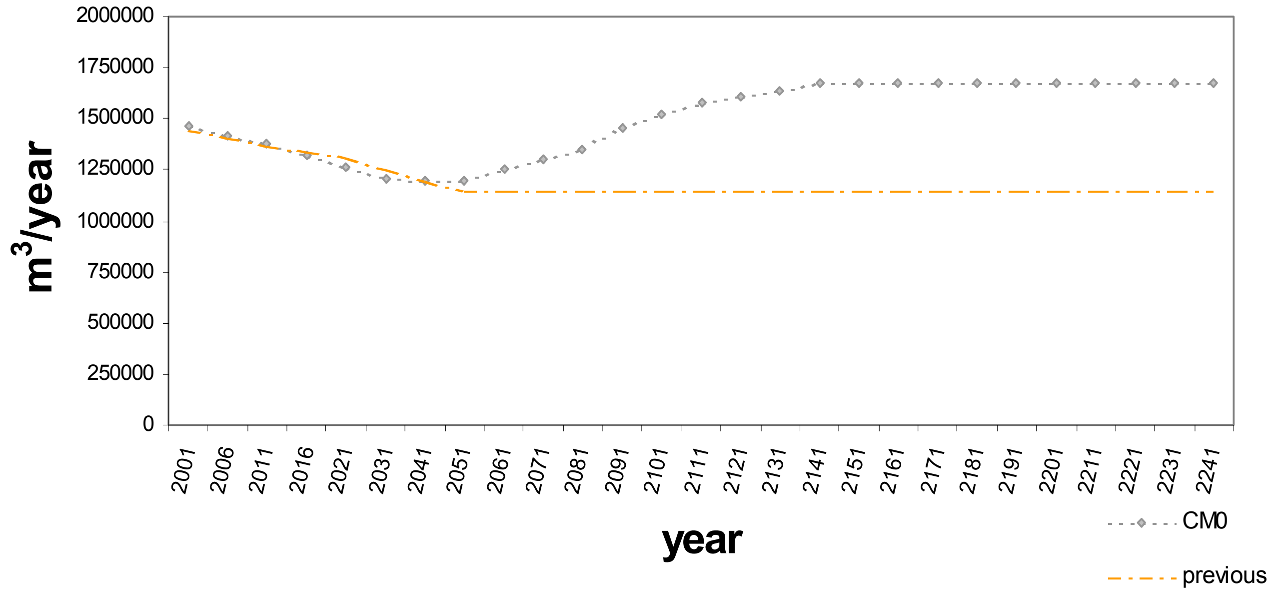

Figure 5 - Previous and Current Harvest Projection

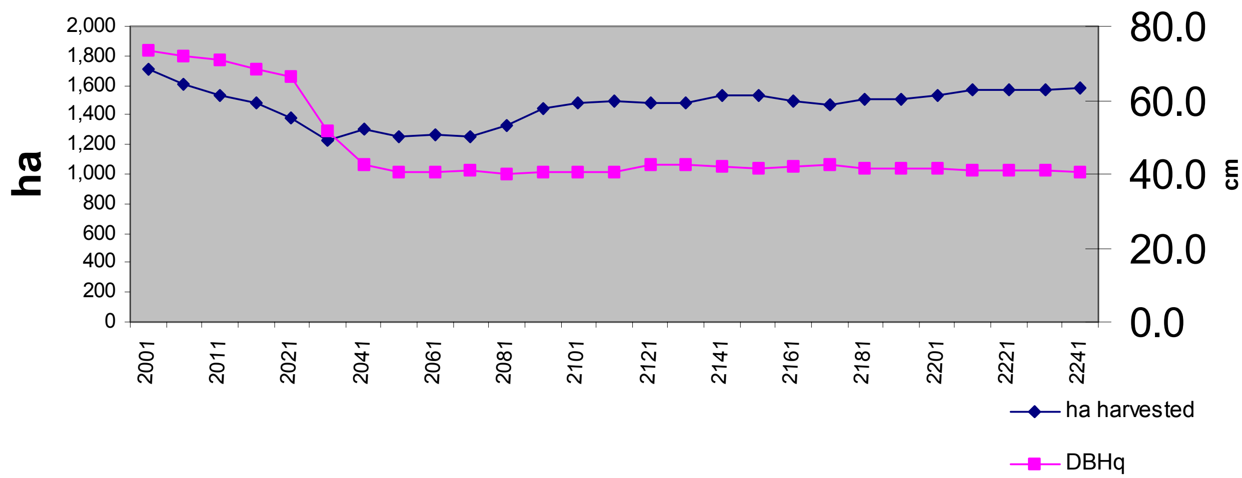

Figure 6 - Current Management - annual area and average stand diameter harvested

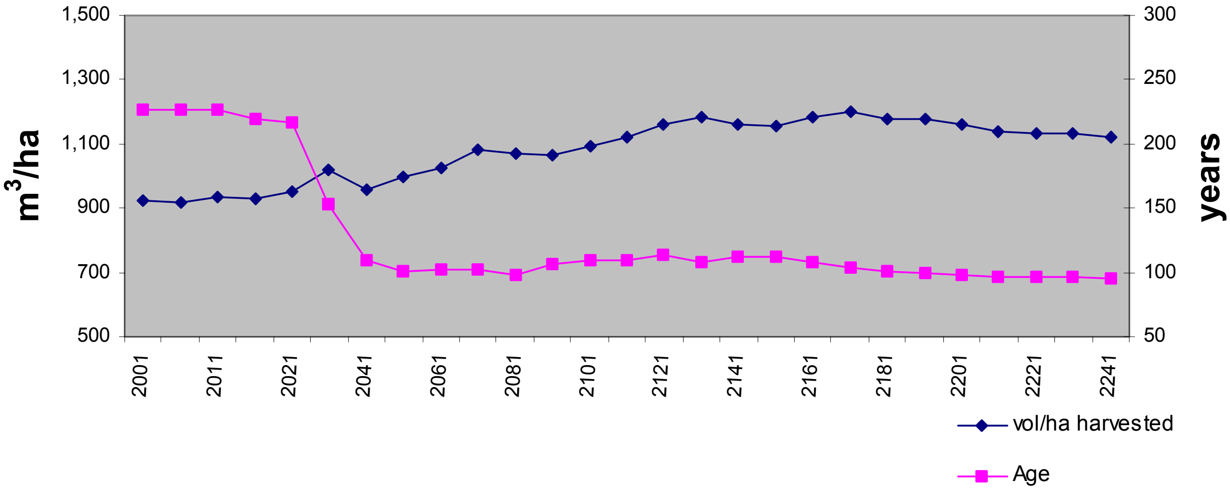

Figure 7 - Average volume/ha and age harvested

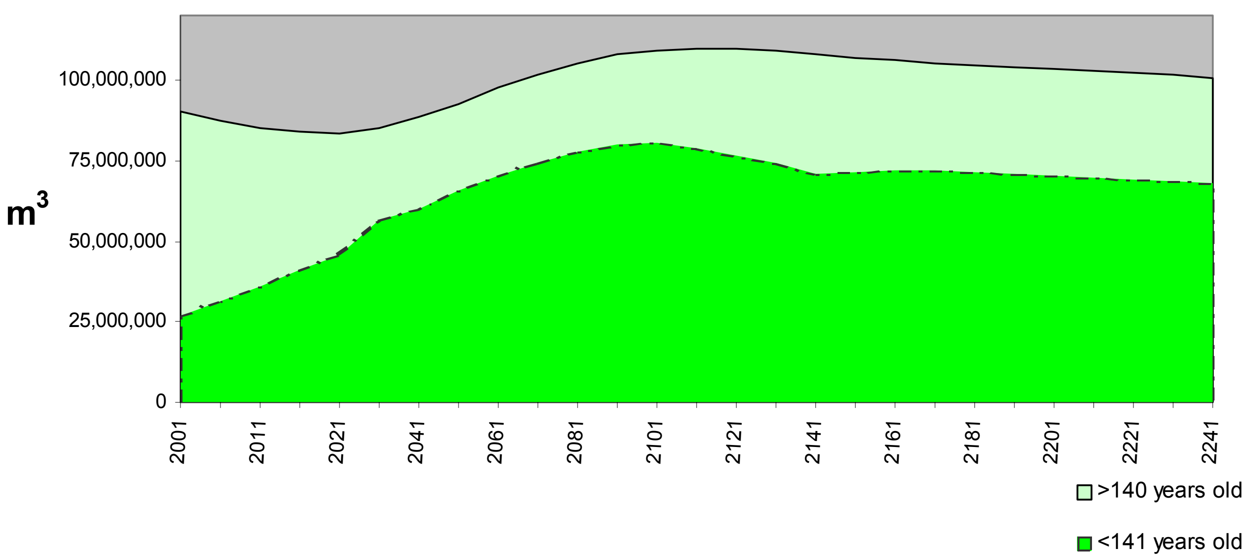

Figure 8 - Growing stock

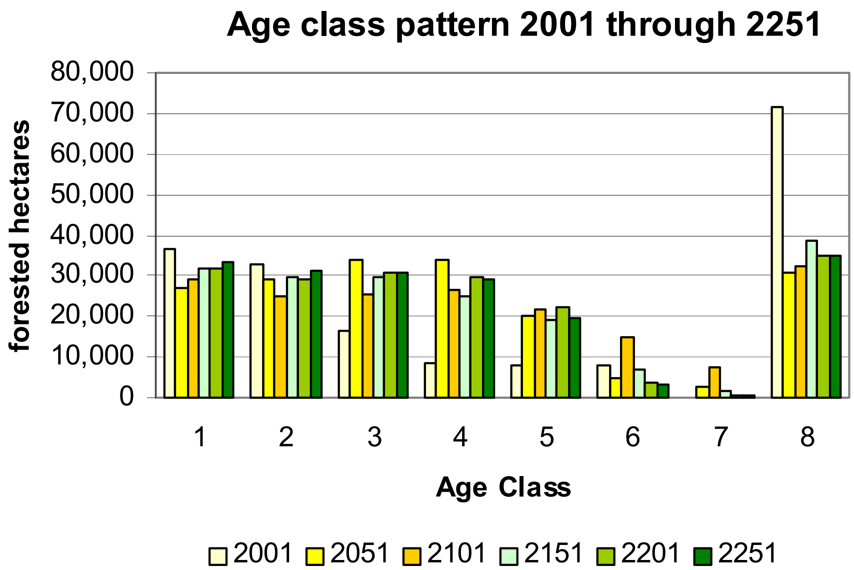

Figure 9 - Age Class Distributions

Mature timber totaling 19,697,000 m^3 (Timber Supply Information Package, Table 4), or roughly 30% of the mature inventory, has been netted out of the analysis for non-timber values. This estimate does not include Vancouver Island Land Use Plan withdrawals from TFL 6 that add about 800,000 m^3 to the withdrawn volume estimate nor the adjacent and nearby withdrawals of 73,500 ha for the Brooks Peninsula and Cape Scott Parks. Conservatively, in excess of 40 million m^3 in association with TFL 6 are protected or projected to be in an unaltered forest condition in perpetuity.

Standing inventories on the timber harvesting land base also provide environmental services. Through the projection horizon, THLB growing stock plus reserved inventory ranges from 83,570,000 to 109,775,000 m^3 (Figure 8). Therefore, within the TFL boundaries alone at least 83,570,000 m^3 (about 6,600% of the associated annual harvest simulated) will be available to provide for non-timber values at any point in time.

A 500-yr run (Figure 10) was completed to ensure that the suggested long term harvest level is stable. Towards the end of the run a number of shortfalls occur suggesting that the proposed LTHL is close to but perhaps slightly above the sustainable level. Slight declines in volume/ha, DBHq, and growing stock also support this. After completion of sensitivity runs with the long term harvest level at 1,670,000 m^3/year, harvest flow was adjusted downward in the long term to 1,640,000 m^3/year which produced stable outputs to 850 years.

An area-weighted summation of maximum mean annual increments of future stand yield curves suggests a long run sustained yield of 1,730,000 m^3/year or an average mean annual increment of 11.8 m^3/ha/year.

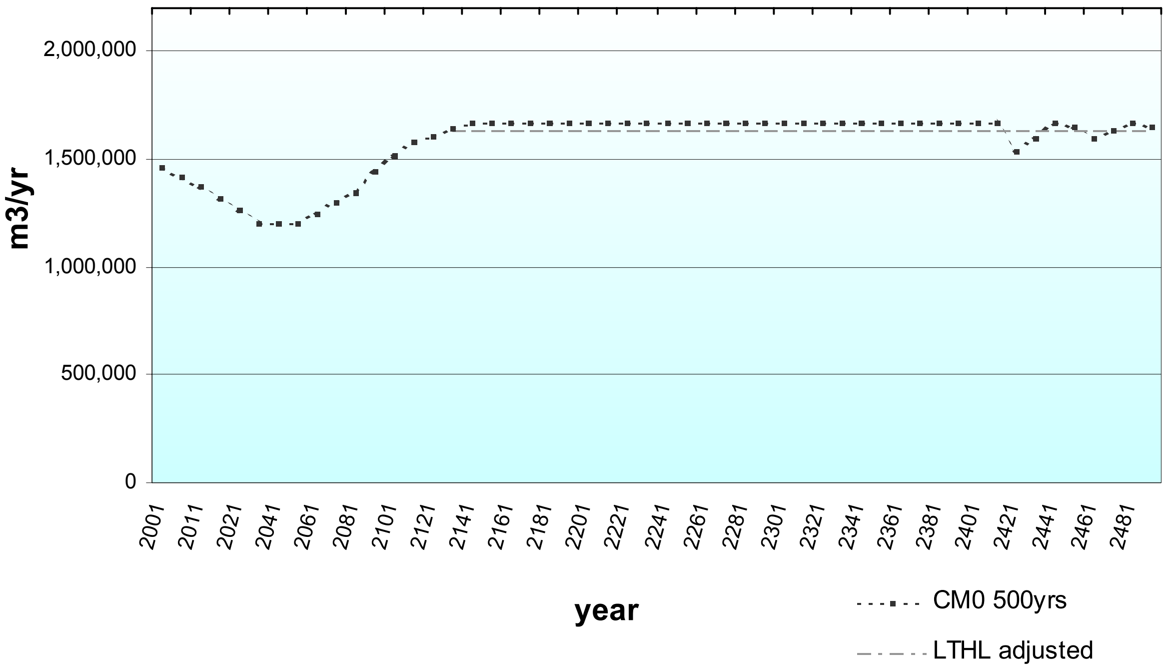

Figure 10 - Current Management to 500 years

3.3 Alternate Harvest Flows

(CM_AAC)[18]

3.3.1 Current Management, maintain present AAC indefinitely

The current AAC of 1,489,000 m^3/year could be maintained for 30 years before falling down to approximately 1,030,000 m^3/year for two decades and then climbing back thereafter (Figure 11).

(CM_LT)

3.3.2 Current Management, Harvest LTHL

The long term harvest level of 1,670,000 m3/yr can be maintained for two decades before dropping abruptly to a mid term level of 937,000 m^3/yr and then recovering by 2081 (Figure 11).

[18] Run naming convention is:

CM = Current Management

Harvest Flow Requests

- **0** = sigmoid step down
- **_AAC** = maintain current AAC indefinitely
- **_10** = -10% per decade step down
- **_15** = -15% per decade step down
- **_LT** = LTHL immediately
- **_AV** = Harvest as stands become available
- **t** = increase request to “fill the trough/dip”
- **u** = uniform increase ahead of trough or dip
- **p** = flow from previous analyses

Sensitivities

+/- indicate changes to constraining factors (generally + = on, - = off)

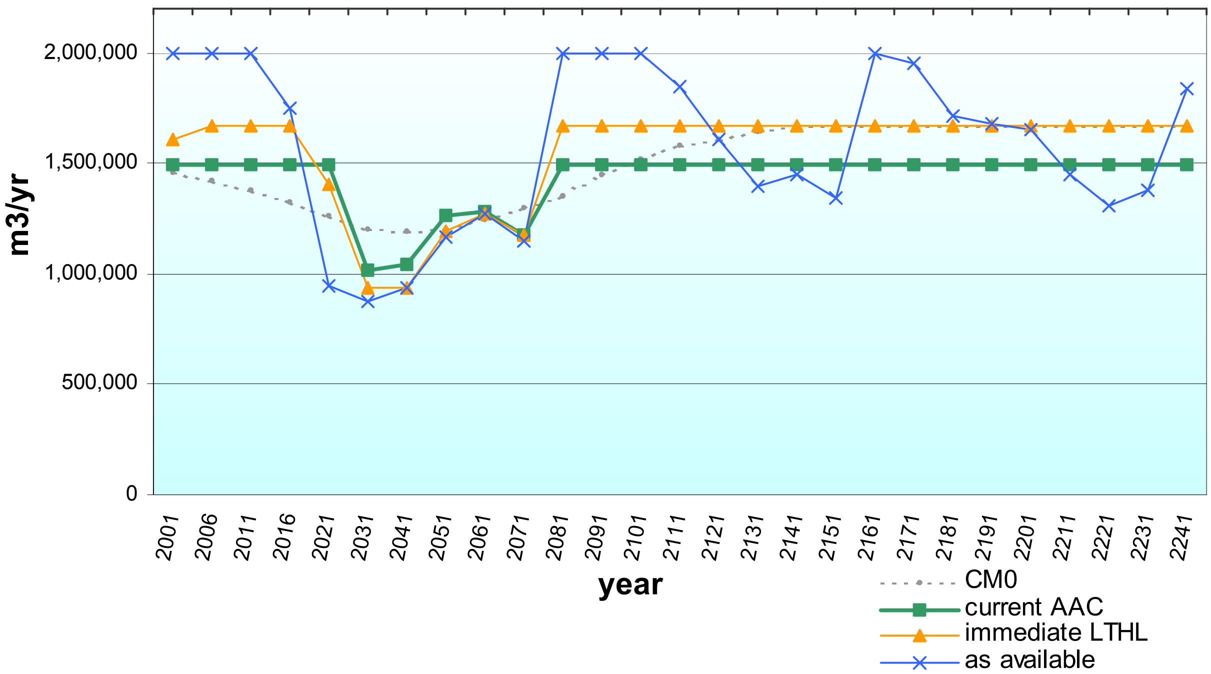

Figure 11- Alternate Harvest Flows

(CM_AV)

3.3.3 Current Management, Cut as Available (request 2,000,000 m^3)

Total volume harvested during the simulation period is 388,380,599 m^3 as opposed to 377,512,542 m^3 for the current management option, suggesting that a regulated timber flow reduces total productivity during the simulation period by about 10,900,000 m^3 or 3%. The high initial cut levels amplify a wave peak and trough that ripples through the simulation (Figure 11) and preclude attainment of a "normal" forest age structure.

4.0 Sensitivity Analyses

For sensitivities that reduced timber supply, no attempt was made to adapt harvest flows as there is still flexibility to adopt many different harvest step downs in the short term (e.g. runs CM0_AAC or CM_15 +45,40,35 in the extremes); hence comparisons of effects would be difficult. For comparison purposes, deficits are stated as m^3/year for the affected period or total m^3 through the first one hundred years of the simulation to 2100.

Where sensitivity impacts were positive, the “floor of the dip” around 2041-2050 was raised until small shortfalls occurred, indicating the limit of the positive effect had been reached. Again, harvest flow in advance of the dip could have been adjusted instead but would have obscured the effect of the changed parameter. As well, efforts to “fill the dip” will reduce the fall down in supply whereas short term harvest level adjustments would tend to maintain, with a slight delay, the currently projected step down.

No attempt was made to adjust harvest flows to capture potential positive impacts of sensitivities in the long term beyond 2100.

(CM0 –sp-fert-impr)

4.1 No juvenile spacing, fertilization, or genetic gains of future stands.

Figure 12 suggests that intensive silviculture treatments have little effect on timber supply until 140 years into the future. As average rotation age is about 100 years, these impacts would be expected to be evident 40-50 years earlier. However a large pulse of harvestable wood becomes available due to past harvesting in the 1970-2000 period and yield improvements from increasing tree planting during this period. During this period, future generations will have flexibility to choose harvest levels over and above those indicated or longer rotations to increase the proportion of older forest and larger trees present. This sensitivity analysis considers only volume impacts on the selected harvest flow; it does not consider potentially important changes in value, harvest cost, or employment opportunities associated with these silviculture programs.

By 2250, these treatments potentially account for 331,016 m^3/year or 20% of projected harvest levels.

(CM0 –sp)
4.2 No juvenile spacing of future stands

Not juvenile spacing future stands had negligible effect on timber supply as the harvest flows are indistinguishable with and without spacing (Figure 12). Juvenile spacing projects undertaken from now on will be too late to advance most stands into the 2041-2060 deficit period; instead these stands would become available for harvest through a period when there is ample growing stock available.

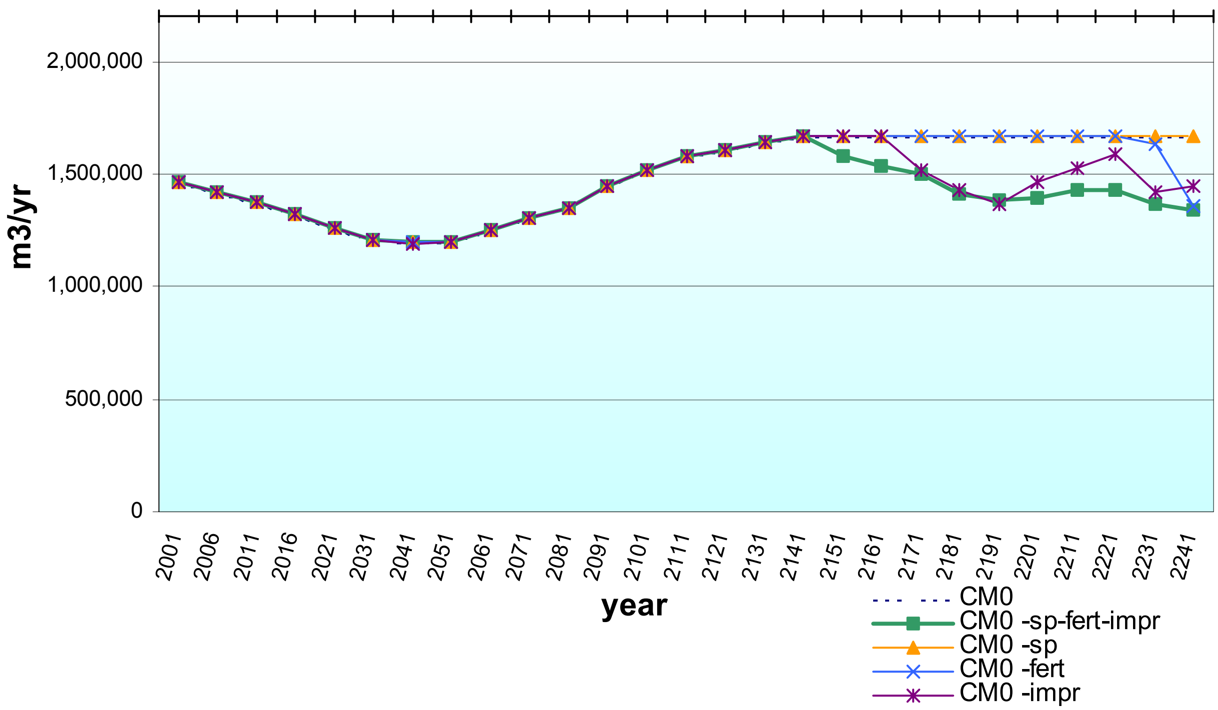

Figure 12 – Without future silviculture treatments or tree improvement

(CM0 –fert); (CM0 –impr)
4.3 No fertilization of future stands
No tree improvement of future stands

Fertilization and tree improvement programs are essential to maintain the forecasted long-term harvest level after 2140 (Figure 12). Note that the effects are not necessarily additive. Tree improvement, as it is applied to more stands, has the greatest impact on future harvest levels; whereas fertilization has a significant impact only in specific periods.

(CM0 –silv)

4.4 No juvenile spacing, fertilization, tree planting, or genetic gains of past or future stands.

Although *CM0 –sp-fert-impr* suggests that current silviculture treatments will not realize a timber supply effect until 140 years into the future, this run (Figure 13) emphasizes that this may change for different harvest flows or as new land base changes occur. Past silviculture treatments have and continue to soften the impact of harvesting restrictions imposed over the past fifteen years. For the first one hundred years of the simulation, silviculture treatments (primarily those to date) produce an additional 10,431,300 m^3.

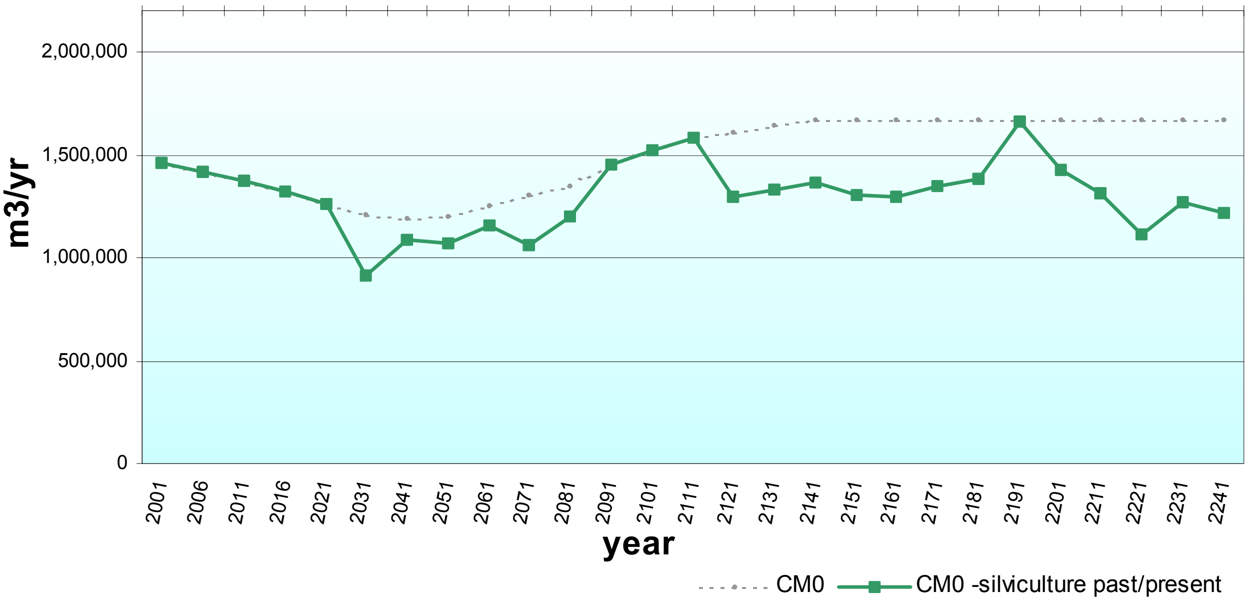

Figure 13 – Effect of past and future silviculture programs

(CM0 –lb)

4.5 Reduce operable land base by removing Terrain Stability Class 4 and 5 (-10.5%)

Through the first hundred years of the simulation, total volume harvested decreased by 8.9% or 10,233,000 m^3 and caused significant harvest disruptions in the 2021-2080 period and the long term beyond 2141 (Figure 14).

(CM0t +lb)

4.6 Increase operable land base by adding marginally uneconomic conventional and heli-logging types (+0.6%).

Through the first hundred years of the simulation, adding marginally economic types increased total volume harvested by 620,000 m^3 or 0.5% (Figure 14). The impact was not assessed beyond 2060 as the effect would be lost in the abundance of available timber 2061-2120.

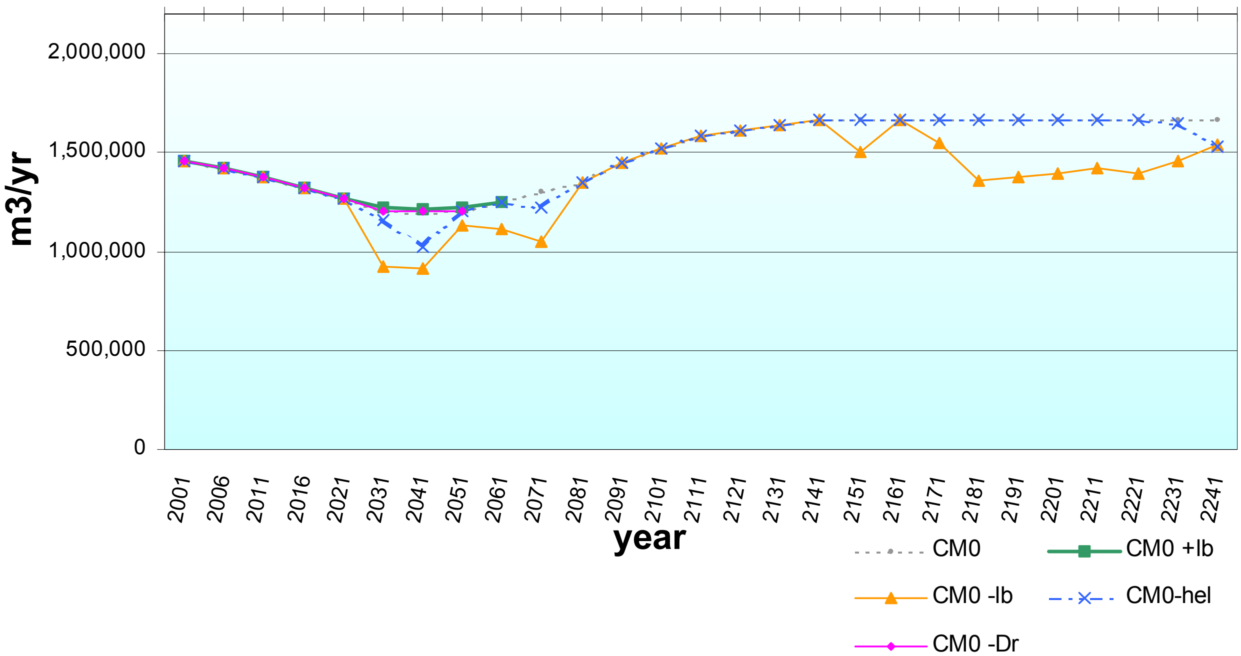

Figure 14 – Land base changes

(CM0 –hel)

4.7 Remove operable helicopter area thus reducing land base (-3.6%).

The result was intermediate between the CM0 and the CM0-lb runs (Figure 14) with a drop in total volume of 2,937,000 m^3 or 2.5% through the first hundred years of the simulation. This run emphasizes the importance of continuing to harvest heli-accessible wood consistently at about 35,000 m^3/year to avoid increased dependence on this harvesting system through the 2021-2080 period.

(CM0 –Dr)

4.8 Remove alder leading stands from THLB (by changing yield curve to zero)

Removing alder stands from the timber harvesting land base had no appreciable effect (Figure 14). Over the first hundred years total harvest volume increased by 46,000 m3. We speculate that retaining alder "reserves" eased adjacency and/or cover constraints to some degree during the 2041-2080 bottleneck period and reduced the area harvested (higher volume stands) allowing slightly more harvesting to occur.

(CM0 +MAImax)

4.9 **Set minimum harvest age to 95% of maximum MAI** for each yield curve.

Harvest levels near 1,500,000 m^3/year can be maintained by reducing minimum harvest age to 95% of maximum mean annual increment (Figure 15). However, average rotation ages drop below 80 years and average stand diameter at harvest to 35cm DBHq. Old forest proportion is reduced and the potential for recruitment is much diminished.

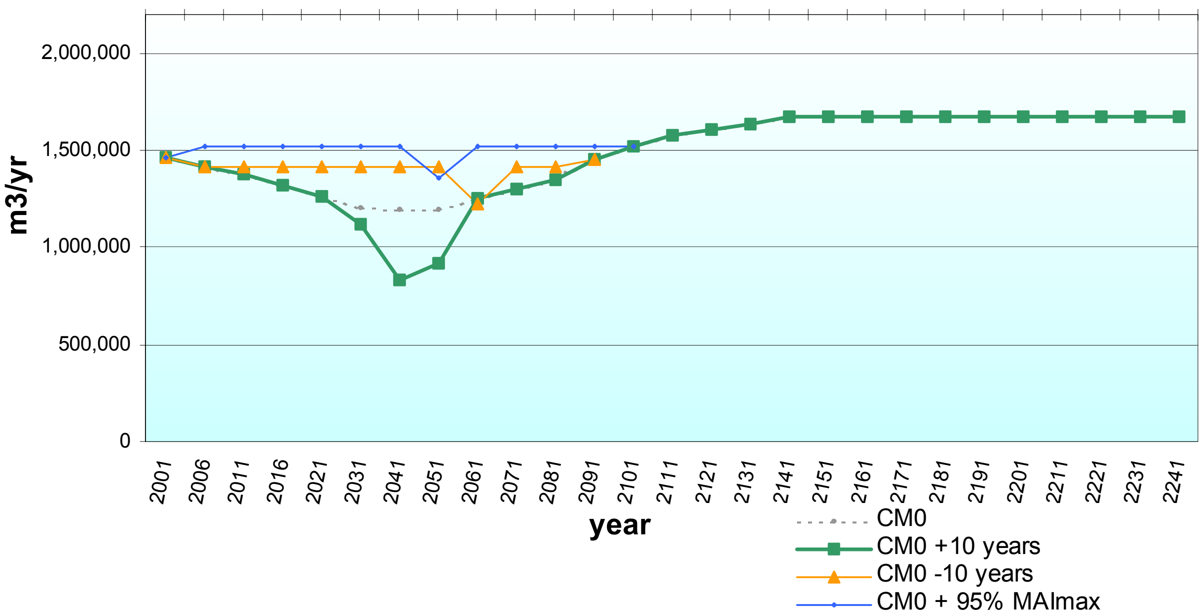

Figure 15 – Minimum harvest ages

(CM0 +45,40,35)

4.10 Increase minimum harvest size to 45, 40, 35cm respectively from 42, 37, 35cm for Productivity Groups 1, 2/3, 4

Attempting to maintain larger product sizes as was assumed in the previous timber supply analysis results in a severe deficit in the period 2031-2060 (Figure 16).

(CM_15 +45,40,35)

4.11 Increase minimum harvest size to 45, 40, 35cm respectively for Productivity Groups 1, 2/3, 4 with 15% per decade stepdown

A 15% per decade stepdown of harvest levels from current AAC (Figure 16) is still insufficient to maintain harvest at the CM0 mid term level of 1,200,000 m^3/yr. Reductions to the mid term level would be needed as well.

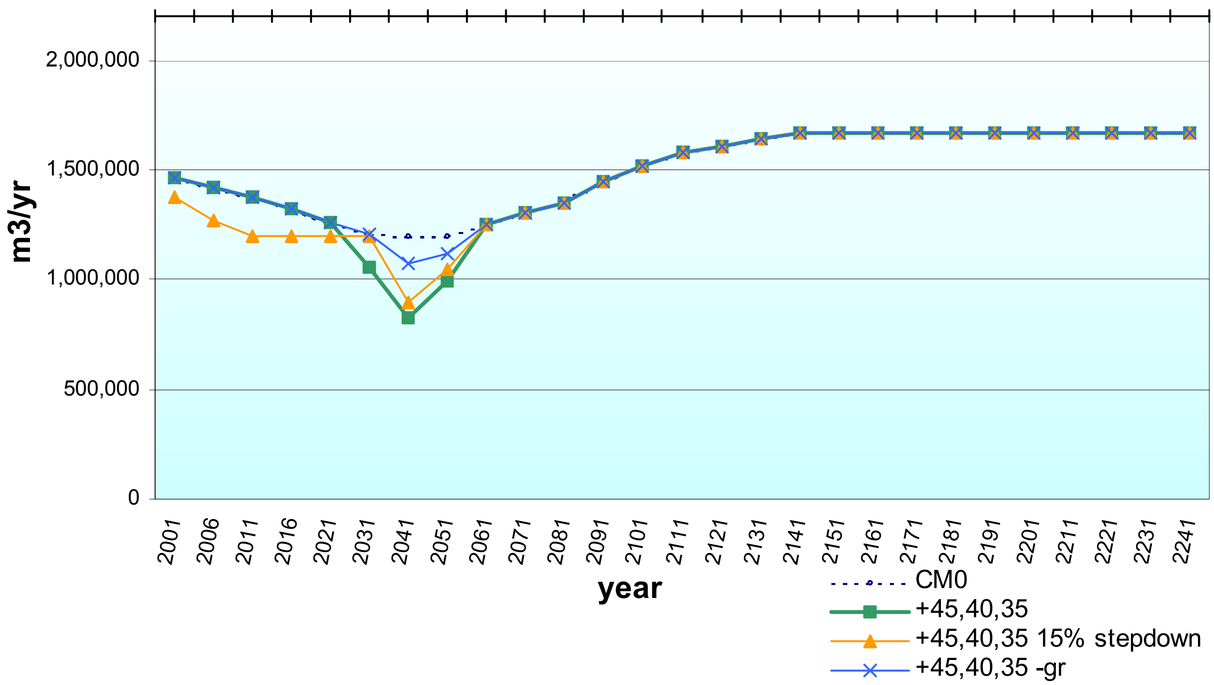

Figure 16 – Minimum Harvest Size

(CM0 +45,40,35 –gr)

4.12 Increase minimum harvest size to 45, 40, 35cm respectively for Productivity Groups 1, 2/3, 4 with no green-up constraint (0m instead of 3m).

This run (Figure 16) suggests that through the short to medium term to 2060, green-up will significantly constrain our ability to increase product size and/or older forest habitat by lengthening rotation ages.

(CM0 +vol); (CM0 –vol)

4.13 Increase yield curve volumes by 10%
Decrease yield curve volumes by 10%

Increasing yields produces an additional 12,048,000 m^3 through the first hundred years and decreasing yields results in a loss of 9,961,000 m^3 (Figure 17).

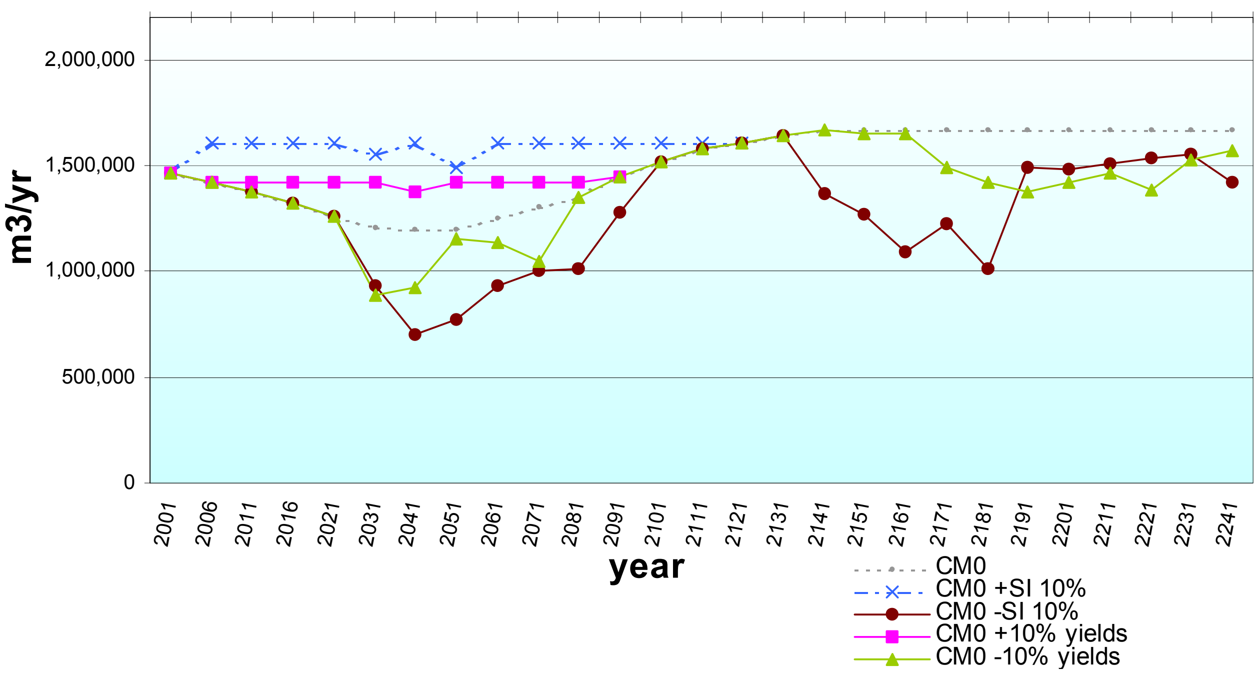

Figure 17 – Yield changes

(CM0 +SI); (CM0 –SI)

4.14 Increase site index 10% for all non-mature yield curves
Decrease site index 10% for all non-mature yield curves

Changes to site index assumptions have similar, but more dramatic, impacts to yield adjustments (Figure 17). Increased site index adds 26,821,000 m^3 and decreased site index causes a loss of 21,521,000 m^3 over the first hundred years of the simulation.

(CM0 +mDr)

4.15 Include alder volumes in mixed (Dr leading) stands as available for harvest

Including the alder component of mixed stands modestly increased harvest by 1,876,000 m^3 (not presented) over the first hundred years of the simulation.

(CM0 +mDr+pDr)

4.16 Include mixed and pure alder volumes as available and minimum harvest at 50 years (Stand Conversion)

This simulation produced a modest increase of 1,907,000 m^3 over the first hundred years (not shown). As conifers make up a portion of the mixed stand volumes, this and the previous run suggest that a small partition of about 10,000 to 15,000 m^3/year for alder components of harvest could be warranted. However, alder harvests are unlikely to be significant in the next five years until second growth harvesting becomes common and local demand develops and/or alder log prices improve. Incidental utilization of alder in the short term should not be discouraged by charging alder volumes against conifer AAC.

(CM0 +st)

4.17 Use percentage seral targets by landscape unit for early seral and mature plus old rather than basic 45/45/10 target for old seral

Early seral restrictions cause a halt of harvesting in the short term (Figure 18).

(CM0 +ost lowdd)

4.18 biodiversity emphasis option old seral targets by landscape unit with guidebook draw down in those units designated low emphasis instead of 45/45/10 method

Adopting old seral targets specific to each landscape unit according to biodiversity guidebook procedures (Figure 18) results in a deficit of 71,000 m^3/year in the 2041-2050 period or about 707,000 m^3 (0.6%) over the first hundred years of the simulation.

(CM0 +gr2)

4.19 Increase green-up heights by 2m

Increasing green-up heights from 3m to 5m (Figure 19) has a significant negative impact during the 2041-2050 transition period resulting in a shortfall of 1,541,990 m^3 or 154,199 m^3/year.

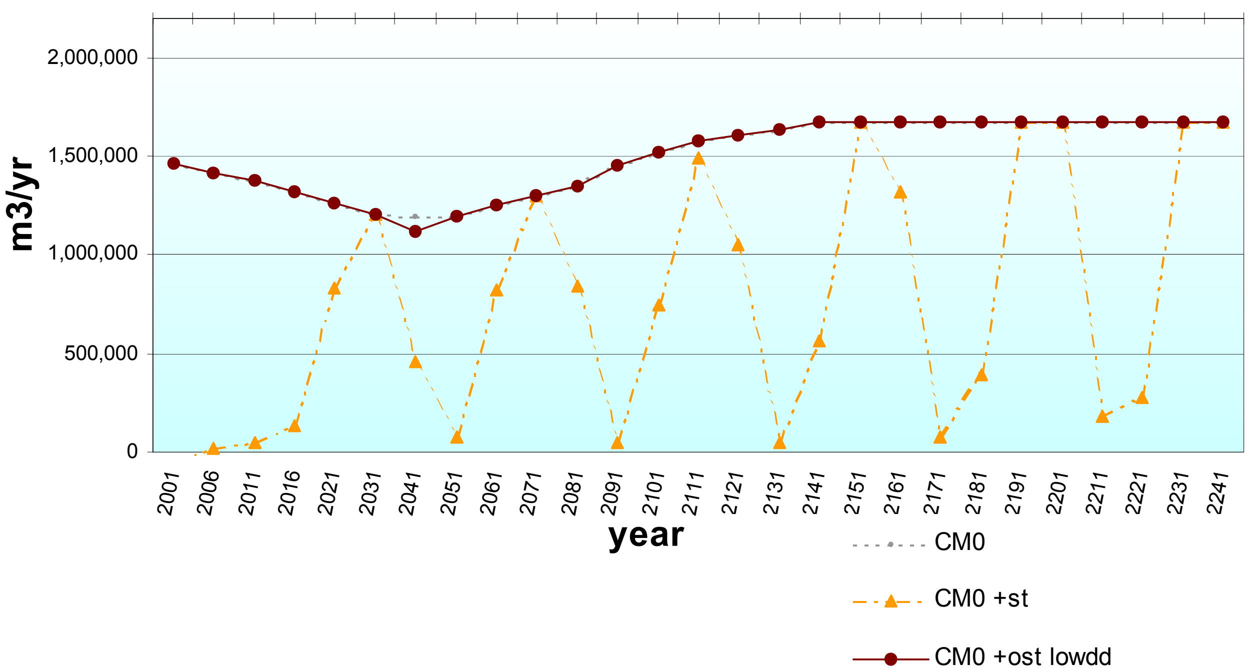

Figure 18 – Seral targets

(CM0t –gr2

4.20 Decrease green-up heights by 2m

Reducing green-up heights to 1 m from 3m (Figure 19) has a slight positive impact on timber supply amounting to 180,590 m^3 in total or an average 6,020 m^3/year during the 2031-2060 period. The implementation of this policy in enhanced zones is unlikely to have a significant positive impact on timber supply.

(CM0t –gr)

4.21 Remove green-up constraint by reducing green-up heights to 0m

This run (Figure 19) illustrates the effect that green-up policy has on mid-term timber supply. With the constraint removed, mid-term harvest levels rise by a total 8,773,743 m^3 or an average 116,983 m^3/yr through the period 2016-2090. This suggests that partial or retention harvesting designed to remove adjacency concerns may have a similar impact.

(CM0t +120)

4.22 Allow 120 ha openings.

Increasing maximum opening size to 120 ha has a similar but less pronounced impact (Figure 19) than removing the green-up constraint but is more powerful than reducing green-up height to 1m (-gr2)

(CM0 +mVQO)

4.23 Change denudation limit to middle of recommended range for each VQO

Reducing the cover constraint from the high end to the middle of the recommended range for each of the visual quality objective classes results in a significant negative impact during the 2041-2050 transition period resulting in a shortfall of 1,165,370 m^3 or 116,500 m^3/year (not shown).

(CM0 -cb)

4.24 Current Management, without compartment balancing

Removal of the compartment balancing constraint had a slight positive impact during the 2041 through 2060 period averaging 5,541 m^3/year (not shown).

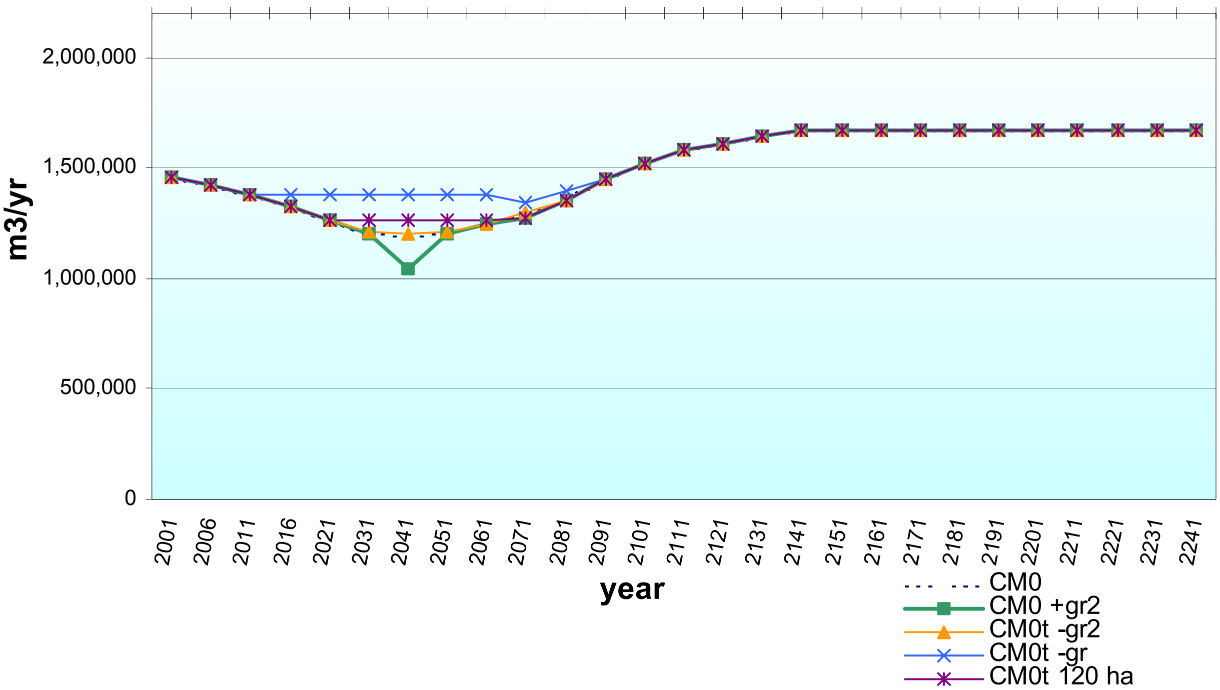

Figure 19 – Adjacent green-up effects

(CM0 +rq)

4.25 Reverse queuing priority order (oldest first)

Changing the queuing order had a negligible effect, increasing harvest by 5,042 m^3/year (deficit 1,097 m^3/year) in the period of 2041-2050 only (not shown).

(CM0 +CT)

4.26 Commercial thinning.

Commercial thinning[19] in the next few decades exacerbates the expected shortfall of timber in the 2041-2060 period by reducing volumes available for harvest during the critical period. Over the first hundred years of the simulation 3,540,000 m^3 are lost (not shown). However, limited trials to further evaluate and improve techniques for commercial thinning in hemlock may be warranted. Ample growing stock becomes available for regular harvest after 2061 and could be the basis of a commercial thinning program through the 2031-2060 period that could supplement harvest levels through the supply dip. Preliminary simulations of an aggressive mid-rotation thinning program starting in 2031 suggest that harvest levels through the transition period could be raised by 50,000 m^3/year or more. If operational problems related to commercial thinning can be worked out and a better idea of which stand types are feasible for thinning systems, future analyses could look at this possibility in detail.

[19] for simplicity, thinning was coarsely modelled as a mid-rotation removal of 30% of basal area with no growth response thereafter.

5.0 Effect of Constraints on Timber Potential

(CM0t -con = CM_un)	**Unrestricted harvesting** of operable area
(CM_un -rip)	Invoke **riparian buffer restrictions**
(CM_un -bio)	Invoke **biodiversity restrictions** (WTP, seral targets)
(CM_un -gr)	Invoke **green-up restrictions**

As mentioned previously, policy constraints to ensure multi-resource sustainability developed over the past two decades have to date and will continue to significantly impact timber potential. This is demonstrated in Figure 20.

The constraints imposed by current policies are primary drivers affecting timber supply through the transition from an unmanaged to managed landscape (Figure 14, Figure 18 and Figure 19). The constraints are of no concern beyond 2081 when projected harvest levels approach and surpass current AAC. As improved yields from the managed landscape kick in, operational flexibility improves, cut levels are easily attained, and all constraints are satisfied.

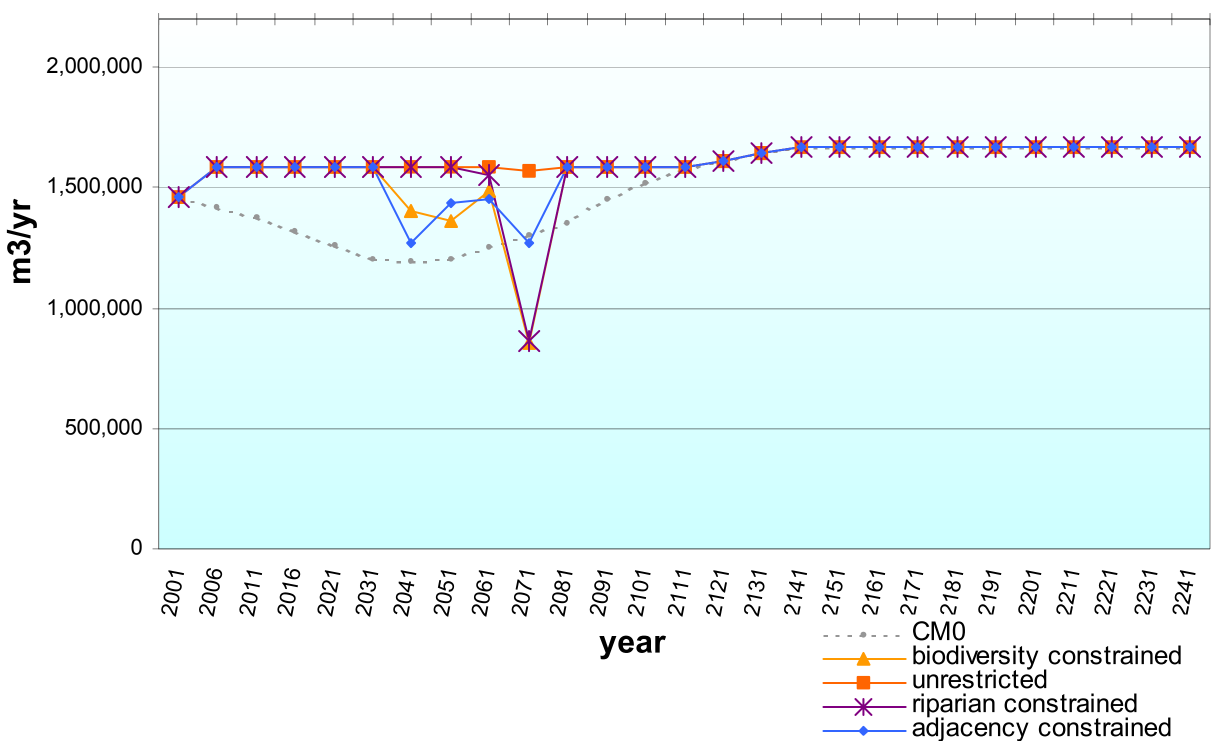

Figure 20 – Timber potential and impact of specific constraints

6.0 Summary of Uncertainties

The Licensee believes this analysis to be a fair, if somewhat conservative, estimation of future timber supply. The following summarizes known uncertainties in this analysis of current management in order of our perception of decreasing relative importance.

+/- A Vegetation Resource Inventory is underway for Tree Farm Licence 6 and will provide revised estimates of old growth volumes.

\+ Current operational adjustment factors are estimated to be conservative.

\+ A Vegetation Resource Inventory is underway for Tree Farm Licence 6 and preliminary field data suggests that growth functions for maturing stands may be underestimated.

\+ Commercial thinning through the dip may allow increased harvest levels, but given that successful implementation remains uncertain an increase in current harvest levels cannot be justified at this time.

\+ Tree improvement gains are likely to be greater than forecasted.

+(/-) Minimum harvest sizes and rotations are conservative relative to some other coastal timber supply analyses and could be reduced to maximize volume production. Undesirable impacts on harvesting costs, log values, and old forest proportions could result.

+(/-) This analysis is based on a "simulation" of timber supply as opposed to an "optimization". The sequence of harvesting, although affected by various queuing methods and constraints, inherently follows the same sequence based on the order of stands in the inventory file. An optimization model would perform multiple runs with sequencing, or other, changes imposed and select the run with the best harvest output. For example, removing all harvest queuing results in a 11,453 m^3/year increase in harvest in the 2041-2050 period, suggesting that queue manipulation would have an impact. We cannot speculate on the increase in harvest level that might be extracted using an optimization technique but it may be of significance. On the other hand, neither is operational harvest planning explicitly optimized – but it will likely become more so as computing tools evolve.

+(/-) Long term growth responses to cedar-salal fertilization cannot yet be proven, but estimates used are conservative relative to research projections of growth response and operationally treated area has exceeded expectations.

+/- Alder succession to conifer stands may take longer than projected. Alder utilization programs are likely to develop and supplement AACs. Conversely, utilization of alder and conversion programs are likely to accelerate conifer production over the yields modelled.

-(/+) Increased partial cutting may reduce stand yields by increasing disease, damage and decay, or shading of regenerating trees. In some situations yield might improve by promoting advance regeneration and reducing time to harvest. Stand level retention will reduce volume/ha harvested in proportion to the amount of volume retained but may relieve adjacency constraints and produce a positive forest level impact. Yield forecasts for these very complex stands remains problematic, particularly for coastal forests where little data exists and models have not been developed or calibrated for partial harvesting.

7.0 Recommendations

7.1 Short Term Timber Supply

The Licensee recommends that for the period of 2001 through 2005 a basic allowable annual cut of 1,459,400 m^3/year, less the estimated annual non-recoverable loss of 7,000 m^3/year, or 1,452, 400 m^3/year be adopted as it meets, in a precautionary manner, sustainability objectives.

7.2 Medium Term Timber Supply

Harvest levels are expected to decline over the coming decades to about 1,200,000 m^3/year by 2041. Harvest levels through this transition period are highly sensitive to policy constraints. Existing and new government policies should be carefully evaluated for their potential negative and positive impacts on timber supply relative to their expected non-timber resource benefits.

7.3 Tactical Interventions

No tactical interventions that can be implemented immediately to improve near term timber supply were readily apparent in this analysis.

Preparations for and further evaluation of a commercial thinning program commencing in thirty to forty years are warranted. To facilitate this, a modest commercial thinning partition of 3,000 m^3/year in excess of the recommended harvest level is suggested to encourage development of locally-adapted thinning methods appropriate to hemlock stands without the disincentive of reducing conventional AAC. Use of this partition should be optional to permit curtailment of thinning during periods of low pulp log prices. A larger, operational program is discouraged in the short term as it would likely exacerbate the coming shortfall period.

To encourage development of innovative harvesting techniques and facilitate sustainable harvesting of marginally economic types (Ohe; Oce) when log prices are high (or costs low), create a “bonus” partition of 14,500 m^3/year in excess of the recommended harvest level that could be accumulated until favourable harvesting conditions exist.

7.4 Suggested Data Refinements for the Next Analysis

- Incorporate Vegetation Resource Inventory data.
- Incorporate partial cutting growth models as they become available.
- Analyse record of harvesting in helicopter accessible (Oh) polygons.
- Analyse record of harvesting in marginally economic (Ohe, Oce) polygons for possible addition of these types to the timber harvesting land base.
- Improve mapping of S3 streams and confirm the riparian net down estimation procedure for smaller streams.

Appendix IV

Silviculture Project History

TFL 6 Silviculture Project History

Year	Denuded (ha)	Planted (ha)	No. Trees Planted	Juvenile Spacing (ha)	Brushing (ha)	Prescribed Burning (ha)	Mechanical Site Prep. (ha)	Fertilization (ha)	Pruning (ha)	Commercial Thinning (ha)
Pre 1965	13793.2	797.7	701400	0.0	174.2	484.9	51.4	0.0	0.0	0.0
1965	1517.5	497.2	361500	20.3	0.0	1514.8	0.0	0.0	0.0	0.0
1966	1756.5	430.9	325300	30.4	0.0	1382.0	0.0	0.0	0.0	0.0
1967	1744.8	674.9	422950	16.2	12.2	1654.2	57.0	0.0	0.0	0.0
1968	1780.8	620.0	444900	0.2	54.6	1363.8	0.0	0.0	0.0	0.0
1969	1742.7	1126.1	989650	0.8	0.0	1195.6	0.0	0.0	0.0	0.0
1970	1469.4	1155.0	751700	17.1	0.0	1177.9	14.6	0.0	0.0	0.0
1971	1751.7	717.8	529350	83.0	30.0	593.2	5.7	0.0	0.0	0.0
1972	1310.2	1066.2	912650	136.1	0.0	1175.1	24.7	0.0	0.0	0.0
1973	1660.7	569.6	600500	105.7	6.1	790.0	8.5	0.0	0.0	0.0
1974	1496.5	459.1	459350	37.9	112.6	646.5	5.7	0.0	0.0	0.0
1975	944.5	911.5	777700	2.2	276.2	10.9	4.5	0.0	0.0	0.0
1976	1875.1	983.6	777050	557.7	390.8	1439.6	0.0	0.0	0.0	0.0
1977	2137.4	1047.8	553900	729.3	726.6	1423.8	0.0	3.5	0.0	0.0
1978	1977.2	692.3	493950	720.4	72.9	23.1	0.0	0.0	0.0	0.0
1979	1879.8	837.4	662850	566.5	127.9	1090.0	1.6	0.0	0.0	0.0
1980	1686.4	566.8	491500	462.0	342.3	374.0	0.0	0.0	0.0	0.0
1981	1378.4	1213.7	1047600	333.4	279.8	231.1	0.0	0.0	0.0	40.9
1982	1659.2	1256.7	1198300	172.2	0.0	335.4	0.0	0.0	0.0	0.0
1983	2313.3	936.6	888000	0.0	373.7	571.7	45.3	0.0	0.0	0.0
1984	2315.2	960.2	882400	132.7	1768.9	421.0	24.0	0.0	26.9	0.0
1985	2234.1	648.2	701800	485.6	712.1	1050.8	118.4	0.0	17.3	0.0
1986	1772.1	1301.1	1347100	303.3	1099.9	1526.2	59.2	200.0	9.0	0.0
1987	2284.8	2258.4	2256650	419.6	2668.4	967.4	115.6	0.0	9.1	0.0
1988	1849.5	1866.9	1844050	65.4	899.7	1008.9	11.1	0.0	2.0	0.0
1989	1173.4	1118.8	1169250	469.9	2053.4	874.7	64.7	838.0	5.5	0.0
1990	1394.3	1382.0	1405700	548.0	1511.7	617.0	35.2	795.0	8.4	0.0
1991	1642.6	1801.9	1491100	827.1	1656.2	347.5	50.2	0.0	63.8	0.0
1992	1467.2	1522.5	1550900	377.1	1429.2	491.7	14.2	0.0	34.9	0.0
1993	1881.7	1524.1	1574650	97.5	1357.5	499.7	86.9	0.0	42.4	0.0
1994	1666.8	1697.2	1712150	829.1	843.9	424.2	92.4	0.0	178.5	0.0
1995	1661.7	2060.6	2003400	802.6	1487.1	76.6	47.7	516.1	350.4	7.5
1996	1697.5	2065.8	2110950	549.7	1475.7	267.8	87.8	1936.2	659.9	0.0
1997	1404.2	1789.4	1944750	539.5	1066.3	175.3	24.3	2323.3	916.5	0.0
1998	1104.7	1333.7	1473600	76.4	1074.0	0.0	25.6	0.0	256.6	7.1
1999	1536.3	983.2	1088000	317.5	759.7	0.0	25.4	3415.9	185.3	0.0
TOTAL	72961.4	40874.9	37946550	10832.4	24843.6	26226.4	1101.7	10028.0	2766.5	55.5

Appendix V

Silviculture Prescription Sample

Silvicuiture Prescription Agreement

Western Forest Products Ltd.

Port McNeill Operation

X Original
___ Amendment #

Printed Date: Sep/25/2000 Page: 1

File ID: 595-900+-6/900+ Licence No.: TFL 6 Timbermark: 6/900+ CP: 900+ Block: 595

A. TENURE IDENTIFICATION

Region	District	TSA	TSB	Licence No.	C/P	Timbermark (s)	Cutting Permit #	Block
VAN	PM			TFL 6	C	6/900+	900+	595

Area Under Tenure	Location	Mapsheet/Opening No.	Licensee
46.4 (ha)	Port McNeill	92L.053 /	Western Forest Products Ltd.

The procedures of the operational planning regulation have been followed for any assessments required for providing BEC and soil disturbance information referred to in the OPR section 39(3)(a).

B. AREA SUMMARY

AREA OF NO PLANNED REFORESTATION (ha) (NPR)

Perm. Access	Rock	Water	Swamp	Other NP	NC > 4 HA.	Res. No Mods	IMM	WTP	Other SU 3	Total NPR Area
2.9						4.2		3.1	0.8	11.0

NET AREA TO BE REFORESTED INCLUDING RESERVES WITH MODIFICATIONS (ha)

SU	SU Area Description	Net Area To Be Reforested
1	MoF Site Series: 01//03. The SU is a mosaic of S1ha (90%) and S2F (10%) ecosystems. The S2F portion is mainly concentrated along the eastern block boundary. Soils are well-drained silt loams with a restrictive compact till layer at 100+ cm. The humus form is a mor approximately 8-15 cm deep. The coarse fragment content is moderate (35-70%) and rooting depth was found to be 40-45 cm. The overstory is dominated by Douglas-fir and western hemlock with a smaller component of western redcedar and amabilis fir. The understory is very open and the moss/herb/shrub layer includes: step moss (40%), oregon beaked moss (20%), lanky moss (5%), red huckleberry (5%), salal (5%), and spiny wood fern (+). Site index is estimated to be 28 and the competing vegetation potential is L.	31.5
2	MoF Site Series: 03//01. The SU is a mosaic of S2F (80%) and S1ha (20%) ecosystems. Soils are moderately well-drained shallow folisols over bedrock. Rooting depth ranges from 0-35 cm and there is a 5-10 cm deep mor humus. The coarse fragment content is low (<35%). The overstory is dominated by slow growing western redcedar, Douglas-fir, and western hemlock. Understory species include: salal (15%), swordfern (+), bracken fern (+), stepmoss (30%), and vaccinium (+). Site index is estimated to be 14 and the competing vegetation potential is M/4.	2.0
3	This SU includes the riparian management zone (RMZ) of stream ID#1. Trees will be removed to create a feathered edge in an attempt to mitigate the impact of potential future windthrow. This SU has the same ecology, soil, topography and vegetation characteristics as SU 1.	1.9

TOTAL NET AREA TO BE REFORESTED : 35.4

TOTAL AREA UNDER PRESCRIPTION : 46.4

C. OBJECTIVES

C.1 MANAGEMENT OBJECTIVE

MANAGEMENT OBJECTIVES STATED IN THE FDP OR HLP(S):

D.R. systems inc. Phoenix Professional 3.4.2 Silviculture Prescription: 3.5.0

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File ID: 595-900+-6/900+ Licence No.: TFL 6 Timbermark: 6/900+ CP: 900+ Block: 595

To harvest timber while maintaining soil productivity and water quality. Re-establish a new series of forest stands with ecologically suitable species. Ensure that fishery and wildlife values are maintained. Operate within the existing recreational and visual guidelines. Enhance biodiversity within opening where opportunities exist and these are operationally feasible. The anticipated rotation length is 85-125 years to produce a mixed stand of conifers of average stand diameter of 35-40 cm which will be utilized for sawlog and pulp. No unusual forest health conditions exist. Maintain the integrity of First Nations cultural and traditional use areas.

C.2 CONDITIONS THAT MUST EXIST AFTER HARVEST OR TREATMENT TO ACCOMMODATE KNOWN FOREST RESOURCES

C.2a WILDLIFE

Deer and bear are common in the area. No wildlife signs were noted during fieldwork.

Prescribed site conditions:
The Wildlife Tree Patch, riparian reserve, and surrounding timber will ensure that wildlife habitat is maintained within proximity of the block.

C.2c FISHERIES

Some streams adjacent to the block have fish value.

Prescribed site conditions:
The riparian management section will address the prescribed site conditions to be met after harvest.

C.2e RECREATION

There are no known recreation resources in the cutblock. The Marble River Provincial Park is located 800 m south of the cutblock.

Prescribed site conditions:
Harvesting the block will have no negative impact on the Provincial Park resources. No special site conditions required.

C.2f BIOLOGICAL DIVERSITY

Biodiversity issues in this block are addressed by the retention of wildlife tree patches. The WTPs will provide for tree and understorey plant species diversity, structural diversity, and wildlife habitat protection. The WTPs will also provide for recruitment of standing dead wildlife trees (snags) and for recruitment of coarse woody debris. As per Biodiversity Guidebook Guidelines, distances between patches or other suitable leave areas outside the block do not exceed 500 m. This block is within the Marble Landscape Unit and has a WTP target of 15%.

Prescribed site conditions:
A 2.7 ha Wildlife Tree Patch (WTP) was established within the 4.2 ha riparian reserve zone (RRZ). The remaining 1.5 ha of RRZ does not meet the criteria of a WTP and therefore does not contribute to the WTP target. Additionally, a 3.1 ha WTP was established outside of the riparian reserve adjacent to block boundaries. These WTPs represent 15.1% of the total area to be harvested. These WTPs will provide wildlife habitat and maintain old-growth features at the stand level. WTPs will be retained at least until other suitable trees can offer equivalent replacement habitat and structural diversity. Falling of hazard trees in Wildlife Tree Patches and reserves is permitted.

C.2g VISUAL RESOURCE MANAGEMENT

The cutblock is not within a known scenic area. No visual impact assessment required.

Prescribed site conditions:
No special site conditions required.

C.2h CULTURAL HERITAGE

During field surveys (SP, cutblock layout) no CMTs or other evidence of traditional use were noted. If CMTs or other related artifacts are located once harvesting commences, operations will cease in the vicinity of the artifact. First Nations will be consulted and appropriate prescriptions developed.

D. ECOLOGICAL INFORMATION AND SITE CHARACTERISTICS

D.1 ECOLOGY AND CRITICAL SITE CONDITIONS

SU	UNIT	AREA	BEC	ELEVATION (m)			SLOPE (%)			SITE SERIES		Site Type
				Low	High	Avg	Min	Max	Avg	Dominant	Related	
1	1	31.5	CWH vm 1	70	190	100	0	50	30	S1ha	//S2F	
2	2	2.0	CWH vm 1	90	125	110	0	50	30	S2F	//S1ha	
3	3	1.9	CWH vm 1	60	100	70	0	40	20	S1ha	//S2F	
4	4	0.8	CWH vm 1	75	125	75	0	40	5	S2F	/S11	

E. MANAGEMENT STRATEGIES

D.R. systems inc. Phoenix Professional 3.4.2 Silviculture Prescription: 3.5.0

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File ID: 595-900+-6/900+ Licence No.: TFL 6 Timbermark: 6/900+ CP: 900+ Block: 595

E.1 RIPARIAN MANAGEMENT STRATEGIES

RIPARIAN RESERVE ZONE

Riparian/ Lake I.D.	Riparian/ Lake Class	Harvesting (Y/N/NA)	SU X Ref.	RRZ Width (m)	Description of the Purpose and Extent of Removal or Modification of Trees and Any Related Forest Practices in Riparian Reserve Zone(s).
1	S2	No		30	
2 (R1)	S3	No		20	
11 (R1)	S3	No		20	
9 (R1)	S4	N/A		0	
2 (R2), 3, 3a, 3b, 5, 8	S6	N/A		0	
9 (R2)	S6	N/A		0	
10	S6	N/A		0	
11 (R2)	S6	N/A		0	
6, 13	W3	N/A		0	

D.R. systems inc. Phoenix Professional 3.4.2 Silviculture Prescription: 3.5.0

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File ID: 595-900+-6/900+ Licence No.: TFL 6 Timbermark: 6/900+ CP: 900+ Block: 595

RIPARIAN MANAGEMENT ZONE

Riparian/ Lake I.D.	Riparian/ Lake Class	Harvesting (Y/N/NA)	SU X Ref.	RMZ Width (m)	Management Strategies for Riparian or Lakeshore Management Areas including: Protecting stream banks (if there is no RRZ), maintaining shade, and debris management if felling and/or yarding across streams. Include either the residual basal area or density for RMZ(s) and LMZ(s).
1	S2	Yes		20	From FC 9 east to FC 1: Fall and yard away. Retain saplings and trees less than or equal to 12 inches in diameter. Residual basal area=30%. Leave trees that lean heavily towards the RRZ, where possible. Windproof select trees with aerial pruning. No special debris management required. From FC 9 south to FC 11: Clearcut RMZ. Residual density = 0 sph. Fall and yard away. Retain saplings where possible. No special debris management required.
2 (R1)	S3	Yes		20	Clearcut RMZ. Residual density = 0 sph. Fall and yard away. Retain saplings where possible. No special debris management required.
11 (R1)	S3	No		20	Stream and RMZ are located outside of cutblock boundaries.
9 (R1)	S4	Yes		30	Stream not dependent on woody debris or streamside trees to maintain channel and/or streambank stability. Stream reach falls within Stream ID# 1 RMA. For the portion of the RMZ that falls within the cutblock: Fall and yard away. Retain saplings and trees less than or equal to 12 inches in diameter. Residual basal area=30%. Leave trees that lean heavily towards stream #1, where possible. Windproof select trees with aerial pruning. Machine clean accumulations of introduced in-stream logging debris. The remainder of the RMZ outside of the cutblock will be fully retained.
2 (R2), 3, 3a, 3b, 5, 8	S6	Yes		20	Stream not dependent on woody debris or streamside trees to maintain channel and/or streambank stability. Clearcut RMZ. Residual density = 0 sph. Fall and yard across is acceptable. No special debris management required.
9 (R2)	S6	Yes		20	Stream not dependent on woody debris or streamside trees to maintain channel and/or streambank stability. A portion of this creek falls within the RMZ of stream #1. In this portion: Fall and yard across is acceptable. Retain saplings and trees less than or equal to 12 inches in diameter. Residual basal area=30%. Leave trees that lean heavily towards stream #1, where possible. Windproof select trees with aerial pruning. Machine clean accumulations of introduced in-stream logging debris. For the remainder of the stream outside of the RMA of stream #1: Clearcut RMZ. Residual density = 0 sph. Fall and yard across is acceptable. Machine clean accumulations of introduced in-stream logging debris.
10	S6	Yes		20	Stream runs adjacent to the cutblock. Stream not dependent on woody debris or streamside trees to maintain channel and/or streambank stability. Clearcut RMZ. Residual density = 0 sph. Fall and yard across is acceptable. No special debris management required.
11 (R2)	S6	No		20	Stream and RMZ are located outside of cutblock boundaries.
6, 13	W3	Yes		30	This is a wetand containing a class L3 lake. Clearcut portion of the RMZ within the cutblock. Residual density=0. Fall and yard away. Retain saplings where possible. No special debris management required.

RIPARIAN COMMENTS

Stream ID# 4 and 7 are non-classified drainages. Unidentified minor watercourses and non-classified drainages are intermittent or ephemeral in nature and do not possess the ability to transport significant debris. Normal harvesting practices are prescribed (ie falling of all trees within the RMZ and cross stream yarding). This prescription presents a low risk to possible instream or downstream impacts. The trees within the RMZs are not needed to meet biodiversity requirements.

A fisheries sensitive zone (FSZ) is located adjacent to stream ID#1. This FSZ falls entirely within the RRZ for stream ID#1. No special actions required.

A RIPARIAN ASSESSMENT HAS BEEN COMPLETED IN ACCORDANCE WITH THE PROCEDURES OF THE OPERATIONAL PLANNING REGULATION AND THE PRESCRIPTION IS CONSISTENT WITH THE RESULTS AND RECOMMENDATIONS OF THE ASSESSMENT.

D.R. systems inc. Phoenix Professional 3.4.2 Silviculture Prescription: 3.5.0

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File ID: 595-900+-6/900+ Licence No.: TFL 6 Timbermark: 6/900+ CP: 900+ Block: 595

E.2 GULLY MANAGEMENT STRATEGIES (COAST)

Management Strategies with Respect to: Debris Management, Protecting Gully Banks, Minimizing Understory Damage, Sediment and Debris Transport Potential, and Felling and Yarding Across Gullies (if applicable).

There are no gullies to be harvested within this block.

E.4 COARSE WOODY DEBRIS MANAGEMENT STRATEGIES

Management strategies to Accommodate CWD Objectives, including Volume and Range of Piece Size, if any.

Logs outside the utilization standards will be left on site. The retention of WTP will provide recruitment of coarse woody debris.

E.5 MANAGEMENT STRATEGIES FOR ARCHAEOLOGICAL SITES

Management Strategies to Manage and Conserve Archaeological Sites

An archaeological impact assessment was not required. During field surveys (SP, cutblock layout) no CMTs or other evidence of traditional use were noted. If CMTs or other related artifacts are located once harvesting commences, operations will cease in the vicinity of the artifact. First Nations will be consulted and appropriate prescriptions developed.

E.6 VEGETATION MANAGEMENT STRATEGIES

Livestock to be used for vegetation management: NO

F. SOIL CONSERVATION

F.1 SITE DISTURBANCE

SU(s)	HAZARD RATING (if logging methods other than cable or aerial are proposed)			SOIL CHARACTERISTICS (if temporary access structures are proposed)	
	Soil Compaction	Soil Displacement	Surface Soil Erosion	Depth To Unfavorable Subsoil (cm - cm)	Type Of Unfavorable Subsoil
1	H	L	H	100+ cm	till/bedrock
2	H	M	H	35 cm	bedrock
3	H	L	H	100+ cm	till

SOIL DISTURBANCE LIMITS

SU(s)	Maximum Allowable Soil Disturbance Within the Net Area To Reforest (%)	Maximum Extent Soil Disturbance Limits May Be Temporarily Exceeded to Construct Temporary Access Structures (%)
1	5.0	5.0
2	5.0	5.0
3	5.0	5.0

Maximum Proportion of Total Areas Under The Prescription Allowed for Permanent Access: 6.3 %

F.2 REHABILITATION TIME FOR TEMPORARY ACCESS STRUCTURES

Maximum Allowable Time To Complete Rehabilitation (measured from completion of harvest): 2.0 yrs.

D.R. systems inc. Phoenix Professional 3.4.2 Silviculture Prescription: 3.5.0

Printed Date: Sep/25/2000 Page: 6

File ID: 595-900+-6/900+ Licence No.: TFL 6 Timbermark: 6/900+ CP: 900+ Block: 595

F.3 MANAGEMENT STRATEGIES FOR TEMPORARY ACCESS STRUCTURES

This prescription is not in conflict with the prohibition, under Section 7(4), 8(3) and 8(4) of the Timber Harvesting Practices Regulation, against constructing excavated or bladed trails.

SU(s)	General Location (also refer to map)	Sediment Delivery Risk (*in community watershed only*)	Max Allowable Height of Cutbanks (m)	Average Height of Cutbanks (m)	Equipment To Be Used (if other than Excavator)
1,2,3	As per map.		2.0	1.0	

G. SILVICULTURAL SYSTEMS

G.1 SILVICULTURAL SYSTEMS

SU(s)	System / Variant / Phase	Group Selection Opening Size Criteria			Residual Stand Structure (other than Single Tree Selection)	
		Min. (ha)	Max. (ha)	Avg. (ha)	BA (m2/ha)	Density (stems/ha)
1-2	Clearcut with reserves					
3	Single Tree Selection (Feathered Edge)					

SU(s)	COMMENTS: Stand Structure and Site Condition	COMMENTS: Leave Tree Species and Functions
1-2	An unlogged Wildlife Tree Patch (WTP) and riparian reserve will be left adjacent to the block (refer to map). There is no harvesting planned in SU 4. Falling of hazard trees in SU 4 is permitted.	The WTP and riparian reserve will contain Hw, Fdc, Cw and Ba. The function of the leave trees in the WTP is to provide structural diversity, mature stand attributes, recruitment of coarse woody debris, a source of live and dead wildlife trees and recruitment of dead wildlife trees (snags).
3	The percentage of Basal Area (BA) remaining after harvesting will vary throughout this SU, but will be, on average, 30% or more of the original BA of this stand.	Leave species will potentially include: Cw, Hw, Ba, Ss, and Fdc. Trees will be selected for retention based on their live crown structure, lean, and position in the stratum. Leave trees will have smaller, sparse crowns that are less prone to windthrow. Retained trees with large crowns will be considered for aerial pruning. The function of the retained trees is to form a feathered edge to disperse destructive winds at the exposed edge of the RMZ, in an attempt of mitigate windthrow in and along the RRZ adjacent to stream #1.

H. STOCKING REQUIREMENTS

FREE GROWING STOCKING REQUIREMENTS FOR SILVICULTURAL SYSTEMS (OTHER THAN SINGLE TREE SELECTION)

SU	Preferred Species/ Free Growing Ht. (cm)	Acceptable Species/ Free Growing Ht. (cm)	Post Spacing Density (sph)		Maximum Coniferous (sph)	Regen Date (years)	Free Growing (years)	
			Min	Max			Early	Late
1	Hw/300 Fdc/300 Cw/150 Ba/175	Ss/300 Plc/125	765	1000	10000	6	9	12
2	Hw/150 Cw/100 Fdc/150	Plc/125	765	1000	10000	6	10	14
3	Hw/300 Fdc/300 Cw/150 Ba/175	Ss/300 Plc/125	765	1000	10000	6	9	12

D.R. systems inc. Phoenix Professional 3.4.2 Silviculture Prescription: 3.5.0

Printed Date: Sep/25/2000 Page: 7

File ID: 595-900+-6/900+ Licence No.: TFL 6 Timbermark: 6/900+ CP: 900+ Block: 595

SU	Target Stocking (TSS) (wsph)	Minimum Stocking (MSSpa) (wsph)	Minimum Preferred (MSSp) (wsph)	Minimum Horiz. Distance (m)	Minimum Pruning Height (m)	Residual Stand Structure BA (m2/ha)	Residual Stand Structure Density (sph)	Height Relative To Competition
1	1000	700	490	2.0				150 %
2	900	600	420	2.0				150 %
3	1000	700	490	2.0				150 %

GENERAL STOCKING STANDARDS COMMENTS

To ensure Free Growing standards are achieved along roadsides, a 1.0 m inter-tree spacing will apply within 10 m of roadsides.

SU 1 and 3 are mosaics of S1ha (90%) and S2F (10%) ecosystems. In the S2F dominant portions of the stratum Ba and Ss are neither preferred nor acceptable species due to expected salal competition. In the S1ha dominant portions of the stratum Plc is neither a preferred nor acceptable species.

SU 2 is a mosaic of S2F (80%) and S1ha (20%) ecosystems. In the S1ha dominant portions of the stratum Plc is neither a preferred nor acceptable species.

In SU 3, openings created by harvesting that are greater than and equal to 0.01 ha will be treated and managed in conjunction with SU 1. Openings less than 0.01 ha within SU 3 will be considered excluded from the stocking standards for this ecosystem and deemed fully stocked with mature trees.

There is no harvest planned in SU 4 at this time and therefore there are no reforestation standards specified.

I. WINDTHROW MANAGEMENT STRATEGIES

The cutblock boundaries have been positioned in their current location in an effort to minimize windthrow risk. Historic windfall has occurred with west winds notably from FC 9 to FC 11 along the old cutblock. The eastern cutblock boundary travels along a rocky ridge. The trees on top of this rock ridge tend to be smaller and wind adapted due to their exposure to wind. This boundary is expected to remain windfirm. The riparian reserves and WTP are in a lower, more sheltered part of the block. An older harvest block on the western edge of the block near FC 9 and FC 10 has exposed this section of the riparian area to some wind and resulted in minor windthrow. The remaining standing trees between FC 9 and FC 10 are somewhat wind adapted due to this previous exposure. The block boundary has been placed along the base of the slope from FC 20 to FC 25 to minimize exposure to wind. From FC 1 to FC 9 there is a higher percentage of western redcedar trees which are expected to remain windfirm along this riparian reserve zone. In addition, a feathered edge in the RMZ next to stream #1 (FC 1-9) and aerial pruning are planned to mitigate windthrow along the RRZ.

D.R. systems inc. Phoenix Professional 3.4.2 Silviculture Prescription: 3.5.0

Date: Sep/25/2000 Page: 8

ID: 595-900+-6/900+ Licence No.: TFL 6 Timbermark: 6/900+ CP: 900+ Block: 595

J. ADMINISTRATION

PRESCRIPTION PREPARED BY (RPF SIGNATURE AND SEAL):

Prescription prepared by:

Amy Beetham
Name (Printed)

R.P.F.'s Prescription Approval:

Amy Beetham
R.P.F.Name (Printed)

Sept. 25/00 — Date Signed (YMD)

3647 — R.P.F. Number

REGISTERED PROFESSIONAL FORESTER
PROVINCE OF
SEAL AMY BEETHAM
BRITISH COLUMBIA
NO. 3647

Amy Beetham
R.P.F. SIGNATURE AND SEAL

MAJOR LICENSEE SIGNING AUTHORITY:

Licence Holder Signing Authority Signature

00/9/26
Date Signed (YMD)

M. G. Brown Operation Manager
Licence Holder Signing Authority Name (Printed)

PRESCRIPTION APPROVED BY:

District Manager's Signature

District Manager's Name (Printed)

Oct 5/00
Date Signed (YMD)

Original Approval Date (if Amended) (YMD

Printed Date: Sep/25/2000 Page: 9

File ID: 595-900+-6/900+	Licence No.: TFL 6	Timbermark: 6/900+	CP: 900+	Block: 595

K. PRESCRIPTION REFERRALS

PRESCRIPTION REFERENCES:

On File	Attached	
__	__	Additional SP Comments
__	X	SP Maps
__	__	Field Data Cards eg. SP plot cards, forest health evaluations, fuel hazard assessment cards, site sensitivity field cards. plantability field cards, etc.)
__	__	Copies of Advertising
__	__	Copies of Referrals
__	__	Comments from Advertising/Referrals
__	__	Assessments on Actions Resulting from Comments
__	__	Tree Acceptability Criteria for Subsequent Surveys
__	__	Terrain Stability Field Assessment
__	__	Archaeological Impact Assessment
__	__	Visual Impact Assessment
__	__	Gully Assessment
X	__	Riparian Assessment
__	__	Forest Health/Pest Incidence Assessment
__	__	Stand and Stock Tables
X	__	Permanent Access Calculation Sheet
	__	Other Attachments

PRESCRIPTION ASSESSMENTS COMPLETE

__ Terrain Stability Field Assessment

__ Archaeological Impact Assessment

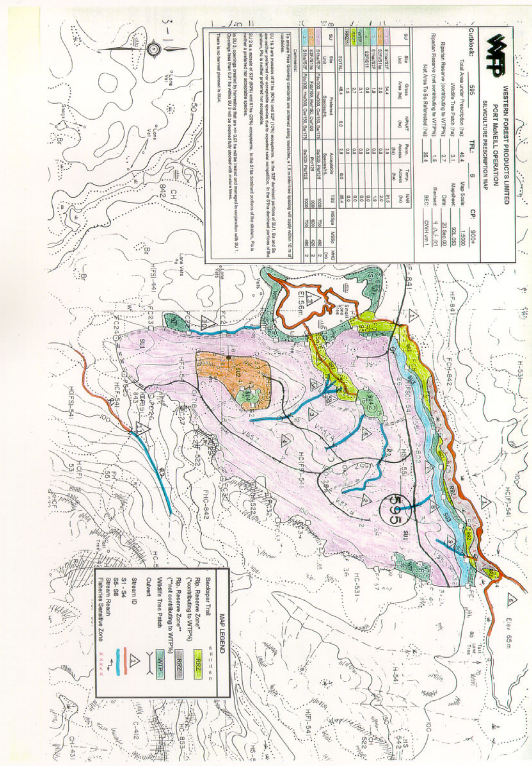
__ Visual Impact Assessment

__ Gully Assessment

X Riparian Assessment

__ Forest Health/Pest Incidence Assessment

D.R. systems inc. Phoenix Professional 3.4.2 Silviculture Prescription: 3.5.0



Silviculture Treatment Regime

Western Forest Products Ltd.

Port McNeill Operation

X Original
__ Amendment #

Printed Date: Oct/13/2000 Page: 1

Licence No.: TFL 6	Timbermark: 6/900+	Cutting Permit: 900+	Block: 595

TENURE IDENTIFICATION

Region	District	TSA	TSB	Licence No.	C / P	Timbermark	Cutting Permit #	Block
VAN	PM			TFL 6	C	6/900+	900+	595

Location	Authority/Opening No.	Licensee
Port McNeill	92L.053 /	Western Forest Products Ltd.

SILVICULTURE PLAN

SITE PREPARATION - Preferred Treatment

Pile roadside accumulations and burn during low hazard conditions if post harvest assessment indicates regeneration objectives cannot be met without treatment.

SITE PREPARATION - Alternative Treatment

Mechanical site preparation of roadsides to prepare 800-1000 plantable spots/ha if post harvest assessment indicates regeneration objectives cannot be met without treatment.

RATIONALE (if required)

Piling and mechanical site preparation are expected to increase the number of available plantable spots in the block in order to achieve stocking standards.

REGENERATION - Preferred Treatment

Plant SU 1 with a Fdc (Hw, Cw) mix. If available, Cw should be targeted along the eastern boundary in the S2F portions of the stratum along with Fdc.
Plant SU 2 with a Fdc/Cw mix. Consider using 410 stock or equivalent in SU 2.
Plant SU 3 with a Fdc (Hw, Cw) mix.
Consider using fertilizer at time of planting, targeting trees planted along roadsides in mineral soil in SU 1 and SU 2.

RATIONALE (if required)

Natural regeneration of Hw is expected to fill-in.

Fertilization of roadside trees at time of planting will assist seedlings in establishing in these harsher microsites, in order to meet free growing objectives.

410 stock is preferable because of shallow soils in SU 2.

BRUSHING - Preferred Treatment

Expect herbicide stem injection/cut stump treatment and/or manual treatment to control alder in all SUs, 7-10 years post-harvest unless survey shows treatments are not required to meet free growing objectives.

RATIONALE (if required)

Red alder has a ready seed source on northern Vancouver Island.

STAND TENDING/PRUNING/FERTILIZATION - Preferred Treatment

No stand tending, pruning, or fertilization treatment anticipated.

RATIONALE (if required)

D.R. systems inc. Phoenix Professional 3.4 Siviculture Treatment Regime: 3.4

Printed Date: Oct/13/2000 Page: 2

Licence No.: TFL 6	Timbermark: 6/900+	Cutting Permit: 900+	Block: 595

PRESCRIBED BY:

Amy Beetham
R.P.F.'s SIGNATURE

REGISTERED PROFESSIONAL FORESTER PROVINCE OF AMY M. BEETHAM BRITISH COLUMBIA NO. 3647

LICENSEE'S AUTHORIZED SIGNATURE

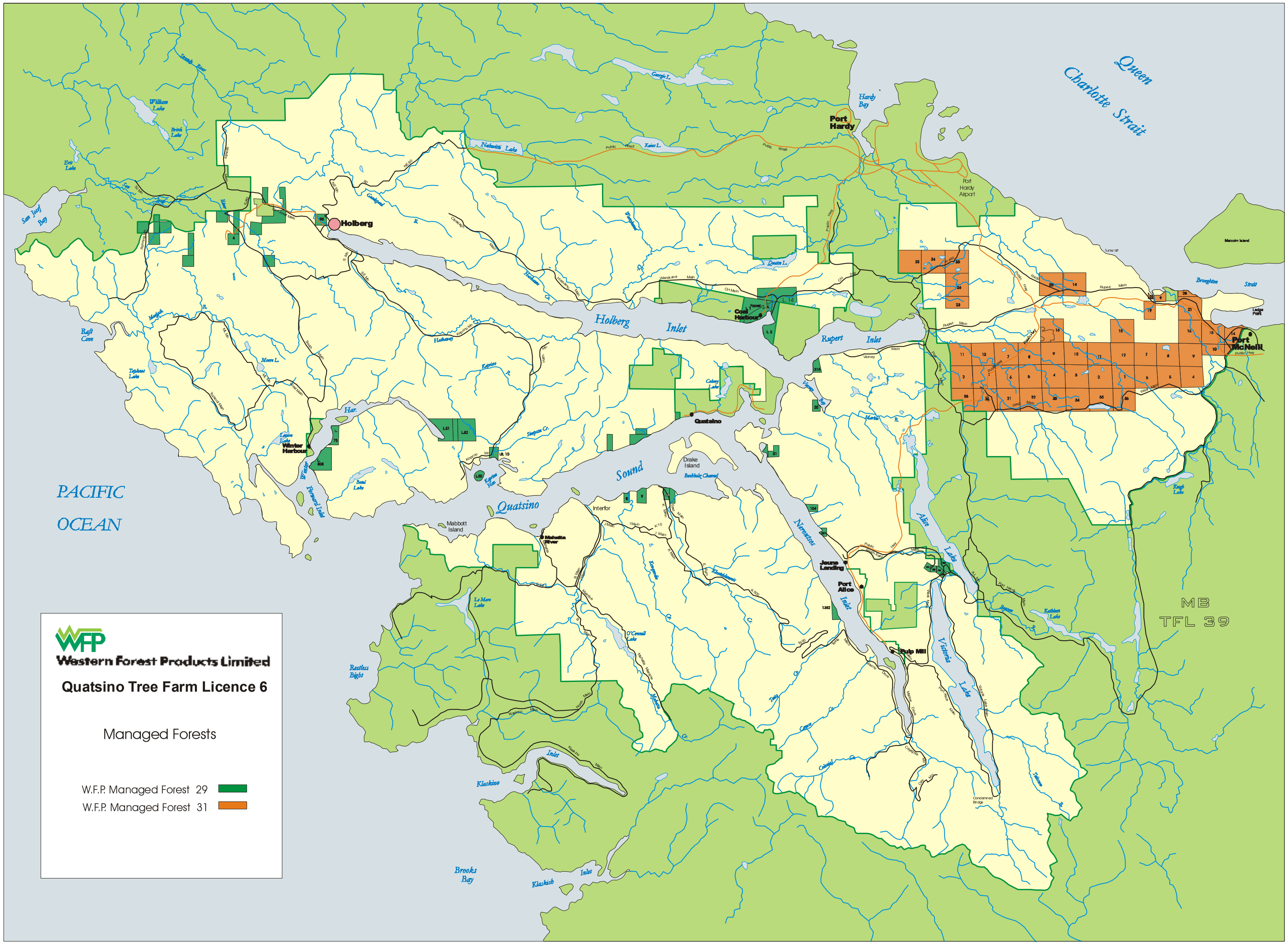
2000 / 10 / 13
DATE SIGNED (ymd)

DATE SIGNED (ymd)

D.R. systems inc. Phoenix Professional 3.4 Siviculture Treatment Regime: 3.4

Appendix VI

Managed Forest No. 29 And 31



Managed Forest 29 and 31

Appendix VII

Sample Fire Preparedness Plan

WESTERN FOREST PRODUCTS LIMITED
PORT McNEILL, NAKA CREEK, VICTORIA LAKE AND COAL HARBOUR
FOREST OPERATION

Consolidated

2000
Fire Preparedness Plan

for

Tree Farm Licence 6 & 25
And Woodlot #72

Definitions

person's area of operation means the area on which or within 500m of which the person is carrying out an industrial activity. [FPC s.83]

industrial activity includes land clearing, timber harvesting, mechanical site preparations and other silviculture treatments, road construction and any prescribed activity [FPC s.75] including engineering, salal picking, yew bark harvest, and blasting [169/95 s.1(2)].

forest includes Crown or private land that is forested in the long-term. [FPC s.75]

worksite means the location of any silviculture, road construction, other activity, or any timber harvesting area of land (cut block) [169/95 1 (1)].

burn area includes, for the purposes of B.C. Regulation 169/95 section 23.8, all of the cutblocks within the Port McNeill Forest Operation where pile burning operations are being conducted[20].

Anticipated Location of Operations

Operations are anticipated during the current fire season at the work sites indicated in Appendix A and in the following section.

Anticipated Worksites and Timings of Operations

Actual timing of operations will vary from those indicated in Table 1 as many operational constraints may interact to change the date of planned operations.

[20] As outlined in presentation by D. Orosz and C. Friesen at the March 4, 1999 Open Fire Regulation meeting at the Port McNeill Forest District Office.

2000 Fire Preparedness Plan – Activities by Road and Projected Timing

Road Name	Area	Activity (Road Construction, Falling & Yarding)	Start - End
TFL 6 Block 2 (PM)			
K1100	106	Road Construction	April - Oct
W80	221A	" "	April - Oct
R100	63	" "	June-Sept
W80	219	" "	April - Oct
W87	227	Yarding & Loading	April – June
W87	228	" "	April – June
W87	229	" "	April – June
K1800	53	" "	April – June
K1800	53A	" "	April – June
K1800	53B	" "	April – June
K1800	106	" "	Sept - Oct
W80	202	" "	Sept - Oct
R600	432E	" "	Sept - Oct
R449	509	" "	Sept - Oct
W80	221A	" "	Sept - Oct
Center Mn	51	" "	July – Oct
Center Mn	51A	" "	July – Oct
K1800	52	" "	July – Oct
R400	504	" "	October
R440	416	" "	June
TFL 6, Block 1 (CH, VL, RI)			
M26	587	Road Construction	April – Oct
M26	588	" "	April – Oct
SE73	120	" "	April – Oct
SE73	124	" "	July – Oct
W25	130	" "	April – Oct
SE73	132	" "	April – Oct
P100	404	" "	June-Sept
CL141	344	" "	April – Sept
CH245	934	" "	June-Oct
V200	574	Yarding & Loading	April - June
V Main	592	" "	Sept – Oct
V130	567	" "	April - June
V Main	596	" "	Sept – Oct
VL325	126	" "	April - June
M26	587	" "	July – Aug
M26	588	" "	July – Aug
P100	404	" "	Sept – Oct
CL141	394	" "	Sept – Oct
CL140	341A	" "	April – July
H1500	371	" "	July – Aug
P200	484	" "	July – Oct
P200	486	" "	July – Oct
TFL 25 Block 3 (NK)			
N120	75A	Road Construction	April – Sept
N121	71A	" "	April – Sept
N700	77	" "	July – Oct
N620	73	" "	July – Oct
N700	68	Yarding & Loading	June – July
T200	107	" "	April – Aug
T200	112	" "	April – Aug
P Main	48H	" "	June - Sept
Woodlot #72			
WN500	B6	Yarding & Loading	April - May
WN500	B6A	" "	April - May

Falling will normally take place 1-3 months in advance of road construction and yarding.

Weather Stations

Several weather stations will be maintained during the current fire season. Initial locations are indicated in Appendix A; however stations will be moved during the course of the season as prescribed burning activities are completed and logging operations move. Data will be collected at least monthly and more frequently as fire hazard increases.

Fire Detection

During normal operating periods, all personnel watch for fires. During shutdowns supervisory staff and forestry/engineering crews are retained to provide detection and initial attack capability.

Fire watches will be maintained during periods of higher fire hazard as indicated in Appendix D.

Blasting, welding, and other hot work will be monitored for at least 30 minutes after completed or longer during periods of high fire hazard as per Appendix D.

Initial Fire Suppression

If a fire is detected within 1.0 kilometres [FPC s83+92] of a worksite, all employees, tools, and equipment working within 1000m of the fire [169/95 s.35(2)] will, as necessary, be used to suppress the fire [169/95 s.35(2)]. If necessary, additional workers will be directed to the fire from forestry/engineering, logging, and road construction crews. As needed, additional tools and equipment will be directed to the fire from tool caches and other work sites within the Forest Operation [169/95 s.35(2)(b,c)].

Voluntary actions will be taken on fires within our Forest Operation which are greater than 1500m from WFP worksites on the understanding that reimbursement will be as specified under “Rates of Compensation“ below. However, reimbursement for air tankers and helicopters will not be expected unless prior Ministry approval is granted [All Licensees letter, 99/03].
When a fire watcher or any employee detects a fire the employee will report the fire by radio or telephone to the foreman or the Operation office and then take action to suppress the fire using the fire tools and equipment at the work site [169/95 s.4].

Reporting

Any unattended or unauthorized fire will be immediately reported to the Port McNeill Operation office. The office will report the fire to the Ministry of Forests at 1-800-663-5555 or to the Port McNeill District office at 956-5000 [FPC s.86,92(1)(a)].

Any authorized fire which spreads beyond the intended area shall be suppressed and reported as above [FPC s.88].

Number of People

The normal number of persons employed at the Port McNeill Forest Operation is about 100, Coal Harbour Operation is about 7 and for Naka Creek is about 17. Additional silviculture workers may be present at certain times of the year for short duration contract work. The logging contractor at Naka Creek maintains his own equipment cache and fire equipment. The following table outlines the approximate permanent workforce by location.

UNIT	Staff	Hourly	Contract
Naka Logging			17
Coal Harbour (Quatsino Forestry)		7	
Port McNeill	12	68	20

Tools and Equipment

A central cache of tools and equipment will be maintained at Port McNeill with a minimum of 22 shovels, 14 pulaskis, 14 hand-tank pumps, and 6 Wajax Mark III (or similar) pumps [169/95 s.11, Schedule 2]. A central cache of tools and equipment will be maintained in Naka Creek with a minimum of 4 shovels, 4 pulaskis, 2 hand-tank pumps, and 1 Wajax Mark III (or similar) pumps [169/95 s.11, Schedule 2].

Chainsaws or other small engines will not be operated unless there is a ½ lb chemical fire extinguisher nearby [169/95 s.13 (1)(b),(3)]. All **machines** will be equipped with one shovel, one pulaski, two fire extinguishers with ULC ratings of 1A 5BC and 3A 10BC each.

At **each worksite** there will be at least one shovel, one pulaski, and one piss-can. If **more than three persons** normally work at a work site then there shall be an additional piss-can for the 4th, 7th, 10th, 13th, 17th, 20th, and 23rd additional person present and one pulsaki or shovel (in roughly equal numbers) for every additional person present [169/95 s5].

At **blasting** and **welding** sites there shall be a second piss-can and shovel in addition to the requirements above [169/95 s.8]. Welding sites shall have two chemical fire extinguishers with ULC ratings of at least 3A 10BC [169/95 s.7].

Where there are normally more than **three workers** at a worksite, a Wajax Mark III pump unit or water tanker (Wajax Mark III plus 550 gallons with ½% foam [169/95 s1(1)] or 1000 gallons without foam [169/95 s.10(6)]) including 1500' of 11/2" hose must be at the work site. If there are normally more than **10 workers** at a work site, two Wajax Mark III pump units or water tankers and 3000' of hose are required.

When cables are used, they are to be running straight and not rubbing on rocks, wood, or any flammable material. Wood and other flammable materials are to be cleared away from cables for at least 3' from tailhold blocks and a full piss can is to be kept close by [169/95 s.15].

Trained Personnel

The following employees have received fire suppression training:

Port McNeill Operation		Coal Harbour	
M. Brown	956-4629	Quatsino Forestry	
P. Wainwright	956-2158	R. Clair	949-7785
E. Gagné	956-3354	J. Clair	949-6809
J. Muress	956-3579	T. Clair	902-0656
W. Anderson	956-2565	H. Johnny	902-0552
J. Broekhuizen	956-4562	R. Nelson	949-8559
G. Anderson	956-3682		
M. DesRochers	956-2141	Naka Logging	
K. Marzoff	956-3951	K. Roberts	282-3239
M. Roscoe	949-6891	B. Flower	282-3444
R. VanderValk	956-4890		
B. Dustin	956-4109		
A. Moffat	956-4904		
W. Tantrum	956-3925		

Key Personnel

WFP Port McNeill Forest Operation **956-3391**

Micky Brown	Manager	956-4629, 949-3928
W. Anderson	Woods Foreman	956-2565, 949-4167
Phil Wainwright	Resident Engineer	956-2158
Eric Gagné	Resident Forester	956-3354
Jim Muress	Road Foreman	956-3579, 949-4104
Grant Anderson	Master Mechanic	956-3682, 949-3901, 949-1022, 949-0272
Jack Broekhuizen	Field Engineer	956-4562
Mike DesRochers	Assistant Forester	956-2141
Chris White	Warehouse	956-4668, 902-1022
Stan Momot	Accountant	956-2266
Lito Pineda	Timekeeper	956-2196

Quatsino Forestry		949-7278	
	Band Office	**949-6245**	
Helen Wallas	Band Manager	**949-2639**	
WFP Vancouver		**665-6200**	
			Home #
Vic Woods	V/P & General Manager		986-2332
Bill Dumont	Chief Forester		924-0146

WFP Northern Division Administration 956-4446

Gary Griffith	Regional Manager	956-2988, 949-3908
Kerry McGourlick	Forestry Manager	956-4939, 949-4194
Bill Day	Divisional Engineer	956-2510

Naka Logging		282-3381	
	Camp	**974-5733**	
Ted Arkell	Manager		923-8999, 286-2511
			830-8440
Ken Roberts	Supervisor		282-3239, 923-6902
			286-2311, 287-6291
Bruce Flower	Chargehand		282-3444, 286-2211
Ed Cyr	Watchman		830-7591, 830-7552

MOF Parksville	Fire Control Centre	1-250-951-4200
	Fax	1-250-954-0823
	Emergency Fire Report	1-800-663-5555
Gary Munro	Manager, Fire Control Centre	1-250-951-4208
Darrell Orosz	Protection Officer	1-250-951-4216

MOF Port McNeill Forest District **956-5000**

		Home #
Jack Dryburgh	District Manager	956-3138
Chuck Van Hemmen	Operations Manager	956-4767

MOF Campbell Forest District **286-9300**

Duty Rosters

2000 Fire Duty Roster
Port McNeill, Coal Harbour, and Naka Creek Forest Operations

During **regular working hours** (M-F 7:00 a.m. – 6:00 p.m.) contact the Port McNeill Operation Office at **956-3391**. After hours and on weekends contact the designated Duty Officer as listed below:

Date	Standby Personnel	Residence, alternate/autotel or cell
April 1 - 20	Jim Muress	956-3579 /949-4104
April 21 - May 12*	Phil Wainwright	956-2158
May 13 - June 2*	Stan Momot	956-2266
June 3 - 16	Grant Anderson	956-3682, 902-1022 /949-3901 /949-0272
June 17 – 29	Jack Broekhuizen	956-4562
June 30 – July 14*	Amy Beetham	956-3260
July 15 - 28	Micky Brown	956-4629 /949-3928
July 29 – Aug 11*	Eric Gagné	956-3354
Aug 12 – Sept 1	Lito Pineda	956-2196
Sept 2 - Sept 22*	Bill Anderson	956-2565 /949-4167
Sept 23 – Oct 6	Mike DesRochers	956-2141
Oct 7 - 31*	Chris White	956-4668, 902-1022

* Denotes a long weekend and bold denotes 3 weeks.

If the designated Duty Officer is not available at the number above, call the
Duty Officer pager at 949-5486.

Quatsino Forestry - Duty Roster

During working hours contact	Quatsino Forestry	**949-7278.**
	Band Office	**949-6245**

Off hour responsibility for the duty roster will be:

	Helen Wallas	**949-8372**

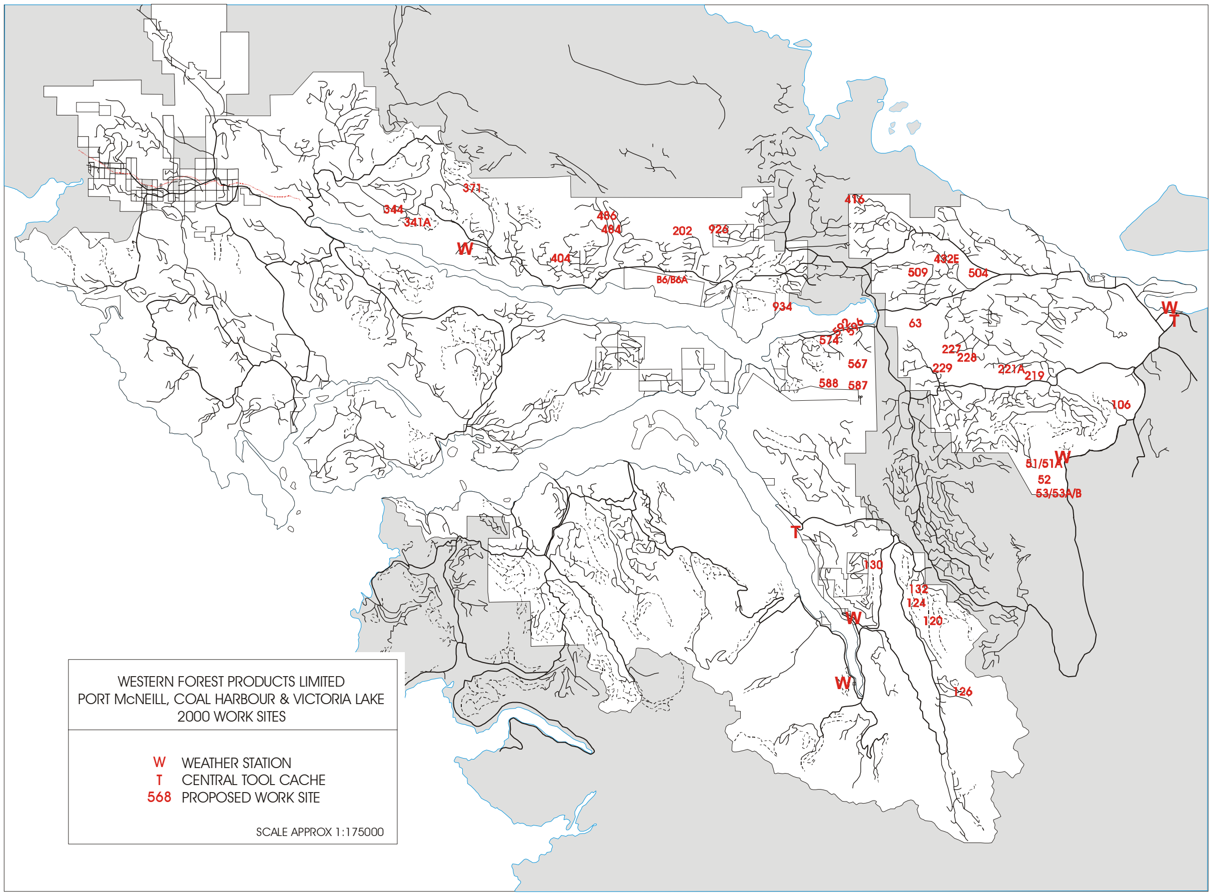
The WFP duty officer should be contacted at the same time.

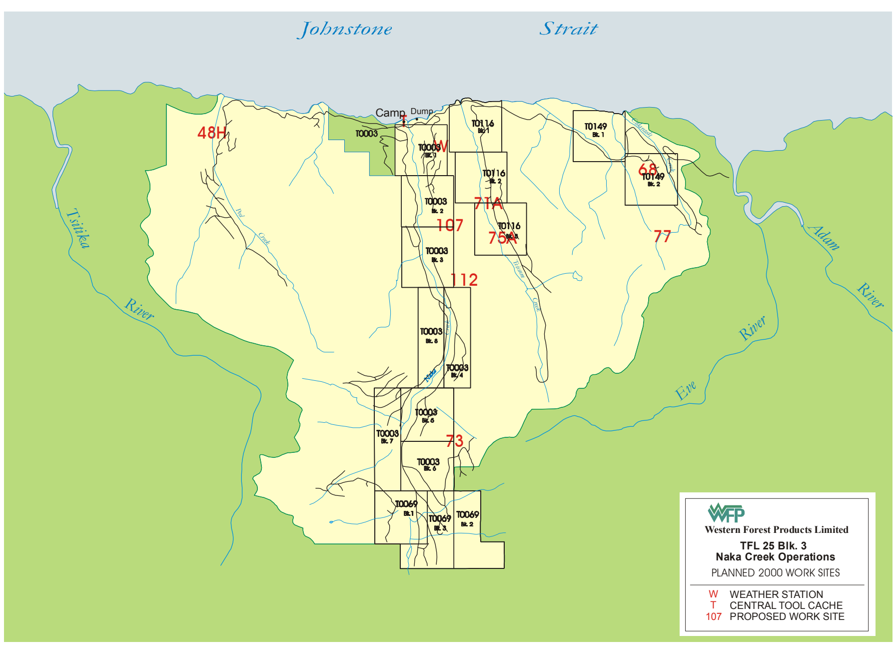
Rates of Compensation

WFP will undertake suppression activities on all fires within our Forest Operation. Those which are no fault of WFP will be suppressed on the understanding that compensation will be forthcoming [FPC s.95] and that labour costs will be reimbursed at current I.W.A. rates including overtime [169/95 s.37(2)] and that equipment costs will be covered at actual rates.

There may be times when it is more efficient for WFP employees to carry out fire suppression activities outside of our Forest Operation than it would be for the MOF to organize a workforce. WFP will supply fire fighting crews and equipment as requested or directed [FPC s.94] on the understanding that compensation will be as above. For requests, the amount of equipment and numbers of men available for suppression work outside of our Forest Operation will depend on the operational situation at the time of the request for assistance.

Appendix A. **1999 Worksites Map**





Appendix B. Equipment List

<u>Port McNeill Forest Operation, Equipment Suitable for Firefighting</u>

<u>WFP</u>

6 hydraulic forwarders
1 850B Bulldozer
1backspar hoe
1 lowbed
10 logging trucks (4 tanker capable)
2 front end loaders

<u>Contract</u>

3 roadbuilding hoes
5 Gravel trucks

<u>Pumps & Hose</u>

16 Wajax MK III pumps
4 Gorman-Rupp pumps
4 Honda pressure pumps
4 Honda volume pumps
2 Midland volume pump
4,200 meters l.5 inch hose
3,300 meters 1.0 inch hose

<u>Tankers</u>

5 2,700 liter pull-tanks
3 l6,400 liter porta-tank
1 l3,000 liter porta-tanks
1 2250 liter tank truck
4 ll,000 liter portable bladders

<u>Quatsino Forestry</u>

1-Hydraulic forwarder

<u>Tankers</u>

1-13,000 litre porta-tank
1-Wajax pressure pump & hose
270 meters 1.5 inch hose
270 meters 1.0 inch hose

<u>Naka Logging</u>

1-Excavator
1-Crawler Tractor

<u>Tankers</u>

1-4,545 litre tank truck
1-9,090 litre tank truck
2-Wajax MK III pumps
900 meters of hose

Appendix C. Risk Classifications

Risk Classification A (High)

- blasting
- chainsaw work (falling & bucking, spacing, girdling, trail building)
- grapple yarding
- skidding
- welding, metal cutting/grinding
- mechanical site preparation

Risk Classification B (Moderate)

- hoe forwarding
- bucking at roadside
- firewood cutting

Risk Classification C (Low)

- bridge building
- drilling (not blasting) & quarrying
- lowbedding
- road construction, maintenance, & excavating
- road surfacing
- log hauling
- log loading
- log dumping
- pruning
- alder control (hand tools only)
- broadcast[1] and individual tree fertilization
- planting
- herbicide applications[21]
- engineering/forestry[1]
- salal/yew bark collection
- use of hand tools

[21] notwithstanding B.C. Reg. 169/95, Schedule 1, use of a helicopter (silviculture-large engine) for aerial spraying or fertilization or general forestry/engineering activities and use of a pump (silviculture-small engine) for power/hose & nozzle herbicide application is deemed, for purposes of the Port McNeill Forest Operation, Risk Classification “C”.

Appendix D. Restriction on Industrial Operations [169/95, Sch. 3]

The Port McNeill Forest Operation is within Danger Region 1.

After **three consecutive days in DGR III (MODERATE**) a fire watch will be maintained for one hour after blasting, falling & bucking, cable yarding, juvenile spacing, welding, hoe forwarding, and roadside bucking; and

After **three consecutive days in DGR IV (HIGH)** blasting, falling & bucking, cable yarding, juvenile spacing, and welding will cease after 1:00 pm until DGR falls below III (MODERATE) or is III for at least two consecutive days. Hoe forwarding, roadside bucking, and other activities may continue during normal operating hours, and

After **two consecutive days in DGR V (EXTREME)** blasting, falling & bucking, cable yarding, juvenile spacing, and welding will cease until DGR falls to IV. After three consecutive days of DGR V (EXTREME) hoe forwarding and roadside bucking will cease after 1:00 pm until DGR III or 3 consecutive days of DGR IV; other activities (C) may continue.

Appendix F. Radio Frequencies

WFP Port McNeill	153.020	simplex
	151.985	repeater transmit

Naka Creek	Tx	Tone	Rx	Tone
Dyer Repeater	166.53	167.9	167.25	167.9
Dyer Simplex	167.55	167.9	167.25	167.9
Naka Link	170.25	110.9	170.25	110.9